



COMPUTER NETWORKS



I

COURSE DISCUSSION



COURSE DISCUSSION

- 1. Course Content**
- 2. Learning Objectives**
- 3. Course Outcomes**
- 4. Prescribed & Reference text book**



COURSE DIVISION

UNITS

CHAPTER

Unit 1

- Introduction to Computer Network
- Physical Layer

Unit 2

- Data Link Layer
- MAC Layer

Unit 3

- The Network Layer
- Congestion Control

Unit 4

- Transport Layer
- Application Layer



CHAPTER -1 INTRODUCTION TO CN

1. Basic of Computer Network & its uses
2. Network Hardware
3. Network Software
4. Reference Models
5. Network Standardization.



CHAPTER 2 - PHYSICAL LAYER

1. Data & Signals
2. Guided Medium or physical medium
3. Unguided or wireless medium
4. Bandwidth Utilization - Multiplexing
5. Switching
6. Mobile telephone System.



CHAPTER 3 – DATA LINK LAYER

1. Data Link Layer Design Issues
2. Error Detection
3. Error Correction,
4. Sliding Window Protocol
5. Example Data link protocols.



CHAPTER 4 – MAC LAYER

1. MAC Sub Layer
2. Channel allocation problem
3. Multiple Access Protocols
4. Ethernet
5. Wireless LANs
6. Data link layer switching.



CHAPTER 5 – NETWORK LAYER

1. Network Layer Design Issues
2. Routing Algorithms



CHAPTER 6 – CONGESTION CONTROL

1. Congestion Control Algorithms
2. Quality of Service
3. The network layer in the internet.



CHAPTER 7 – TRANSPORT LAYER

1. Elements of Transport protocols
2. UDP
3. TCP
4. Performance issues.



CHAPTER 8 – APPLICATION LAYER

1. Domain Name system,
2. Electronic Mail,
3. WWW.



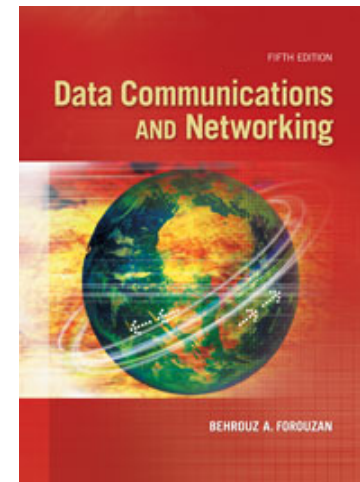
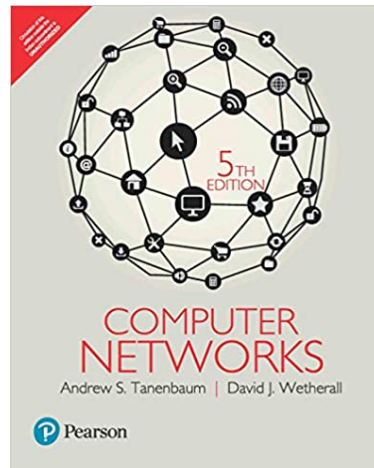
LEARNING OBJECTIVES & OUTCOMES

1. Identify the different components and their respective roles in a communication System
 - Upon Completion of the course, the students will be able to.
2. Design an enterprise network employing the common LAN technologies and be able to evaluate the advantages and disadvantages ☒
 - To master the fundamentals of data communications and networks by gaining knowledge of data transmission concepts.
3. Describe the importance and functions of the OSI layers Physical, data link, network and transport layer
 - Understanding the operation of physical and data link layer.
 - Learning the algorithms used to design data networks.
 - Understanding the principles of transport and application layers



PRESCRIBED & REFERENCE TEXT BOOKS

1. Andrew S Tanenbaum, David J Wetherall "Computer Networks", Pearson Education, 5th Edition, Elsevier Inc, 2014.
2. Behrouz A. Forouzan, "Introduction to Data Communication & Networking" 5th Edition, McGraw Hill Education Pvt Ltd 2013



THANK YOU



1. Introduction

BASIC OF COMPUTER NETWORKS & ITS USES



LECTURE -2

- Basic of computer network & Its uses

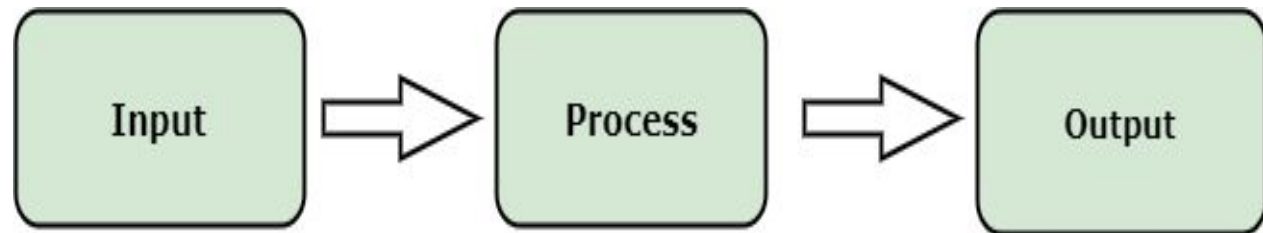
1. Computer
2. Computer system
3. Network
4. Computer Network



COMPUTER

- A **computer** is an electronic device that manipulates information, or data. It has the ability to **store**, **retrieve**, and **process** data.

IPO - Principle



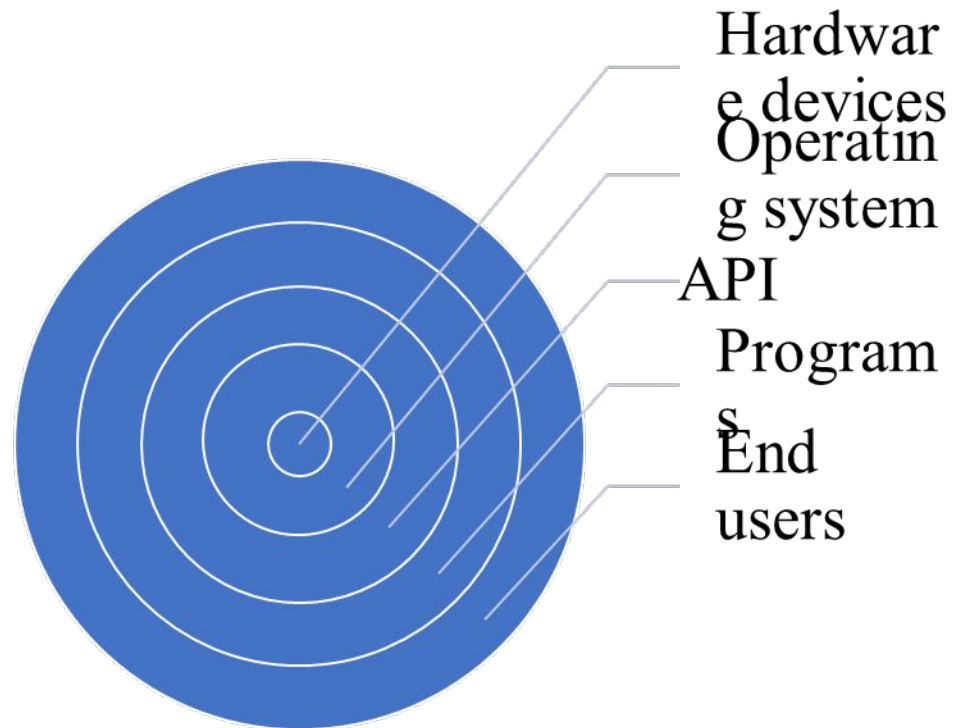
COMPUTER SYSTEM

1. It is a collection of entities(hardware, software and human ware) that are designed to receive, process, manage and present information in a meaningful format.



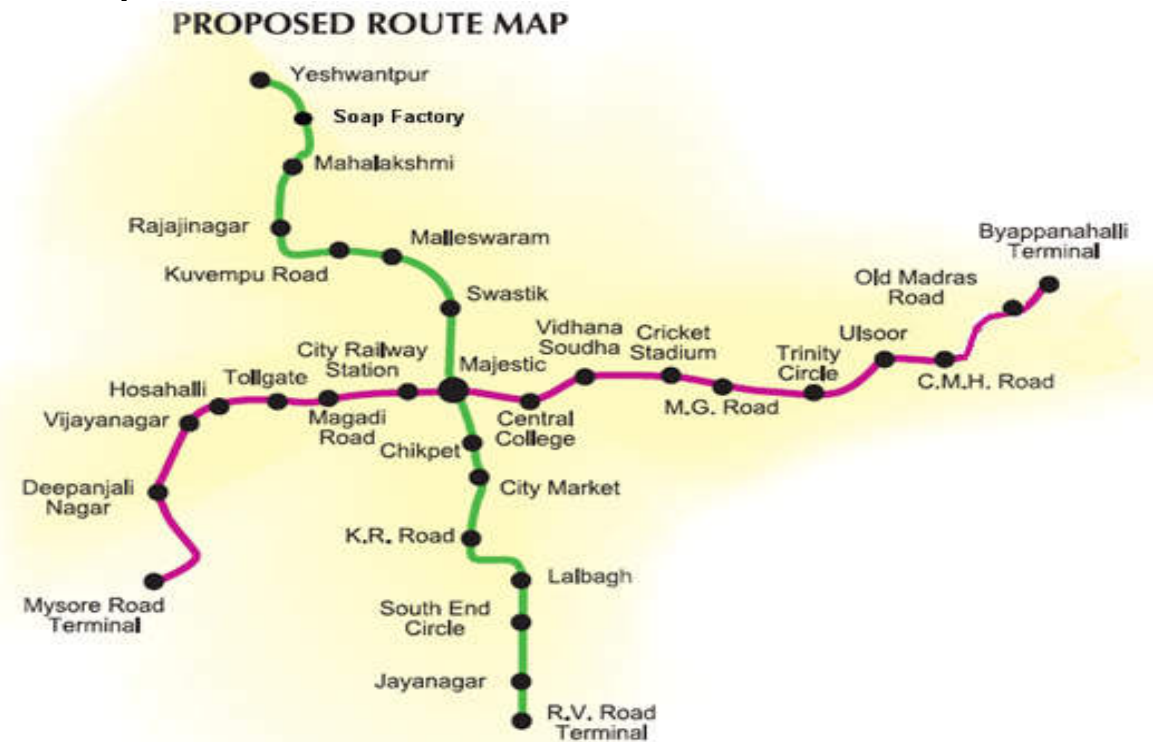
COMMUNICATION WITHIN COMPUTER SYSTEM

1. Software or Hardware alone will not serve purpose, need to channelize communication between them



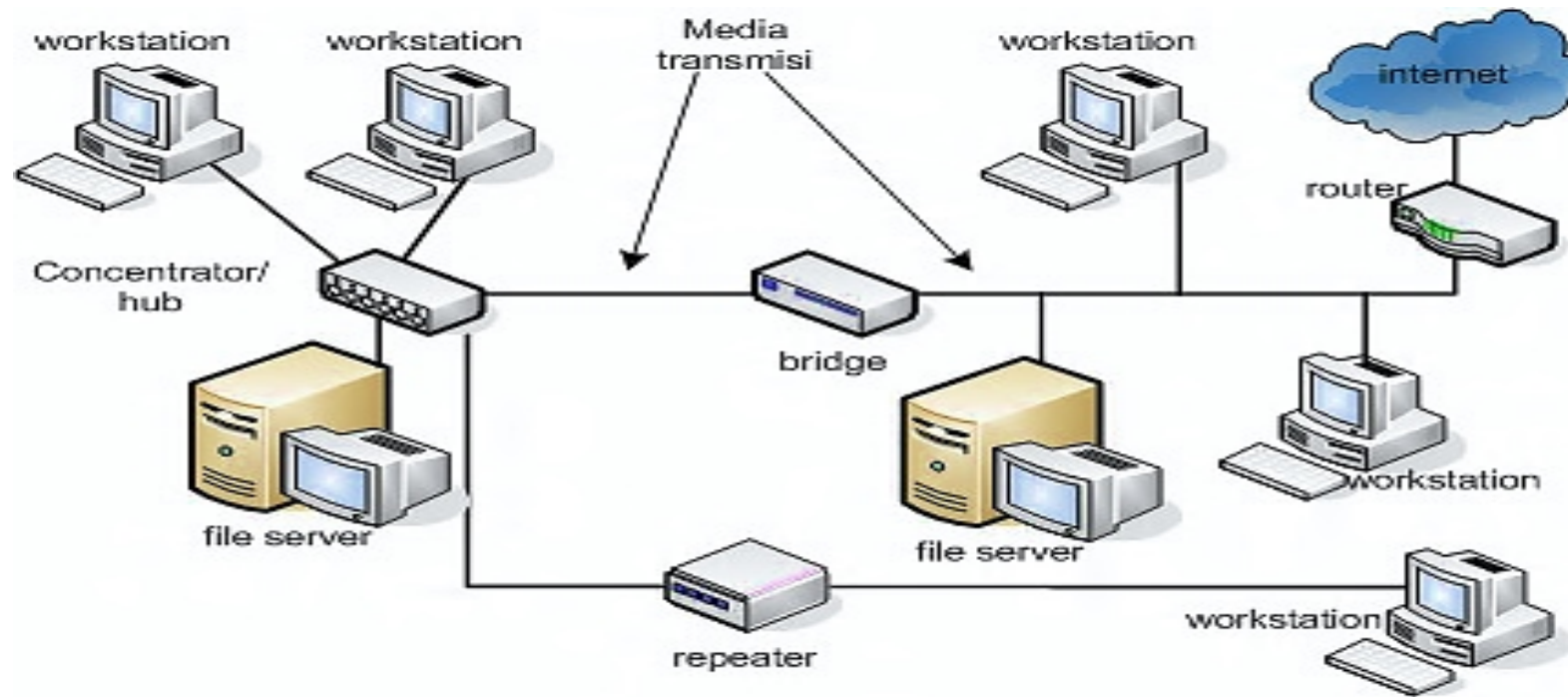
NETWORK

1. An arrangement of intersecting horizontal and vertical lines.
2. Ex: A group or system of interconnected people or things;
3. A complex system of roads, railroads



COMPUTER NETWORK

1. A computer network is a set of computers that are connected together so that they can share information or to establish communication.



SUMMARY

- Lecture-1 : Basic Of Computer Network & Its uses

Q.1 What is computer ?

Q.2 What is computer system?

Q. 3 How the communication is carried out within computer system?

Q.4 What is network?

Q. 5 What is computer network?



ACTIVITY-1

- Lecture-1 : Basic Of Computer Network & Its uses

Recommended Reading:

1. Computer Generation .
2. Types of memory storage.
3. Other communicatable devices.



| COMPUTER NETWORK &

1. Introduction

ITS USES



LECTURE – 3

Basic of computer network & Its Uses

Recap

1. Computer 
2. Computer system
3. Network
4. Computer Network

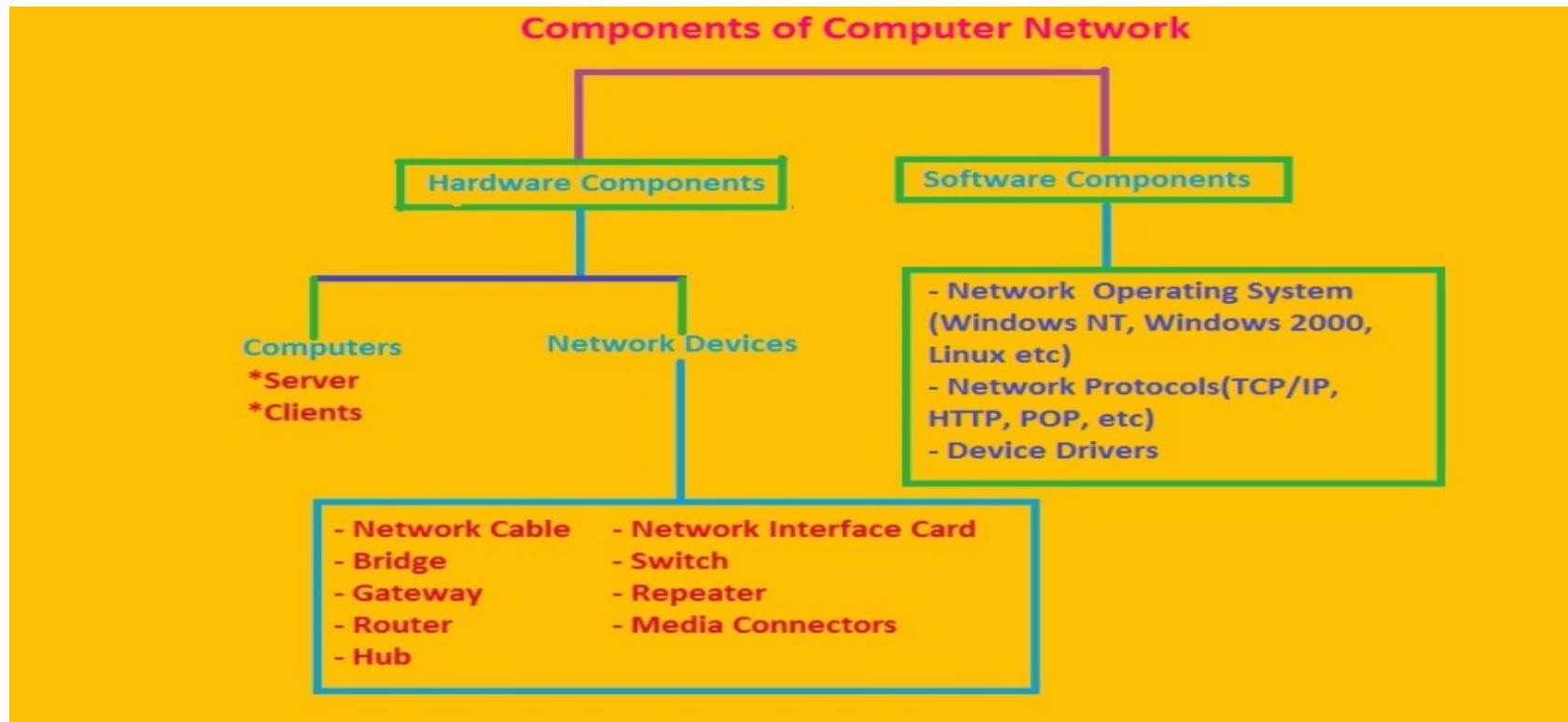
Objectives

- Components of Computer Network
- Why Computer Network - Purpose ?
- What are the uses of Computer Network – Applications ?



COMPONENTS OF COMPUTER NETWORK

1. Computer networks share common devices, functions, and features including servers, clients, transmission media, shared data, shared printers and other hardware and software resources, network interface card(NIC), local operating system(LOS), and the network operating system (NOS).



WHY COMPUTER NETWORKS?

1. - Sharing files
- Sharing devices: such as printers, scanners and photocopiers
- Communicating : using email, video, instant messaging and other methods
- Sharing software : and operating programs on remote system
- Allowing network users to easily access and maintain information



WHAT ARE THE USES OF COMPUTER NETWORKS?

1) Business Applications

- a) Resource Sharing
- b) Server-Client model
- c) Communication Medium
- d) E-Commerce

2) Home Applications

- e) Access to remote information
- f) Person-to-person communication
- g) Interactive entertainment



SUMMARY

- Lecture-2 : Basic of Computer Network & its uses

Q. 1 What is computer?

Q. 2 What is computer system?

Q. 3 What is computer network?

Q. 4 What are the components of computer Networks?

Q. 5 What are the applications of computer networks?



ACTIVITY - 2

- Lecture-2 : Basic Of Computer Network & Its uses

Website:

<https://www.studytonight.com/computer-networks/uses-of-computer-networks>

Quiz :

http://highered.mheducation.com/sites/0073376221/student_view0/chapter1/quizzes.html



NETWORK HARDWARE

1. Devices-1



LECTURE –4

Network Hardware

Recap



1. **Components of Computer Network**
2. **Why Computer Network - Purpose ?**
3. **What are the uses of Computer Network – Applications ?**

Objectives

- **Network – set of devices + links**
- **Relationship between – link, channel, medium**
- **Types of connection – point-to-point connection+ Multipoint connection**
- **Transmission Technology – Broadcast Links + Point-to-Point links.**



TYPES OF NETWORK DEVICES

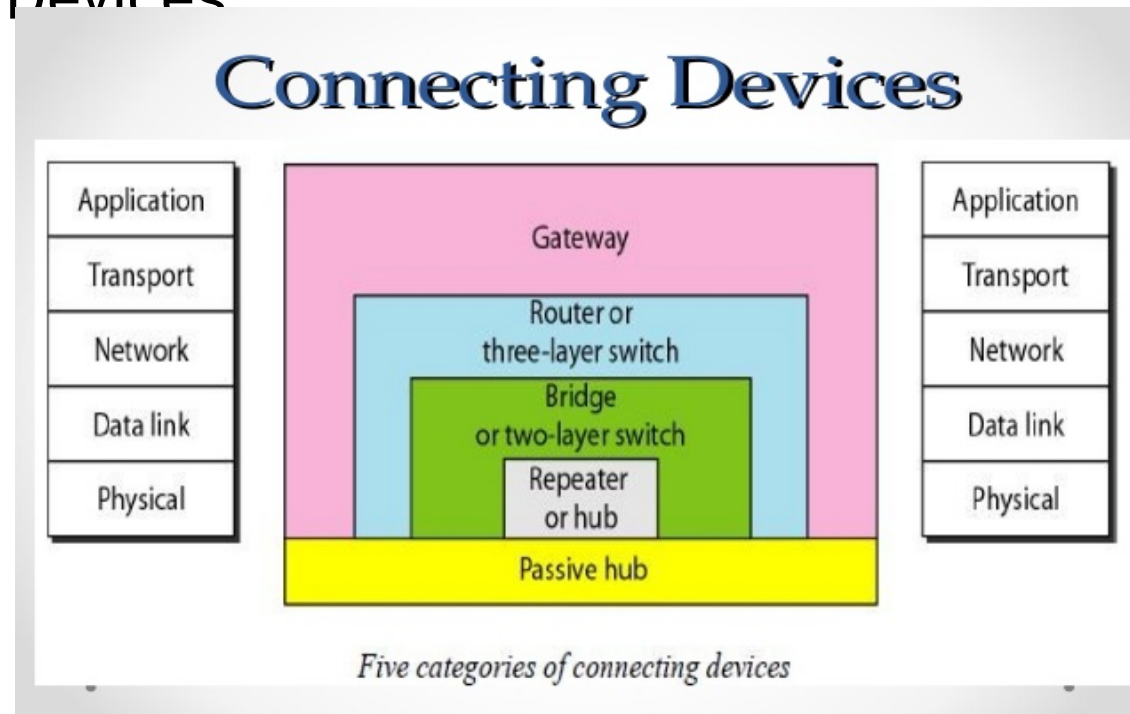
1. Communicating Devices

Communicating Devices	
Networking Devices	Internetworking Devices
1. Hub - Physical layer device 2. Repeaters – Physical layer 3. Bridges – Data layer devices 4. Switches - 2nd and 3rd layer device	1. Routers - Network layer / 4th layer device 2. Gateway – 5th layer device



TYPES OF NETWORK DEVICES

1. Connecting Devices



LINKS

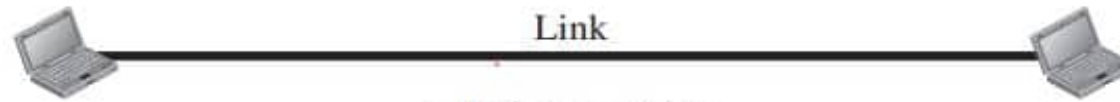
1. A **Medium** serves as the **means of communication** .
2. A **Link** is a communication pathway that transfer data from one device to another.
3. A **Channel** refers to the **means of transmission** of a message between the sender and the receiver.

1. A medium, which is an **abstraction**, can be oral , written or non verbal.
2. Channel, on the other hand could be a letter, a report, a book, a memorandum, a fax, an email, the television, the telephone, etc.

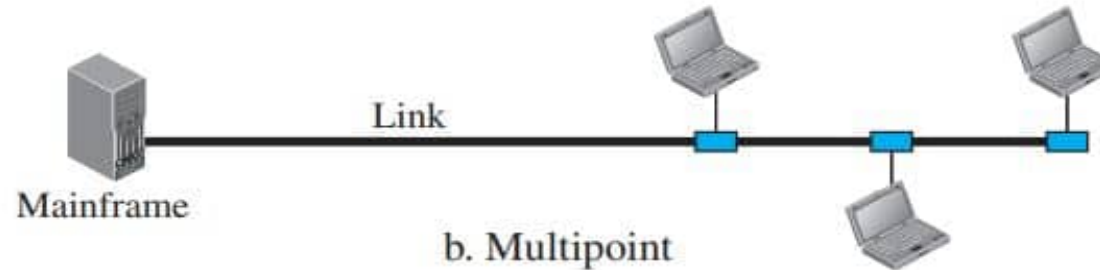


TYPES OF CONNECTIONS

Types of connections: point-to-point and multipoint



a. Point-to-point



TRANSMISSION TECHNOLOGY

1. Point-to-Point Network & Broadcast Network

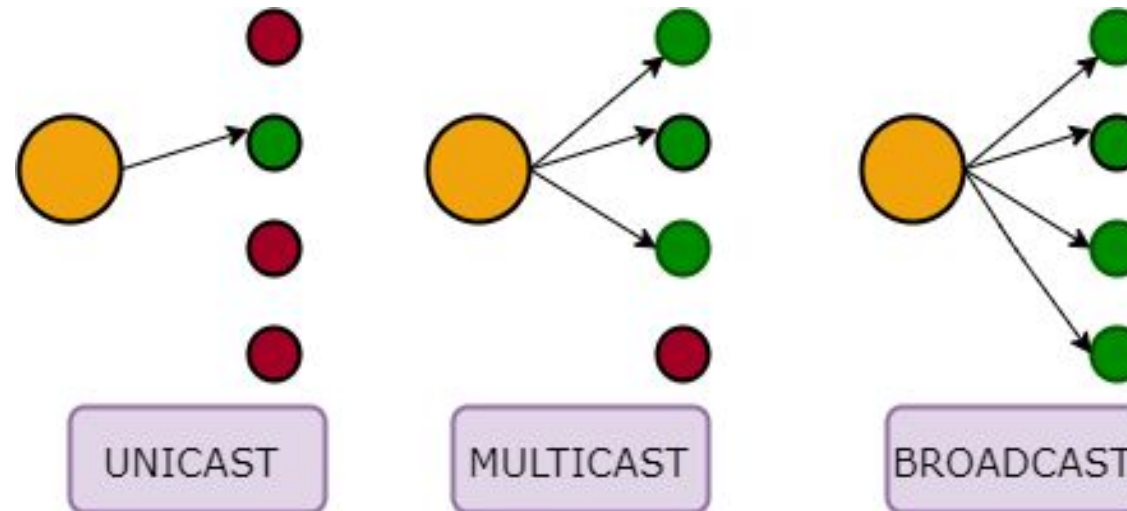
Sl. No	Point-to-point Link	Broadcast Link
1.	It connects individual pairs of machines.	The communication channel is shared by all the machines on the network.
2.	Dedicated Channel	Shared Channel
3.	Packets sent reaches to intended machine only	Packets sent by any machine are received by all the others. (An address field within each packet specifies the intended recipient.)
4.	Ex: Telephone, Leased line.	Ex: Radio, Television



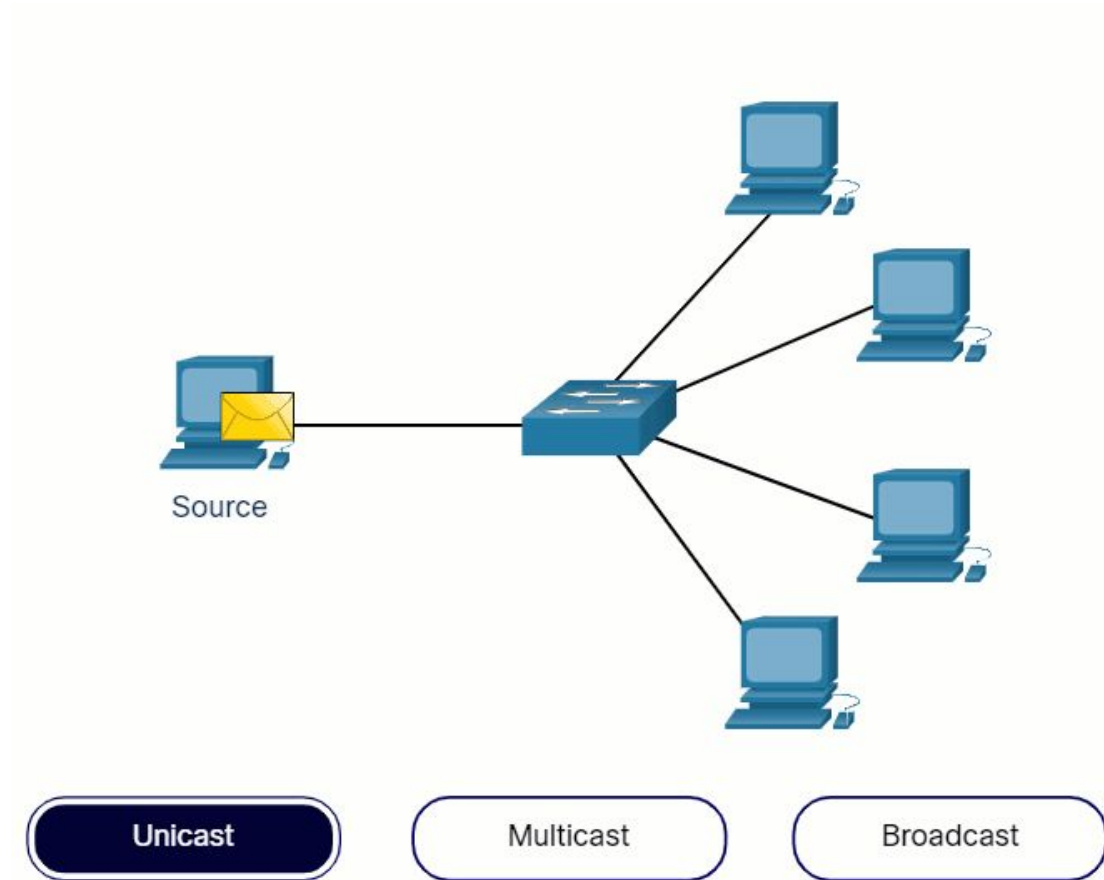
BROADCAST NETWORK

- Network type - 1

1. **Casting** – the way to stream data from source point to nor of destination point.



IN PICTORIAL FORM...



TRANSMISSION MODES

- Communication mode.
- Transferring of data between two devices.
- Direction of

Simplex Mode

- Unidirectional
- used by simplex system.
- Ex: loud speaker, TV, Radio, monitor etc

Half –Duplex Mode

- Transfer both direction but only one at a time.
- Ex: Walkie-Talkie

Full-Duplex Mode

- Bi-directional : at a time send and receive data.
- Ex: Telephone Network



SUMMARY

- Network Hardware

Q.1 What is Node?

Q.2 What are the components of computer network hardware?

Q.3 What are the types of computer network devices?

Q.4 What is medium, link, channel?

Q.5 What are the differences between computer medium , link and channel?

Q.6 What are different types of connection /transmission technology?

Q.7 What is casting? What are the different types of casting?

Q.8 What is transmission mode? What are the different types of Transmission mode?



ACTIVITY - 3

- Lecture 3: Network Hardware

Reading: Network topology

<https://www.comparitech.com/net-admin/network-topologies-advantages-disadvantages/>

Video: Network Device (Animation)

1. <https://www.youtube.com/watch?v=eMamgWlIRFY>
2. https://www.youtube.com/watch?v=1z0ULvg_pW8
3. <https://www.youtube.com/watch?v=ck3gx9HB9-k&list=PLSNNzog5eydtmcbcbc1b8pVRkgre3vNUy>



NETWORK HARDWARE

1. Devices-2



LECTURE –5

Network Hardware

Recap

1. Network -  of devices + links
2. Relationship between – link, channel, medium
3. Types of connection – point-to-point connection + Multipoint connection
4. Transmission Technology – Broadcast Links + Point-to-Point links.

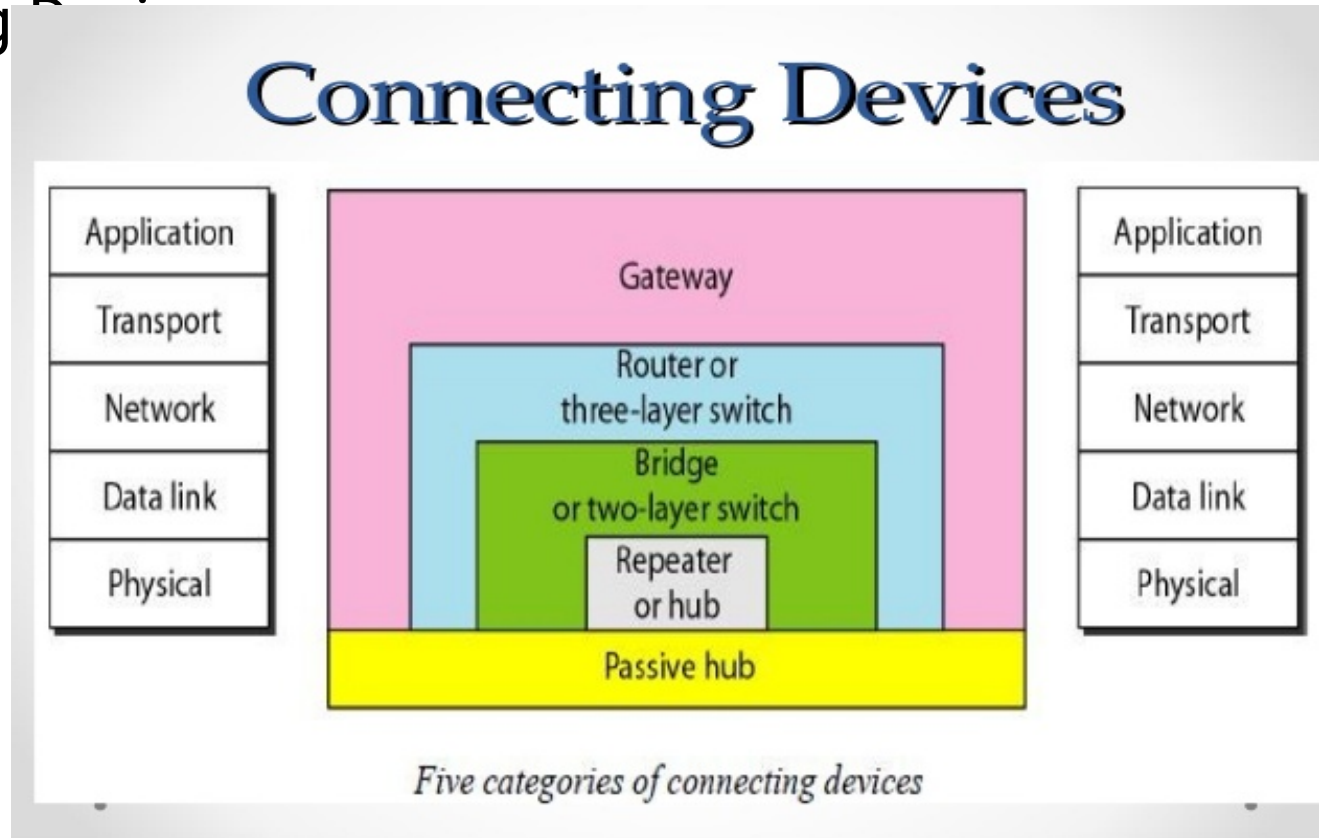
Objectives

- Hub
- Repeater
- Bridge
- Switch
- Gateway



TYPES OF NETWORK DEVICES

1. Connecting Devices



LAYER 1- NETWORK DEVICE



Passive Hub
device



Active Hub
device

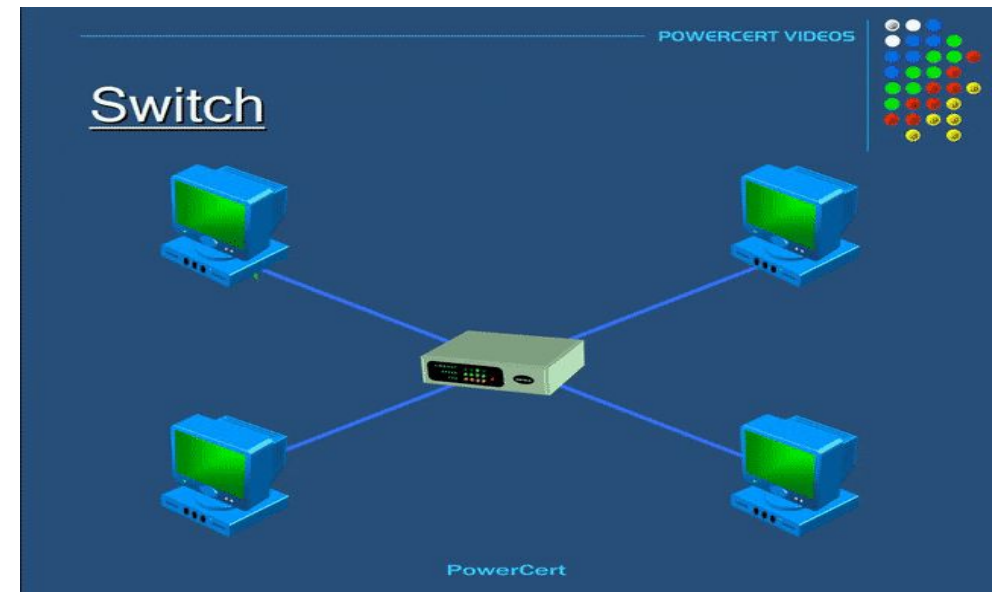


Repeater

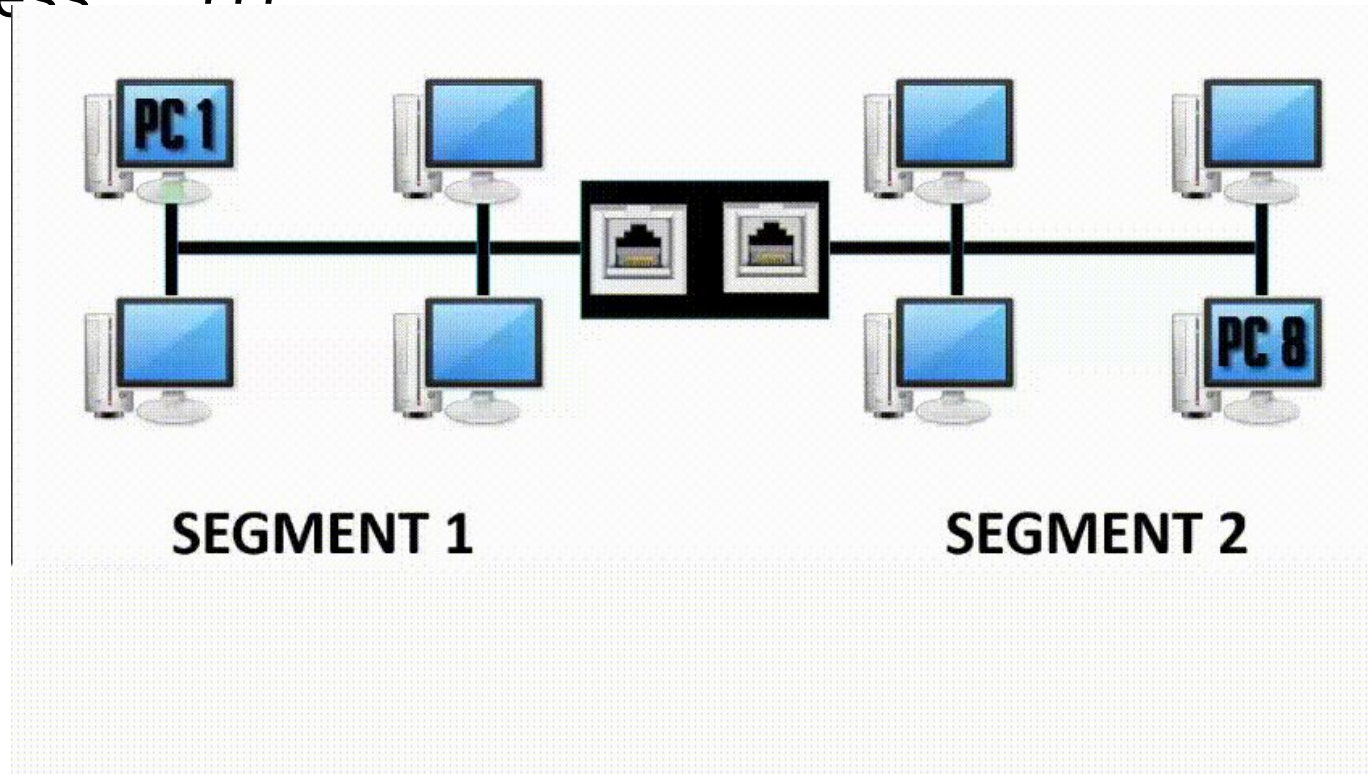


LAYER 2- NETWORK DEVICE

1. Bridge & Switch



Guess ???

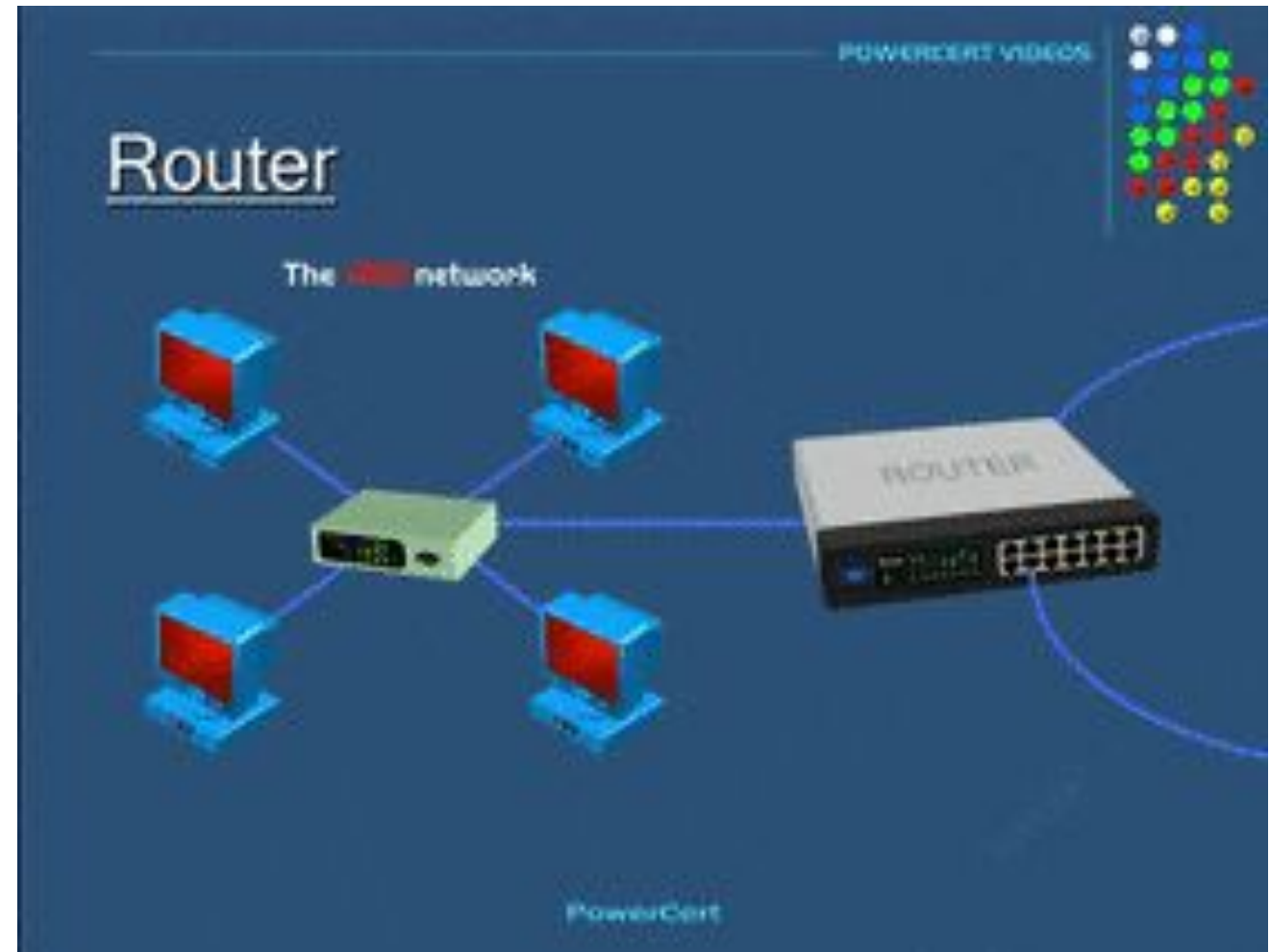


LAYER 3 – NETWORK DEVICE

1



Router



GATEWAY

1. Software layer Network device



Gateway



SUMMARY

- Lecture 4 : Network Hardware

Q. 1 What is network device?

Q. 2 Why network device?

Q. 3 How many types of network devices are there?

Q.4 What is the concept behind 1,2,3,upper layer networking devices?



ACTIVITY - 4

- Lecture 4 : Network Hardware

Reading: Bridge

<https://www.elprocus.com/what-is-a-bridge-in-computer-network-working-types-its-functions/>

Difference between Switch & Router

<https://blog.networkbulls.com/top-10-major-difference-between-routers-and-switches>

<https://www.guru99.com/router-vs-switch-difference.html>

Video: Network Device (Animation)

1. <https://www.youtube.com/watch?v=eMamgWlIRFY>
2. https://www.youtube.com/watch?v=1z0ULvg_pW8
3. <https://www.youtube.com/watch?v=ck3gx9HB9-k&list=PLSNNzog5eydtmcbbcb1b8pVRkgre3vNUy>



NETWORK HARDWARE

1. Devices-3



LECTURE –6

Network Hardware

Recap



1. Network devices
2. Hub
3. Repeater
4. Bridge
5. Switch
6. Gateway

Objectives

- Topology
- Bus Topology
- Ring Topology
- Star Topology
- Mesh Topology



NETWORK TOPOLOGY

Network topology is the arrangement of the elements (**links, nodes, etc.**) of a communication network.

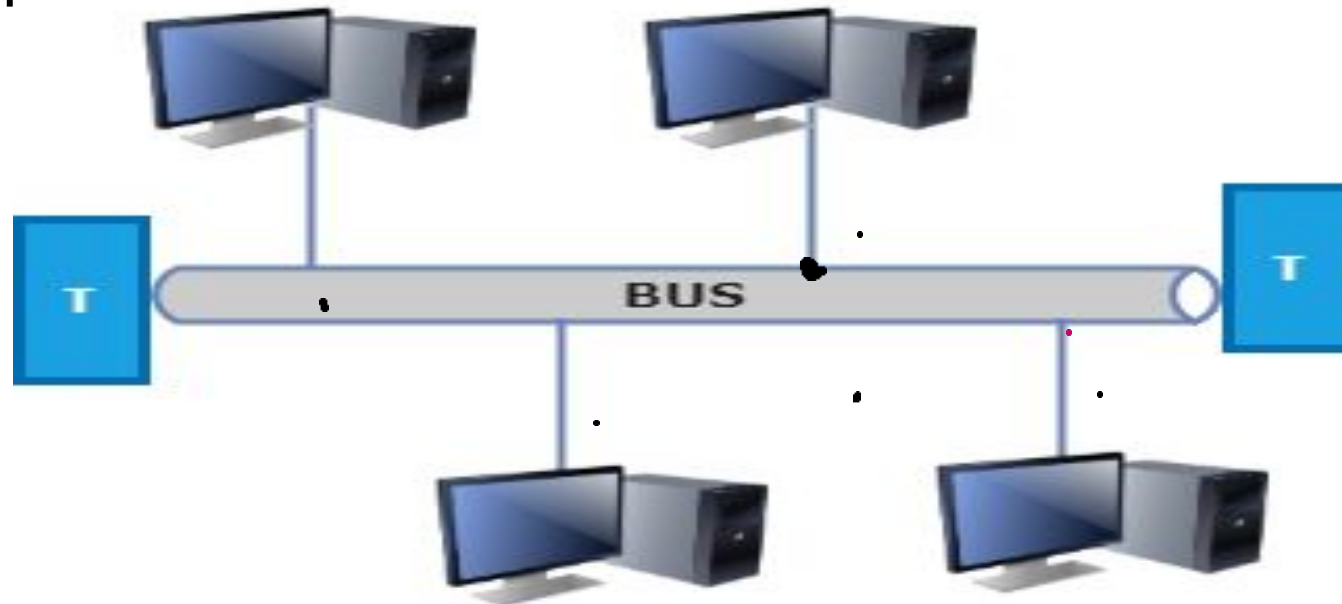
Physical topology is the placement of the various components of a network (e.g., device location and cable installation)

Logical topology illustrates how data flows within a network.



BUS TOPOLOGY

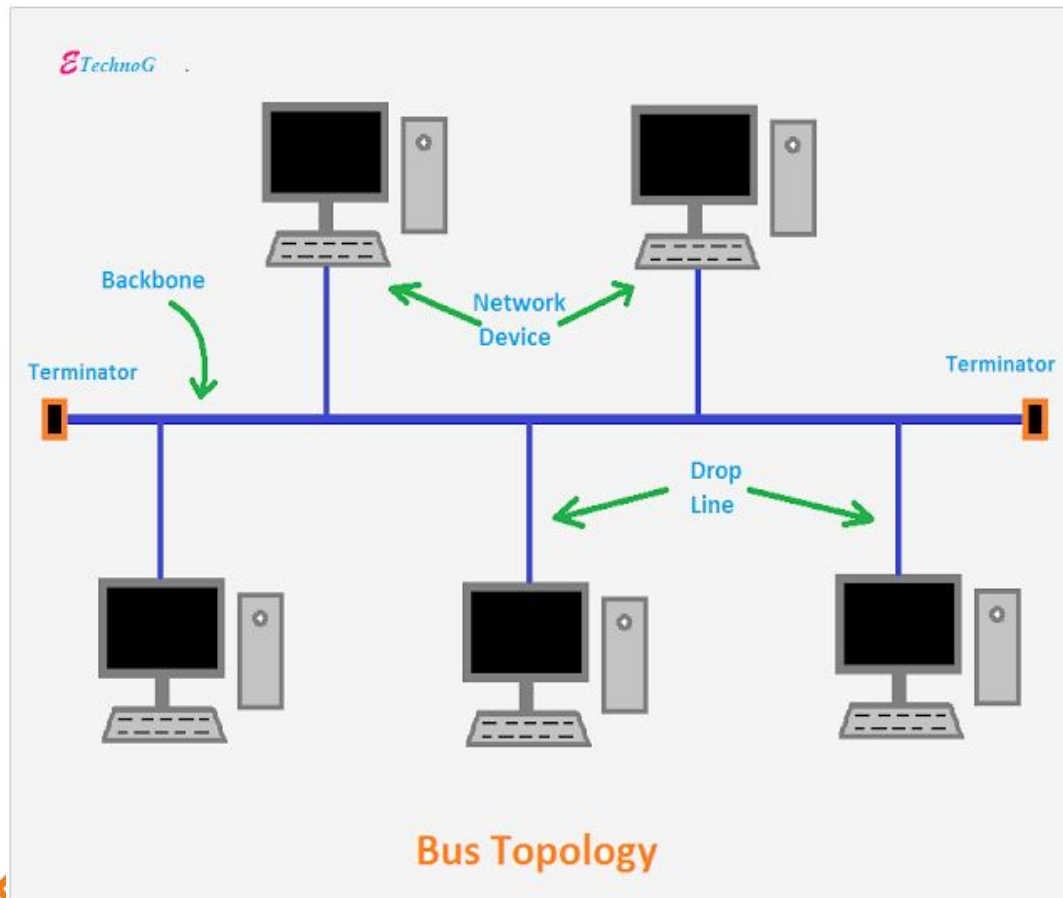
1. T



Nodes
Coaxial cable
Terminators
DropLines
Network device



IN BRIEF.....



Bus Topology



By Natcha Phohan

BUS TOPOLOGY

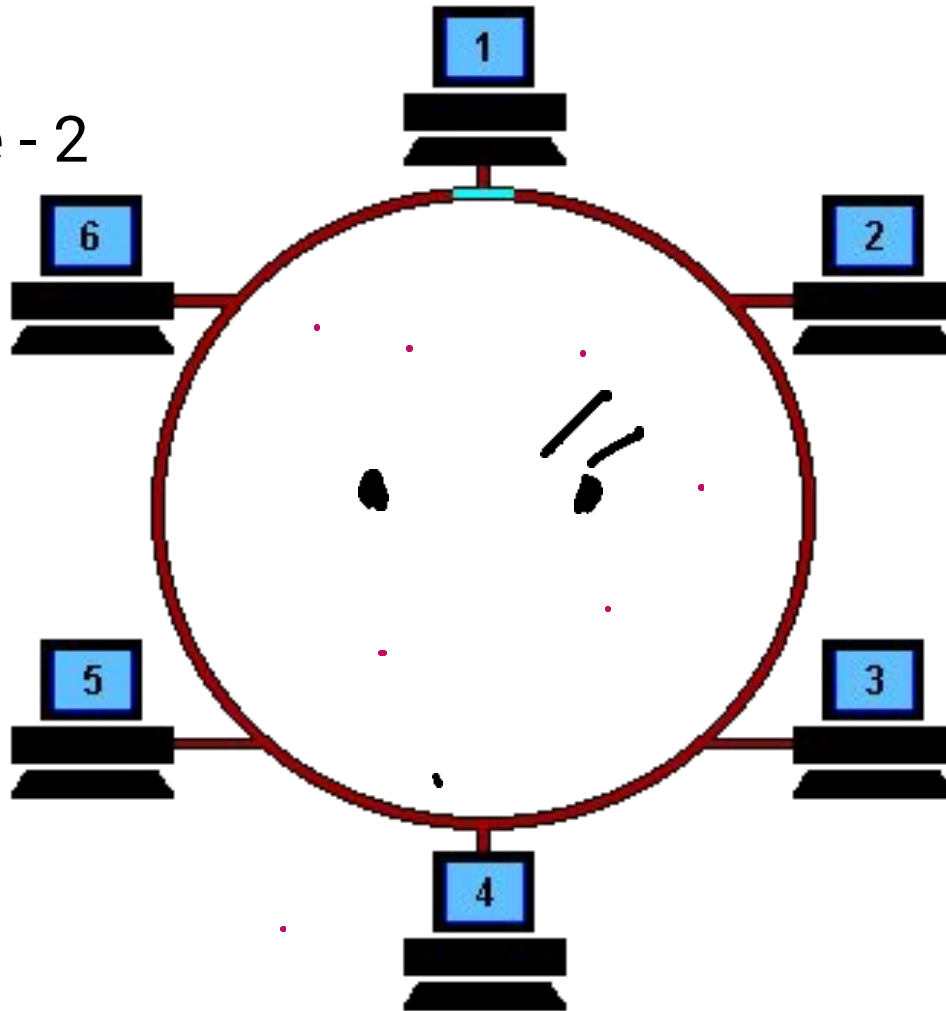
1. T

Advantages	Disadvantages
1. Installation: Easy 2. Cost: Cheap 3. Connection: Every computer is connected to single cable through drop lines. 4. Transmit: Signals/ data stream travels from one end to another end of cable in one direction no bi-directional. 5. Sharing: cable sharing is carried out.	1. Maintenance : High cost 2. Robust: No 3. Performance: Slow 4. Troubleshoot: Difficult 5. Security: Less secure 6. Termination : proper loop termination required.



RING TOPOLOGY

1. Type - 2



Nodes
Coaxial cable
Terminators
DropLines
Network device

Work-Stations
Tokens - Early & Delay Token



RING TOPOLOGY

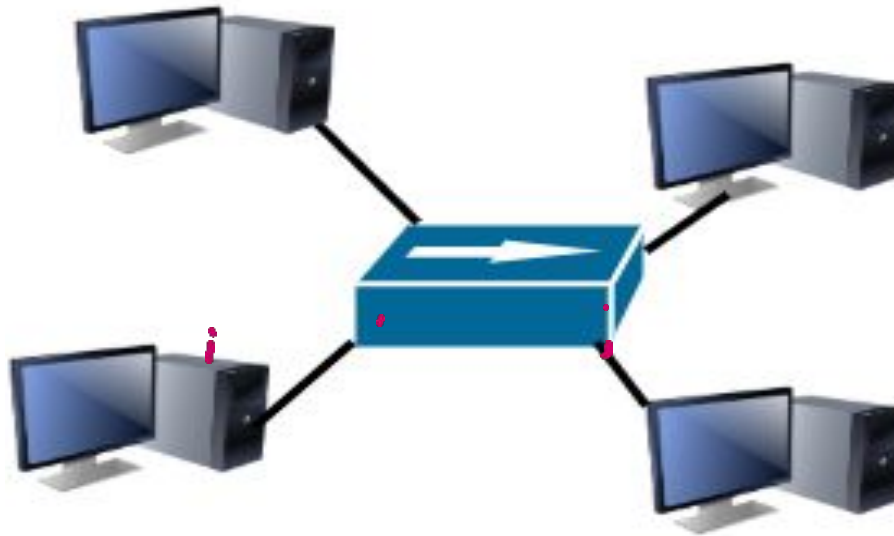
1. Type

Advantages	Disadvantages
1. Installation: Easy 2. Cost: cheap 3. Connection: Every computer is connected to single cable through drop lines with closed loop. 4. Transmit: Unidirectional 5. Data Transmission: Transmit the data, only when station hold the token 6. Sharing: cable sharing . 7. Performance: quite good than bus topology 8. Termination : performed by monitor station 9. Data transfer: High 10. Collision : minimum collision.	1. Performance: lesser than star topology 2. Robust: Less 3. Troubleshoot: Difficult 4. Impact : get disturbed on addition or removal 5. Security: Less



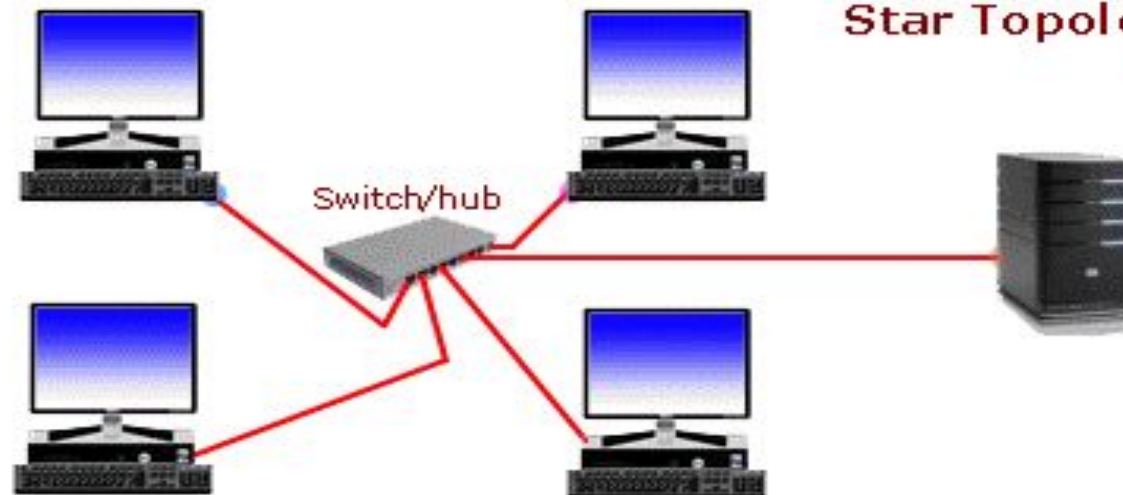
STAR TOPOLOGY

1. Type - 3



Nodes
Coaxial cable
Terminators
DropLines
Network device

Star Topology



STAR TOPOLOGY

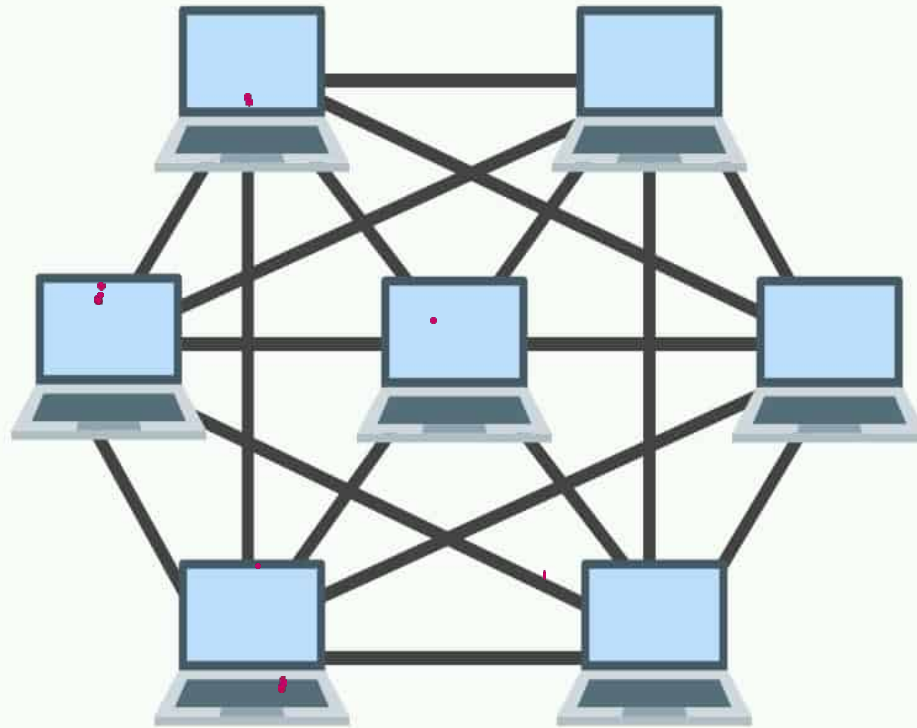
1. Type-2

Advantages	Disadvantages
1. Installation : Easy 2. Cost: less expensive as each device require only one I/O port & needs to be connected with hub with one link. 3. Impact : If one link fails other links works fine. 4. Fault detection: easy as link identification is easily.	1. Central management : if hub goes down network goes down. 2. More cables required.

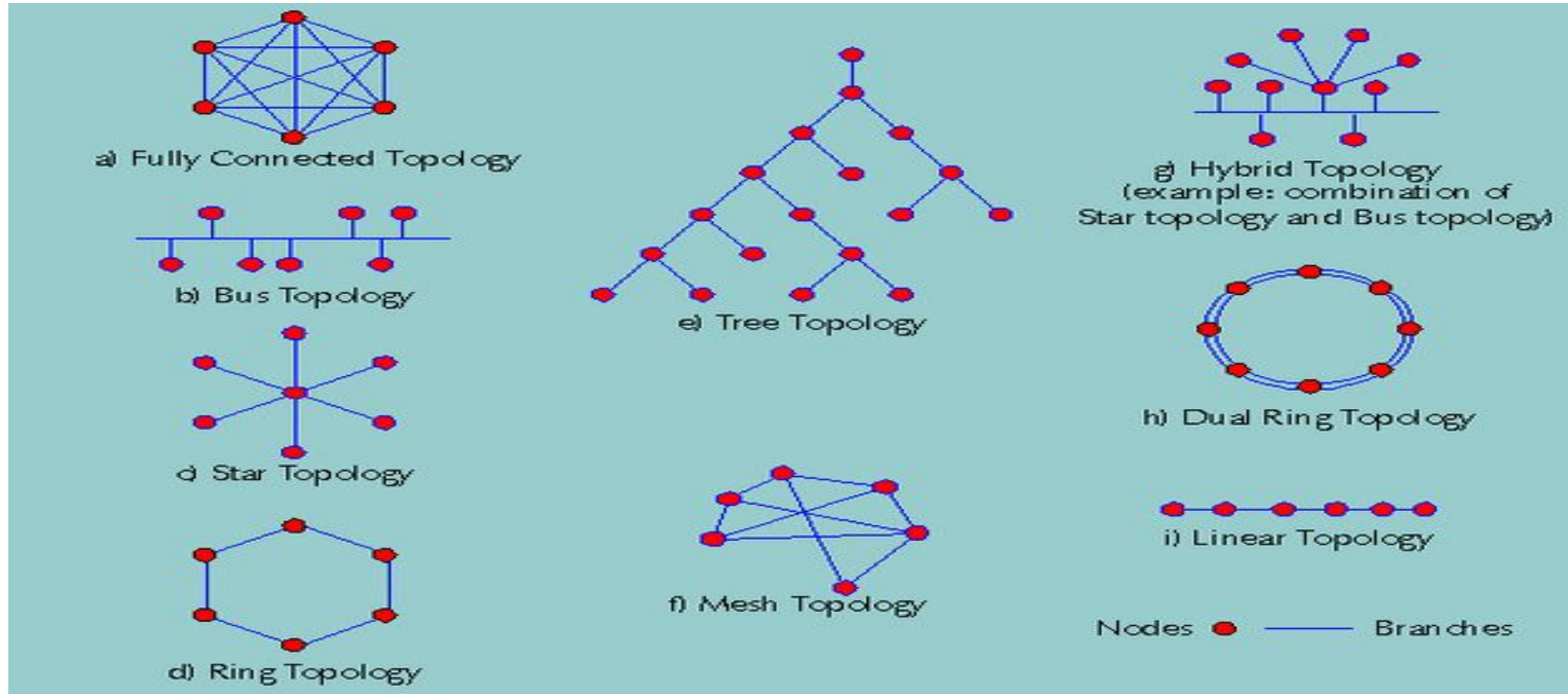


MESH TOPOLOGY

1. Type - 4



OTHER TYPES OF TOPOLOGY



SUMMARY

• Lecture 5: Network Hardware

Q.1 What is topology?

Q.2 Why topology?

Q.3 What are the different types of network topology?

Q.4 Which Topology is best and why?

Q.5 Comparison between Topology



ACTIVITY - 5

- Lecture 5: Network Hardware

Reading: Network topology

<https://www.comparitech.com/net-admin/network-topologies-advantages-disadvantages/>



NETWORK TYPES

1. Under various scenarios



LECTURE –7

Network Types

Recap

1. **Topology** 
2. **Bus Topology**
3. **Ring Topology**
4. **Star Topology**
5. **Mesh Topology**

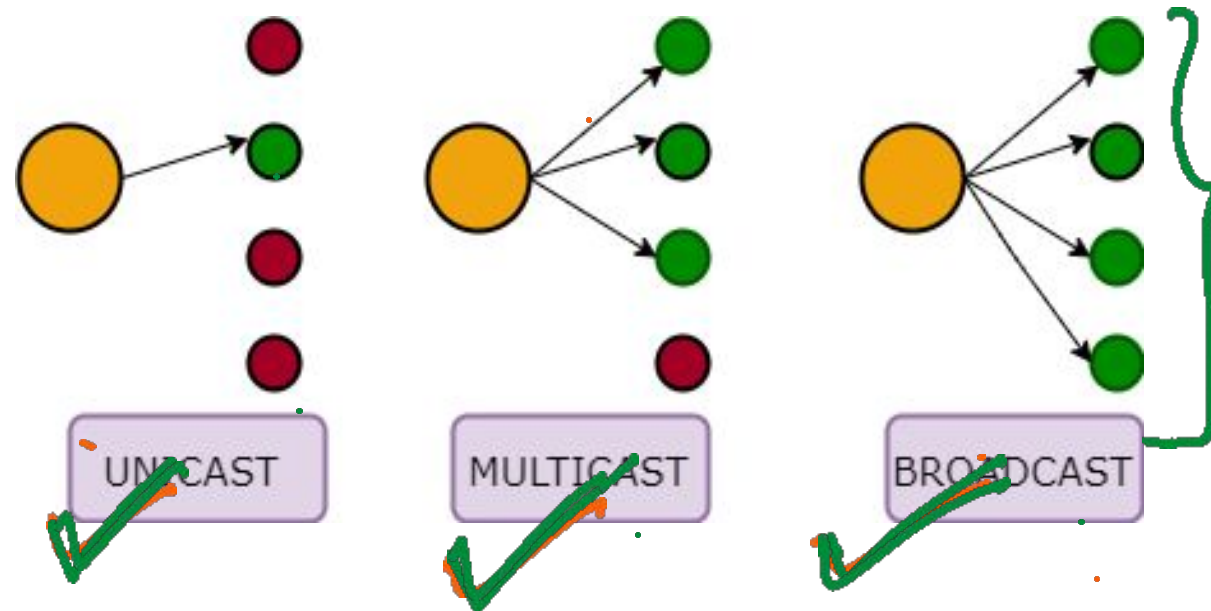
Objectives

- Revisit on casting – unicasting, multicasting, broadcasting.
- Based on request & service – client-service model & peer-to-peer model
- Based on scale or coverage area – PAN, LAN, MAN, WAN.
- Based on communication within network (Intra-network) and between network (Inter-network)



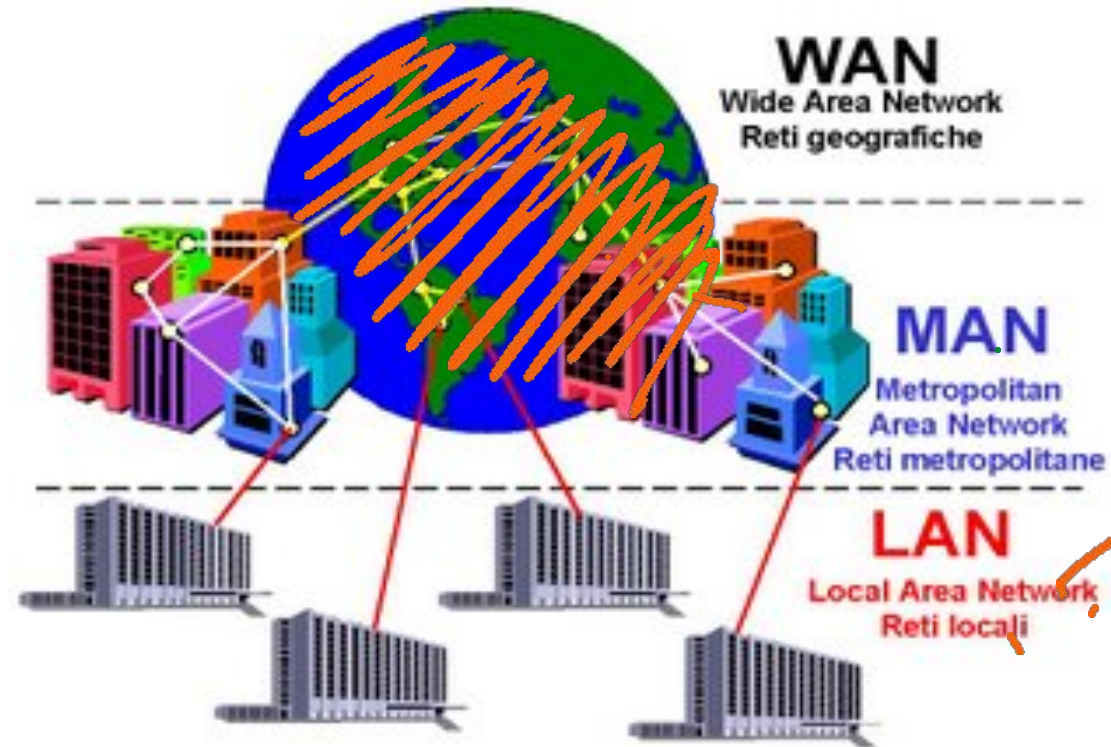
BROADCAST NETWORK

1. **Casting** – the way to stream data from source point to nor of destination point.



NETWORK TYPES BASED ON SCALE

1. LAN,MAN,WAN



NETWORK TYPES BASED ON SCALE

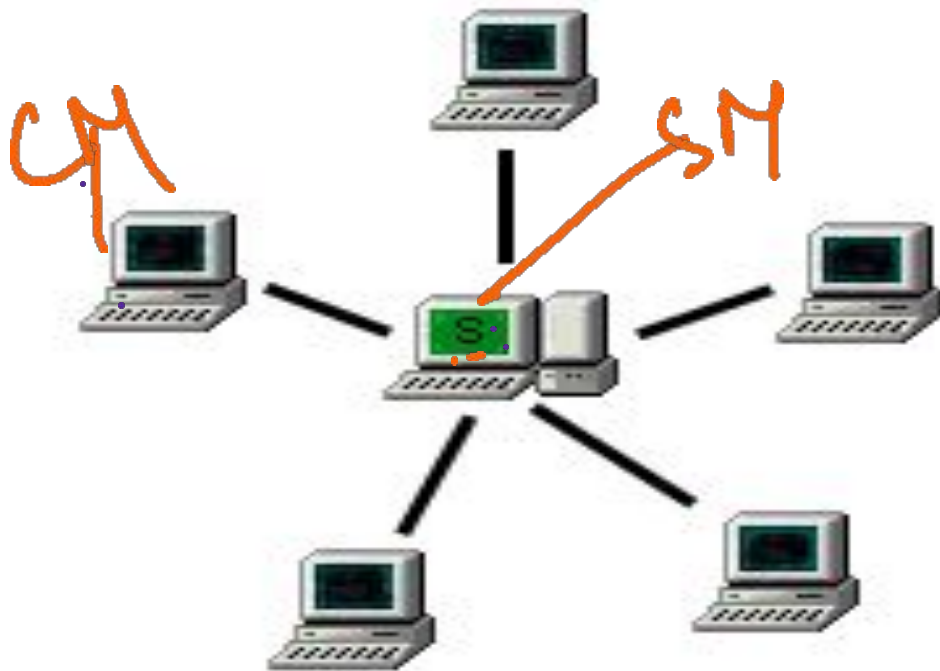
1.

Features	PAN	LAN	MAN	WAN
Expansion	Personal Area network	Local Area network	Metropolitan Area Network	Wide Area Network
Owned	Private	Private	Private or Public	Private or Public
Span	Meters	Kilometres	City	World
Delay	Short	Relative Short	Moderate	High
Congestion.	<(Lesser than)	<(Lesser than)	< (Lesser than)	High
Fault Tolerance	> (Greater than)	> (Greater than)	> (Greater than)	Low
Speed	> (Greater than)	> (Greater than)	Average	Low



NETWORK BASED ON SERVICES & CONTROL

1. *Client* Server Based Network

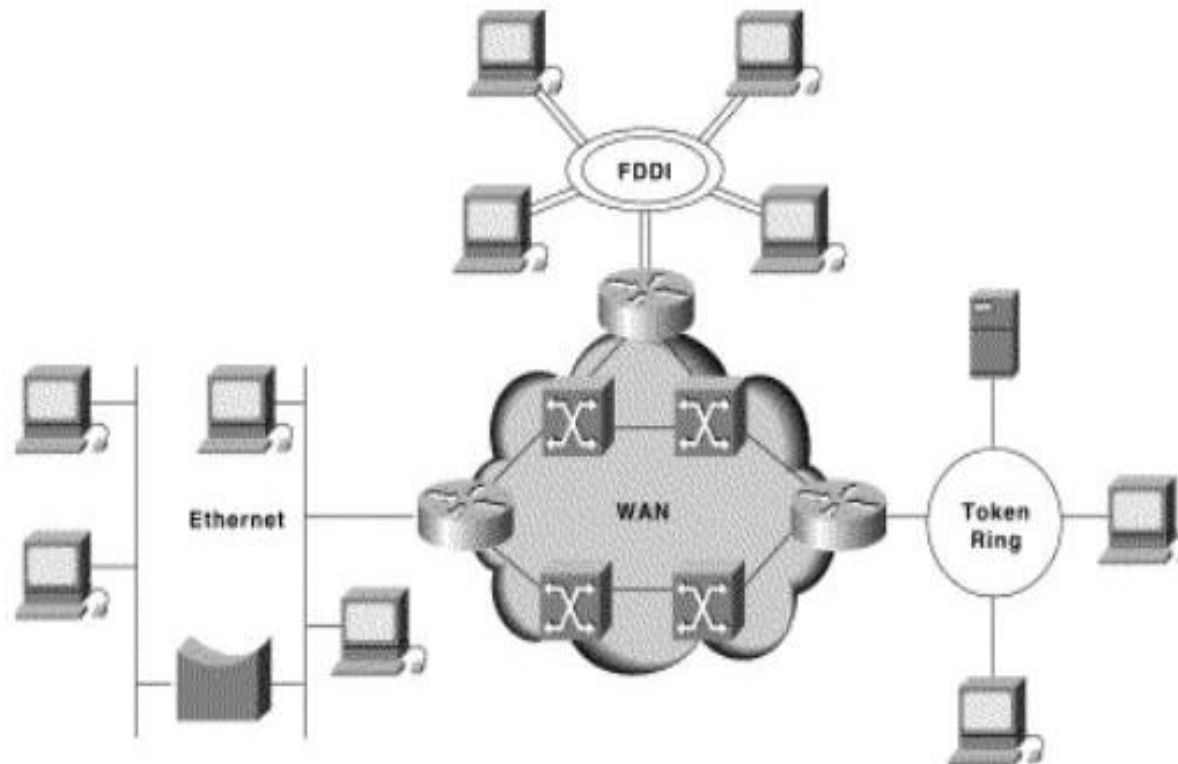


✓
Peer to Peer Network



INTER-NETWORK

1. Netw



SUMMARY

- Lecture 7: Network Hardware

Q.1 What type network you find based on ownership?

Q.2 Network based on casting.

Q.3 Network based on services and control

Q.4 What are the types of network models based on scale and its comparison?



ACTIVITY

- Lecture 7 : Network Hardware

Quiz:

http://highered.mheducation.com/sites/0073376221/student_view0/chapter1/quizzes.html



NETWORK SOFTWARE

1. Services, Interfaces, Protocols



LECTURE –8

NETWORK SOFTWARE

Recap



1. Revisit **sting – unicasting, multicasting, broadcasting.**
2. Based on request & service – client-service model & **peer-to-peer model**
3. Based on scale or coverage area – PAN, LAN, MAN, **WAN.**
4. Based on communication within network (Intra-network) and between network (Inter-network)

Objectives

• Types of Software-

• Application software

• Network Protocol

• Drivers of the NIC

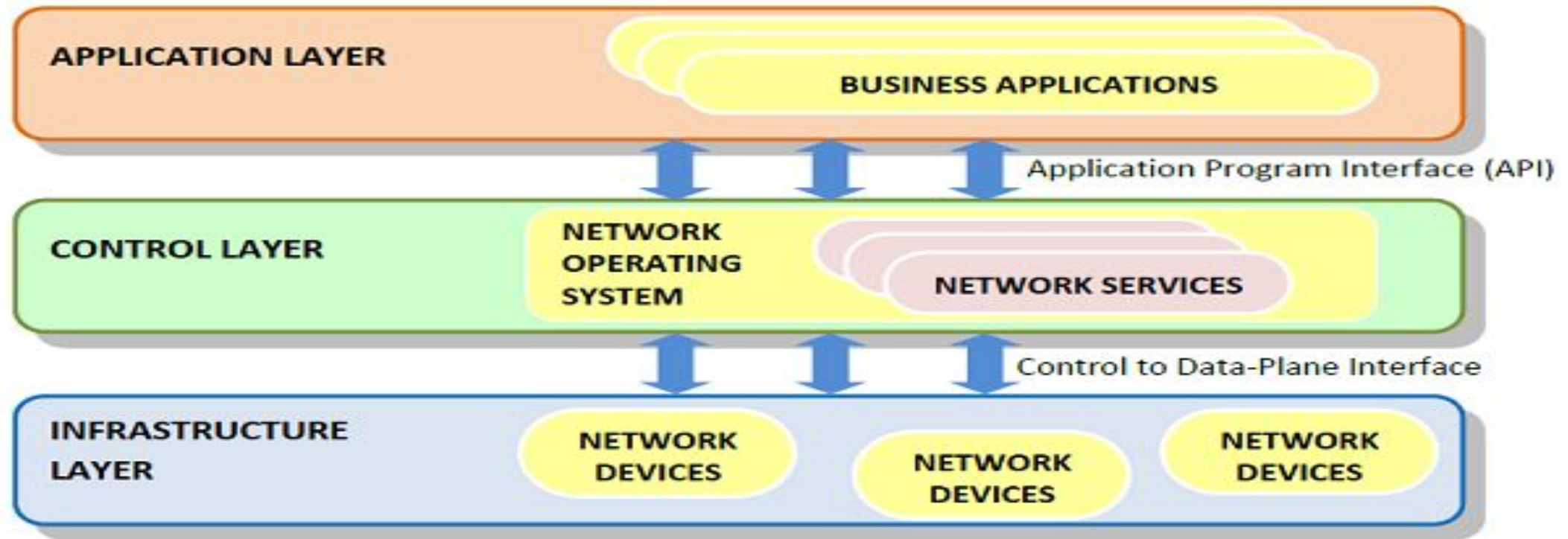
2. Network Protocol

3. Connection-oriented Vs

Connectionless-oriented Service.



NETWORK SOFTWARE



LAYERS, PROTOCOL, INTERFACES

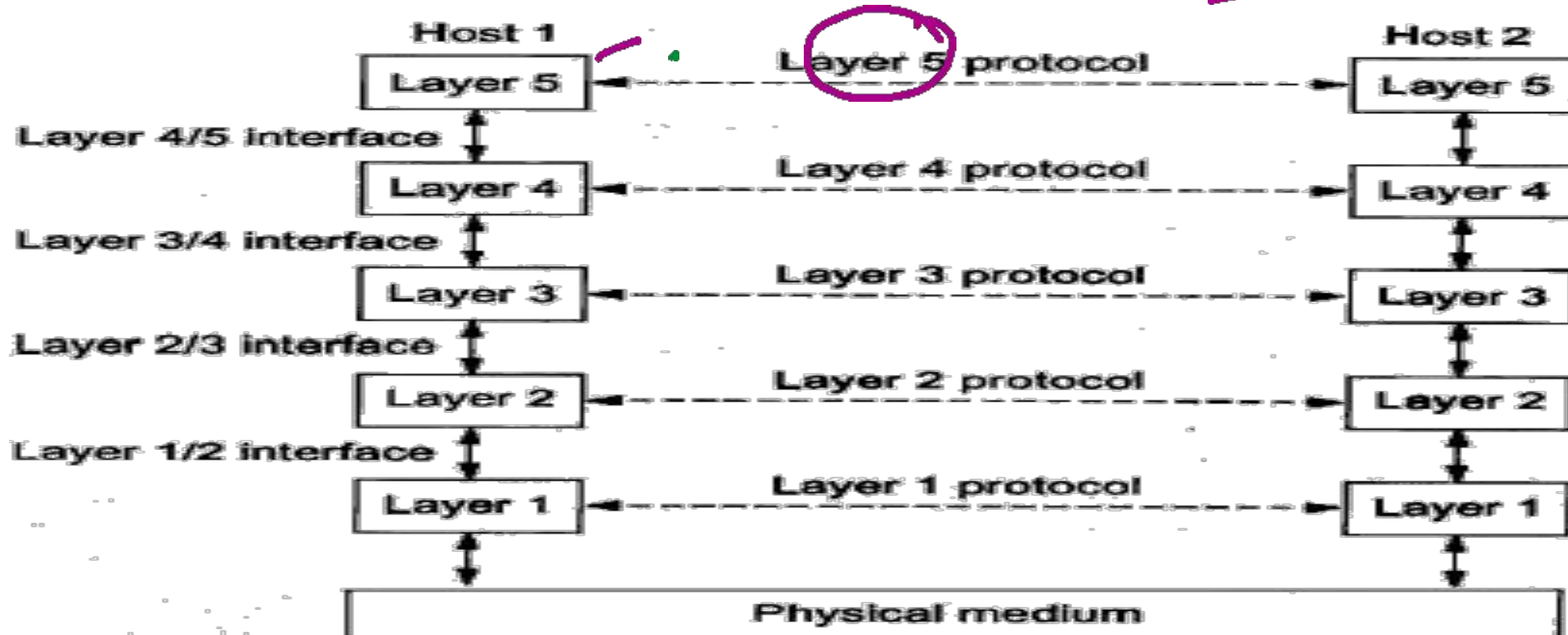
Protocol: is an agreement between the communicating parties on how communication is to proceed.

Interface: Between each pair of adjacent layers.

Services: A service is formally specified by a set of primitives .



LAYERS, PROTOCOL, INTERFACES



NETWORK PROTOCOL STACK

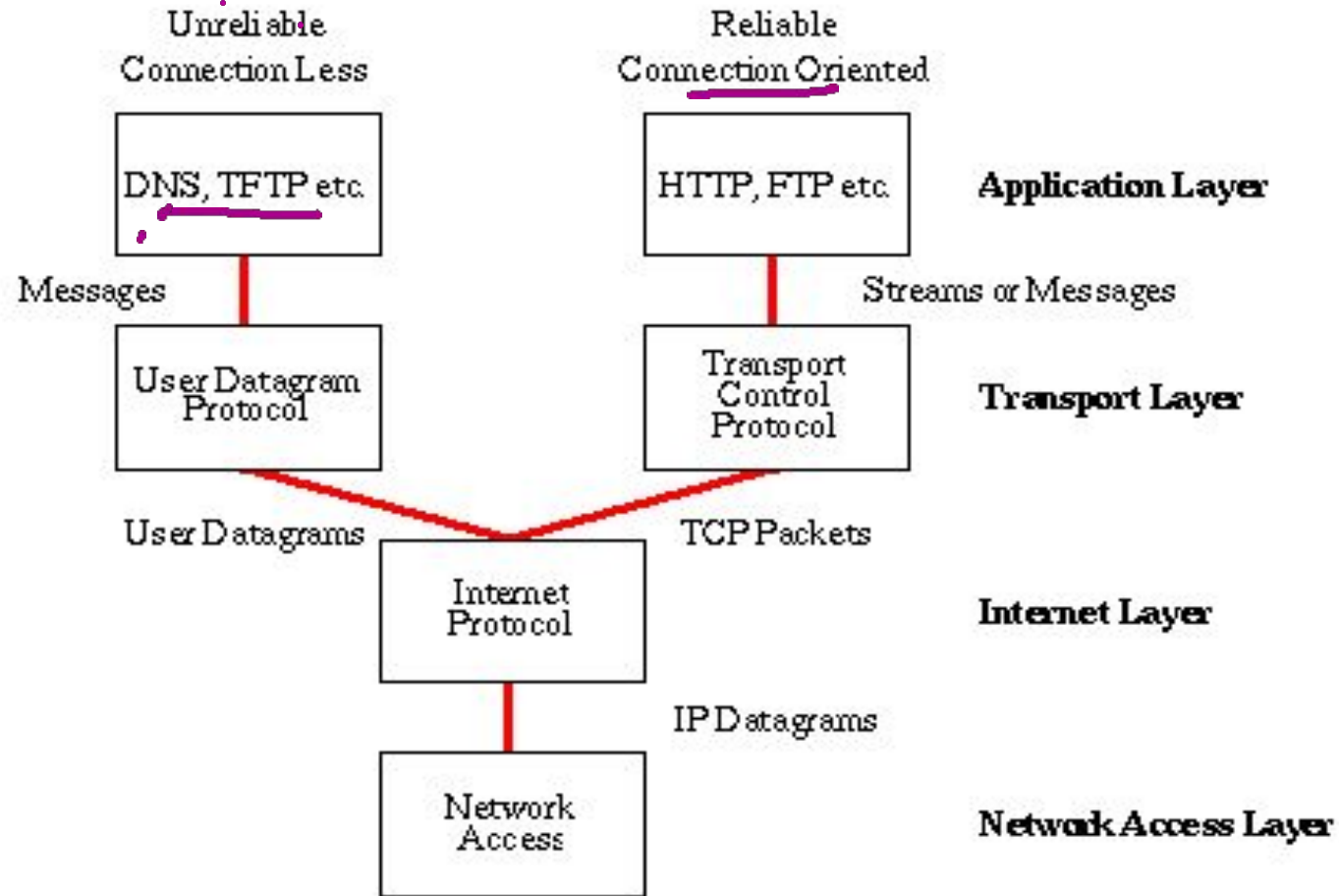
1. **Protocol Stack:** A list of protocols used by a certain system, one protocol per layer

TCP/IP Model	OSI Ref Model	Protocols
Application Layer	Application Layer	FTP, HTTP, Telnet
	Presentation layer	JPEG,MPEG
	Session Layer	NFS,SQL,PAP
Transport Layer	Transport Layer	TCP,UDP
Internet Layer	Network Layer	IPV4,IPV6
Network Access Layer	Data Link Layer	ARP,CDP,STP
	Physical Layer	Ethernet, Wi-fi



CONNECTION-ORIENTED VS CONNECTIONLESS SERVICE

1. Als



CONNECTION-ORIENTED VS CONNECTIONLESS SERVICE

Task / issue
past law

Connection-oriented communication :

It is a network communication mode in telecommunications and computer networking, where a **communication session** or a **semi-permanent connection** is established before any useful data can be transferred, and where a stream of data is delivered in the same order as it was sent.

Connectionless communication:

It is a data transmission method used in packet switching networks in which each data unit is individually addressed and routed based on information carried in each unit, rather than in the setup information of a prearranged, fixed data channel as in connection-oriented communication.



CONNECTION-ORIENTED VS CONNECTIONLESS SERVICE

Connection-Oriented and Connectionless Services

Connection-oriented		Service	Example
		Reliable message stream	Sequence of pages
		Reliable byte stream	Remote login
Connectionless		Unreliable connection	Digitized voice
		Unreliable datagram	Electronic junk mail
		Acknowledged datagram	Registered mail
		Request-reply	Database query

Six different types of service.



CONNECTION-ORIENTED SERVICES

1. **Service primitives:** Communication between adjacent protocol layers (i.e. within the same communications node) are managed by calling functions, called **primitives**, between the layers. There are various types of actions that may be performed by **primitives**

Sl.No	Primitives	Meaning
1.	LISTEN	Block waiting for an incoming connection
2.	CONNECT	Establish a connection with a waiting peer
3.	ACCEPT	Accept an incoming connection from a peer
4.	RECEIVE	Block waiting for an incoming message
5.	SEND	Send a message to the peer
6.	DISCONNECT	Terminate a connection.



CONNECTION-ORIENTED & CONNECTIONLESS-ORIENTED COMPARISON

Criteria	Connection-Oriented	Connection-Less
Connection	Prior connection need to be established	No prior connection need to be established
Resource Allocation	Resources need to be allocated	No prior allocation of resources is required
Reliability	It ensures reliable transfer of data	Reliability is not guaranteed as it is best effort service.
Congestion	Congestion is not at all possible	Congestion can occur likely
Transfer Mode	It can be implemented either using circuit switching or virtual circuits.	It is implemented using packet switching (Datagram).
Retransmission	It is possible to retransmit the possible lost data bits.	It is not possible
Suitability	It is suitable for long and steady communication	It is suitable for bursty transmission
Signalling	Connection is established through process of signalling	There is no concept of signalling
Packet travel	Packet travel from source to destination in sequential manner.	Packets move in random fashion.
Delay	There is more delay in transfer but once connection established fast delivery	No delay as connection is not required.



SUMMARY

• Lecture 8 : Network Software

Q. 1 What is protocol, interface, services?

Q. 2 What is connection services and connectionless services?

Q. 3 What are the differences between connection services and connectionless services?

Q. 4 What is protocol stack?

Q. 5 Discuss the relationship between layers, interface, protocol and services.



ACTIVITY - 7

- Lecture 8: Network Software

1. <https://www.youtube.com/watch?v=oJT2Iza0I00>



REFERENCE MODEL

1. OSI & TCP Model



LECTURE –9

Reference Model

Recap



1. Types of Software-

- Application software
- Network Protocol
- Drivers of the NIC

2. Network Protocol

3. CO Services Vs CL Service.

Objectives

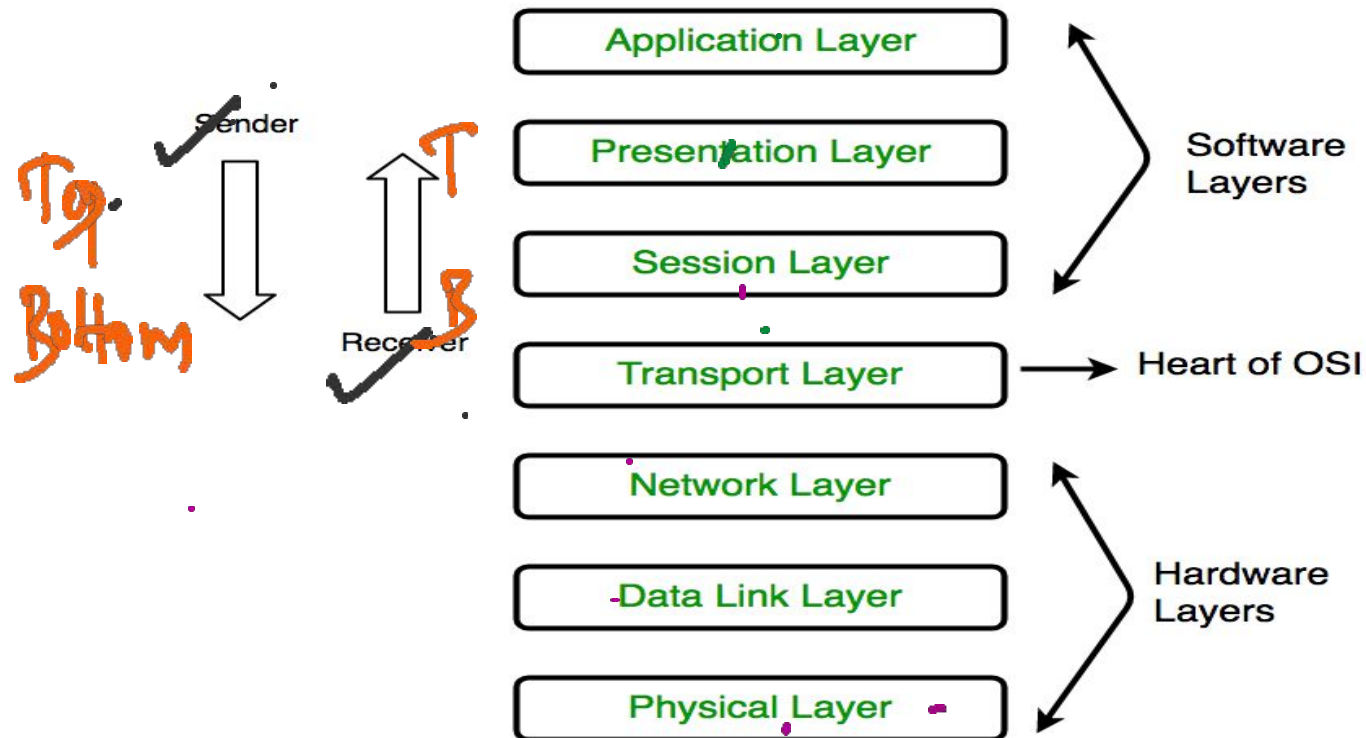
- OSI Reference model
- TCP/IP Reference model
- Direct vs Indirect communication
- Physical/vertical vs Logical/horizontal communication
- Data Encapsulation & Stripping
- Data Units, Message delivery types, Address types



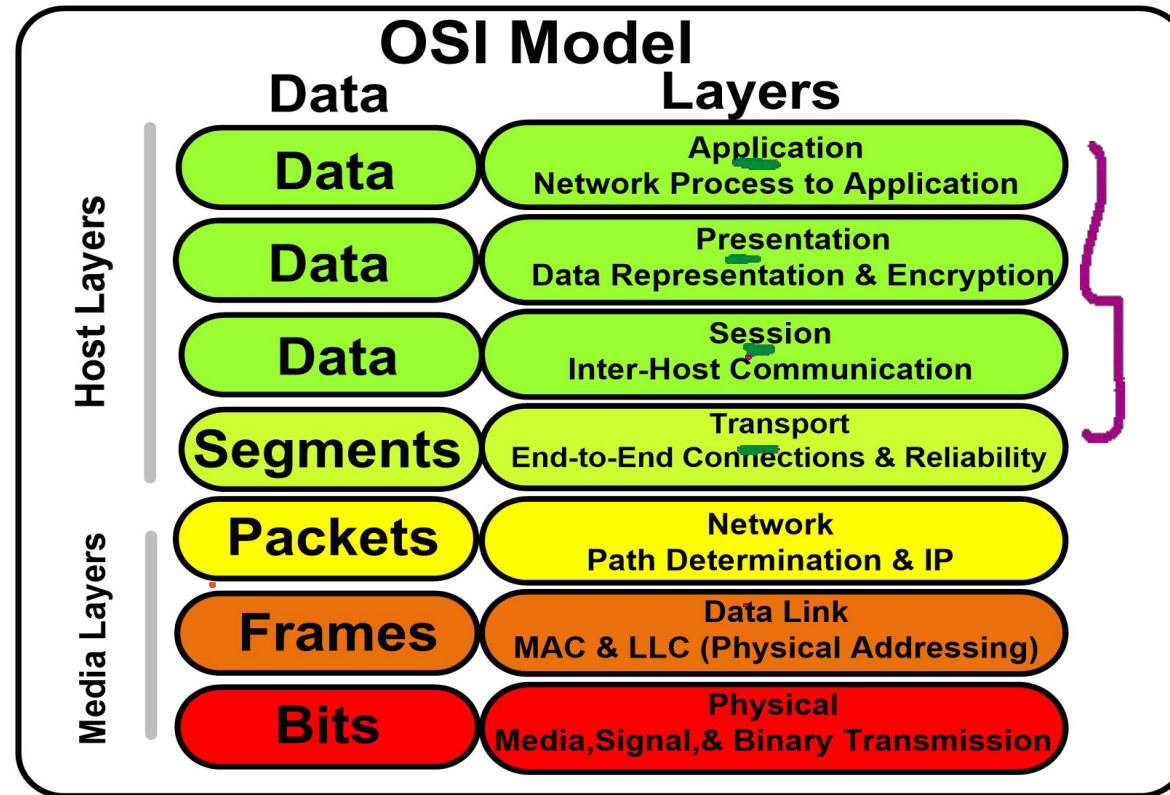
NETWORK ARCHITECTURE

⇒ LAYERED ARCHITECTURE

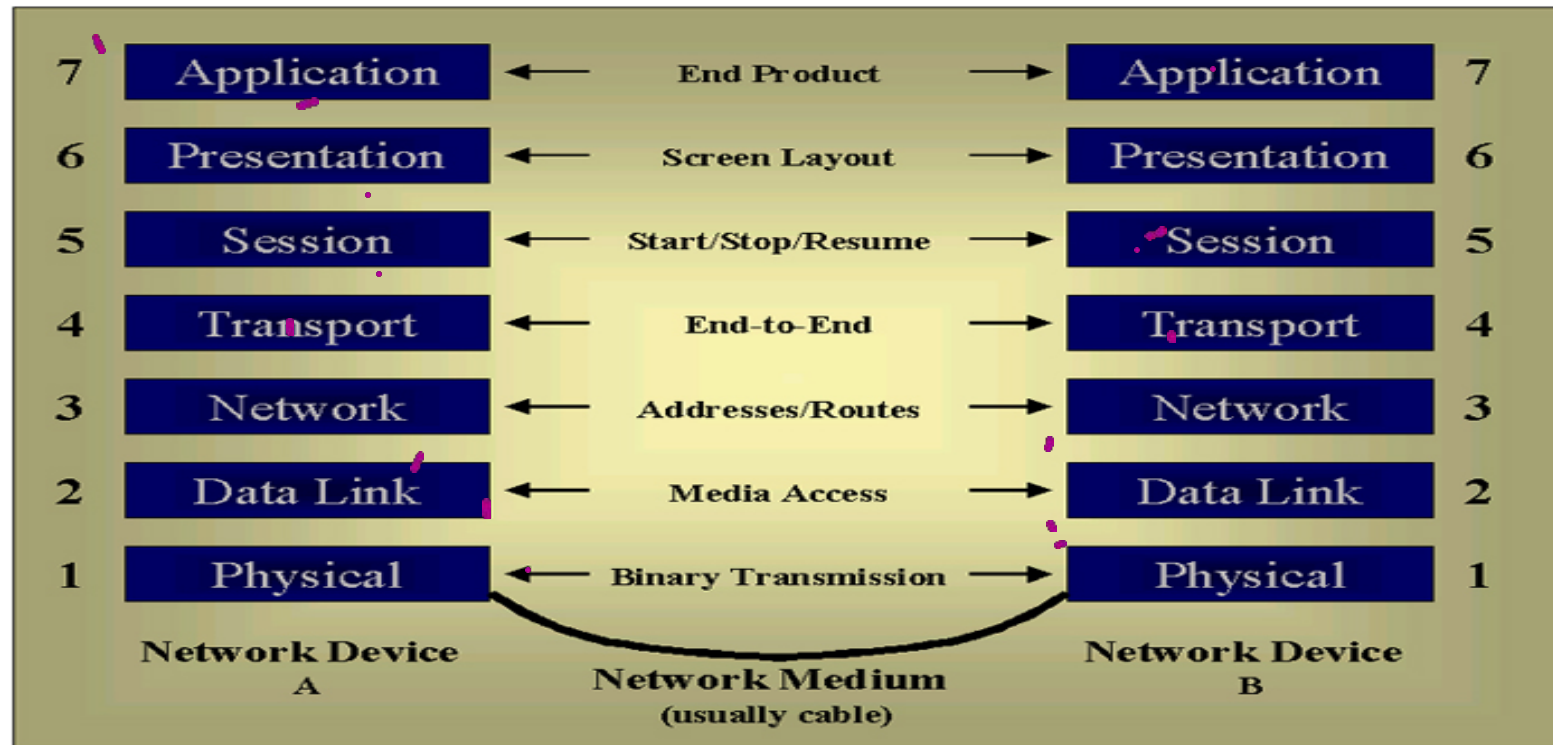
1. NA: Set of Layers + Set of Protocols + (style in which request and service established)



OSI REFERENCE MODEL

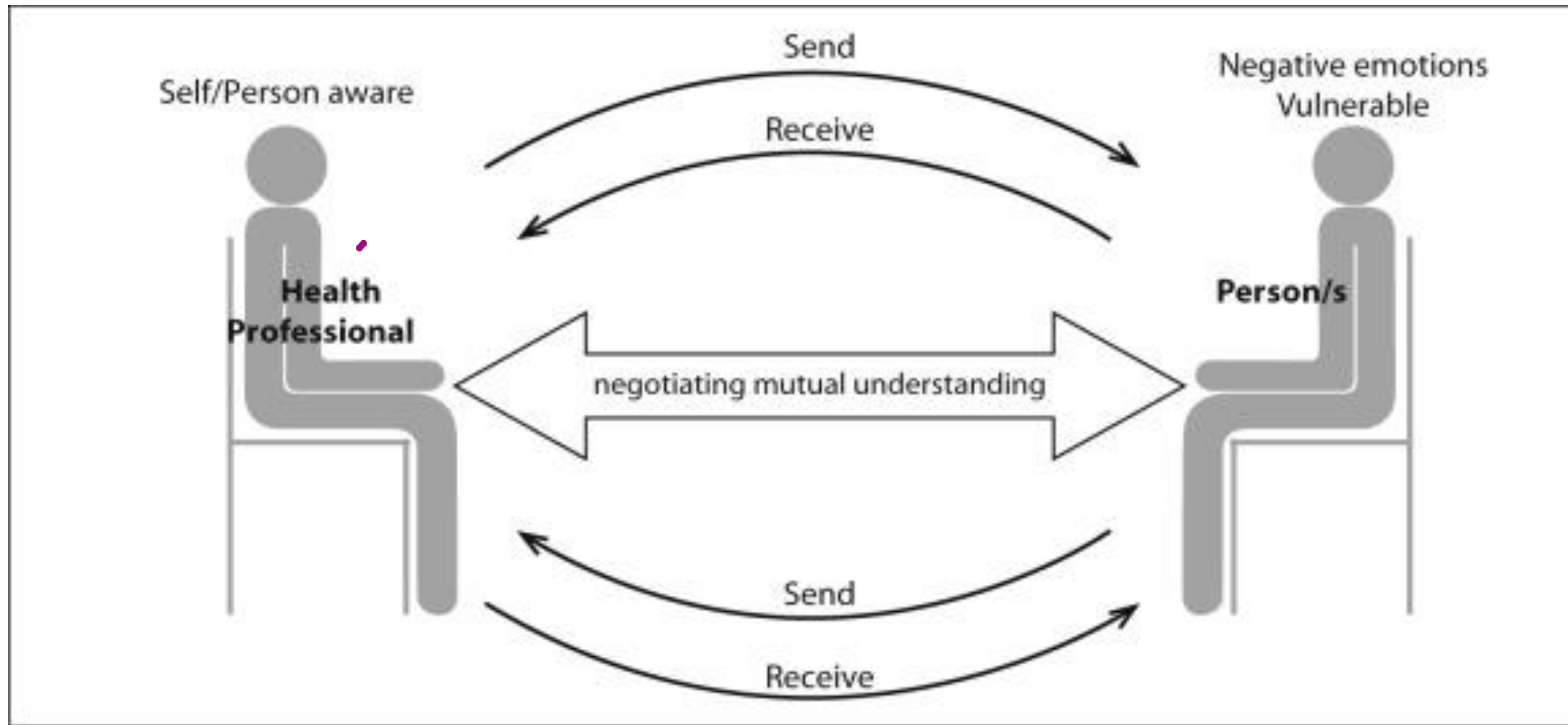


COMMUNICATION BETWEEN SENDER AND RECEIVER

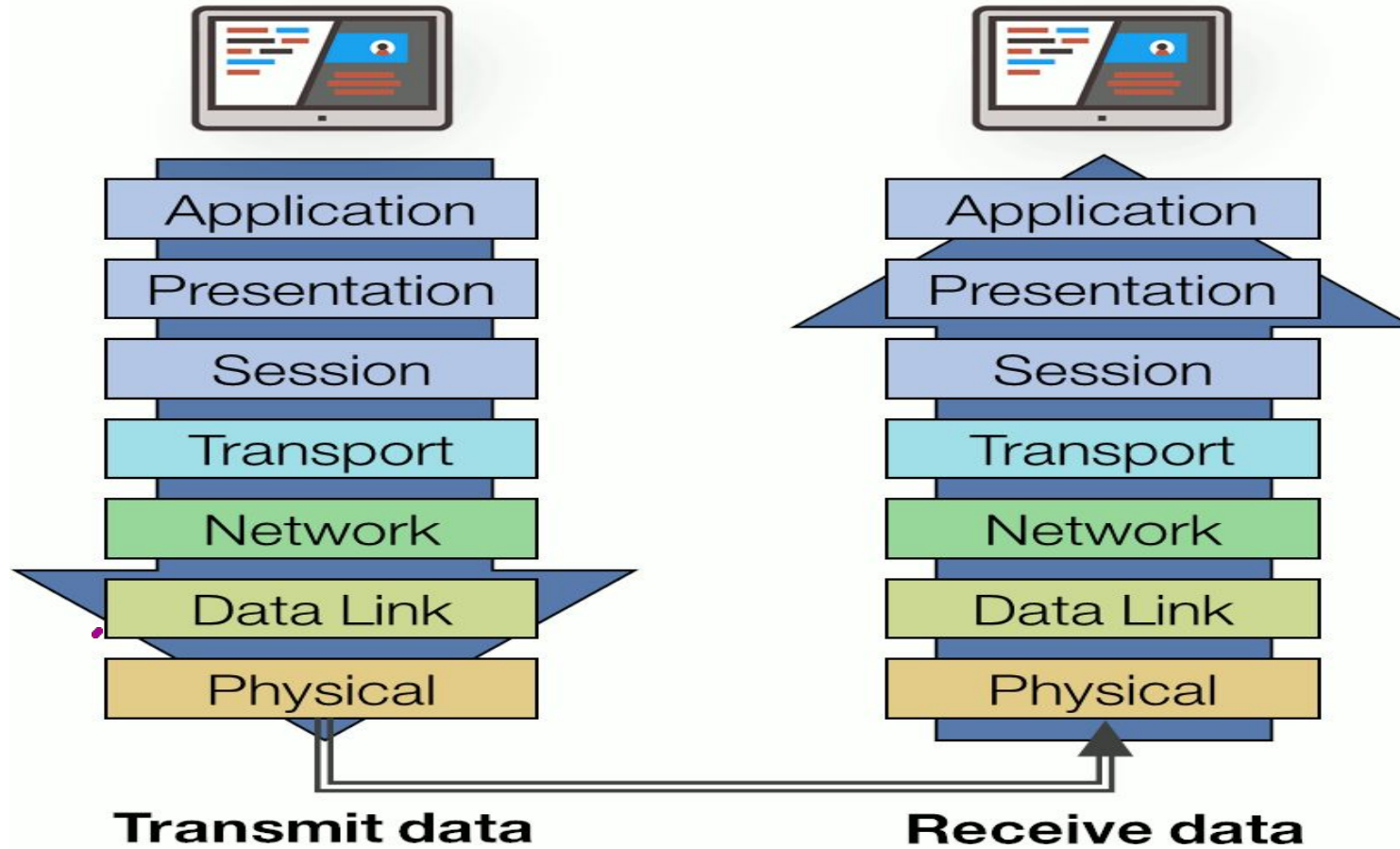


COMMUNICATION BETWEEN SENDER AND RECEIVER –

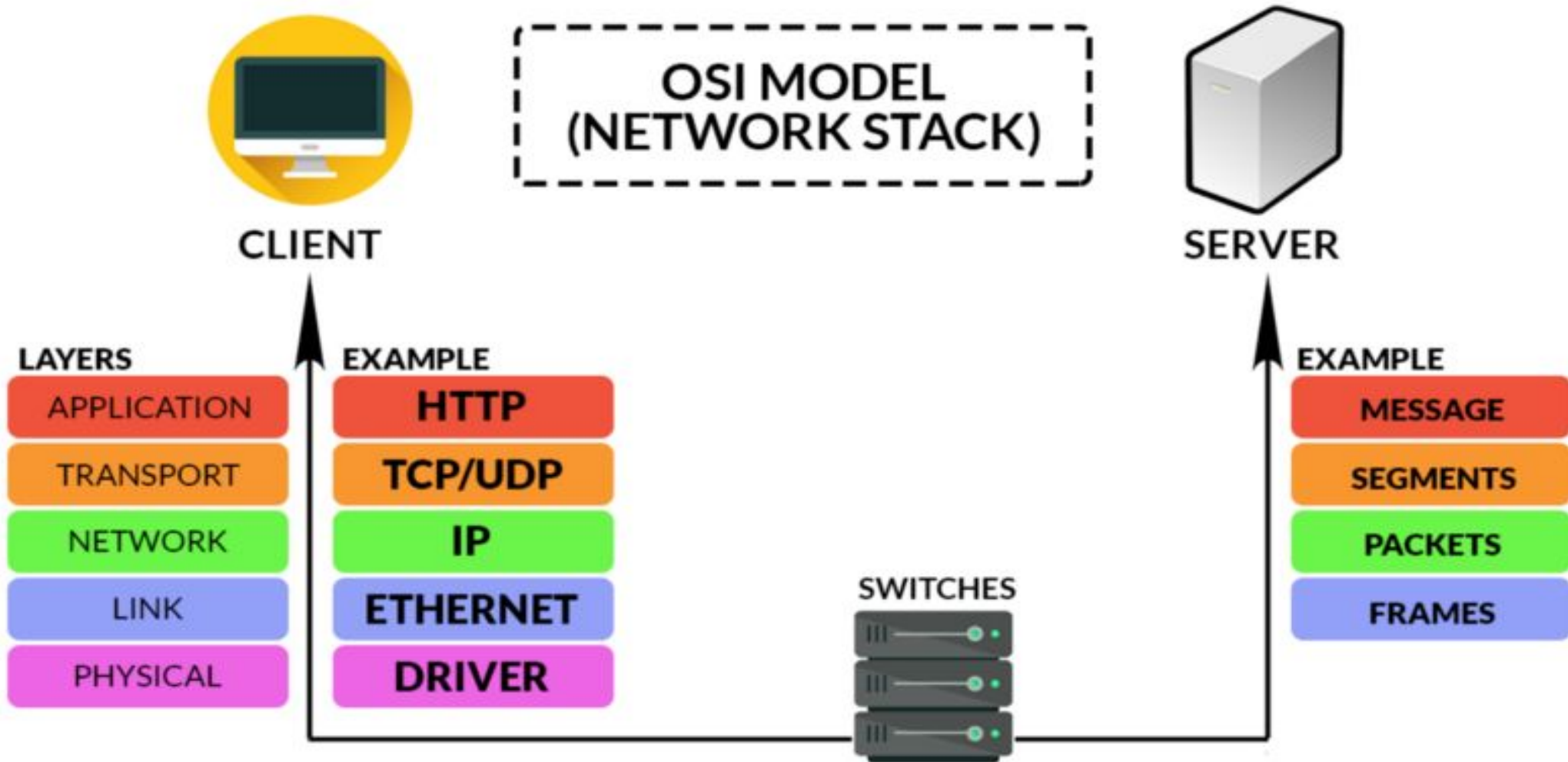
1.



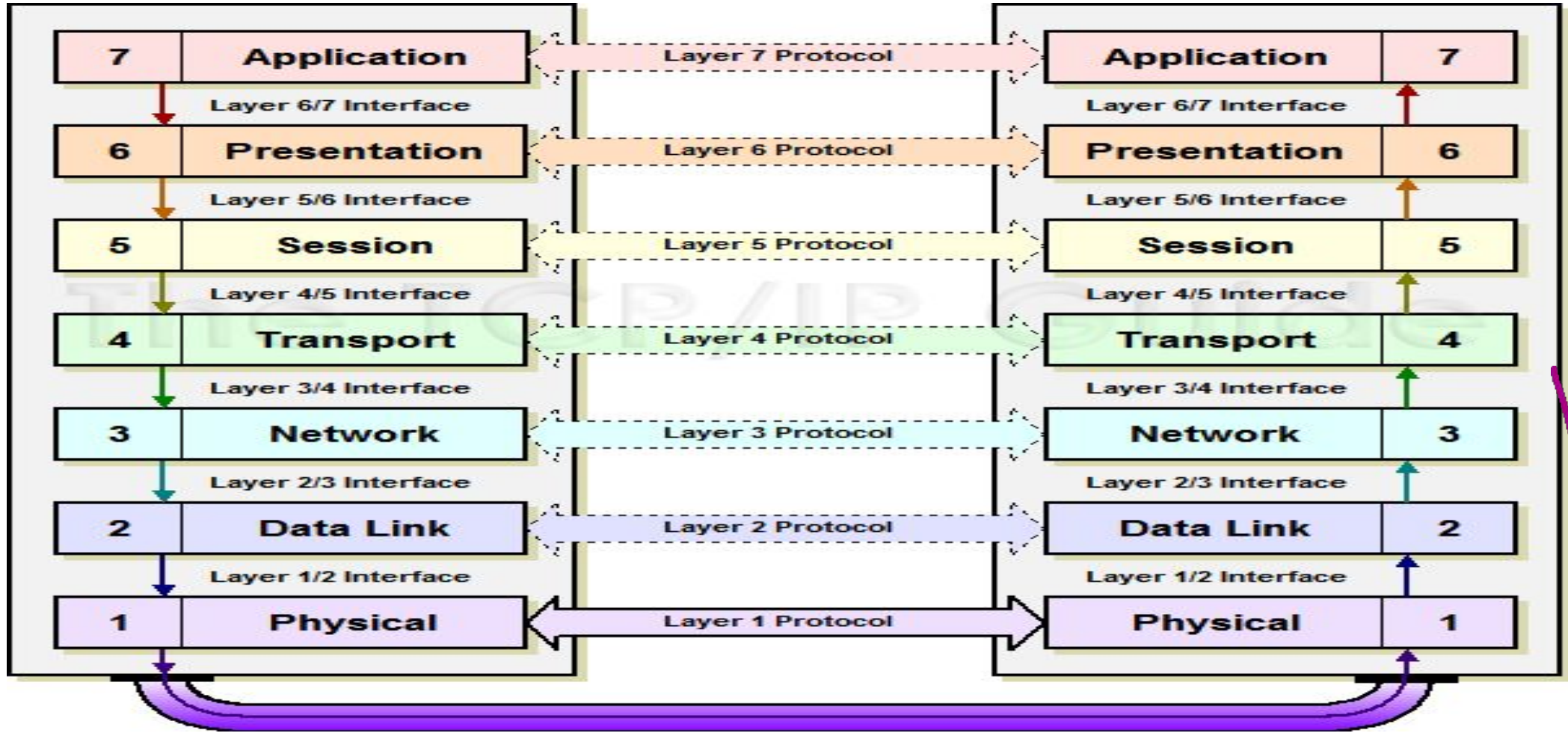
NON- HORIZONTAL COMMUNICATION



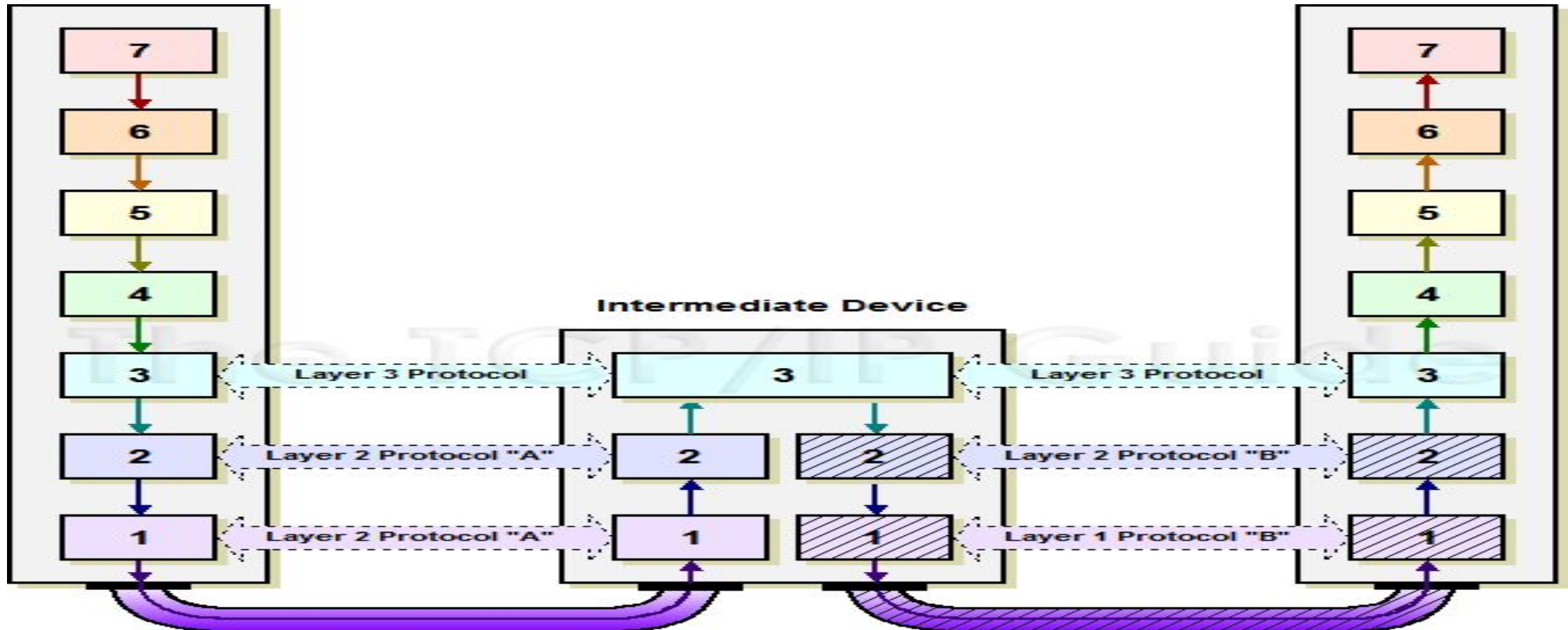
INDIRECT COMMUNICATION



ACTUAL VS LOGICAL/VIRTUAL COMMUNICATION



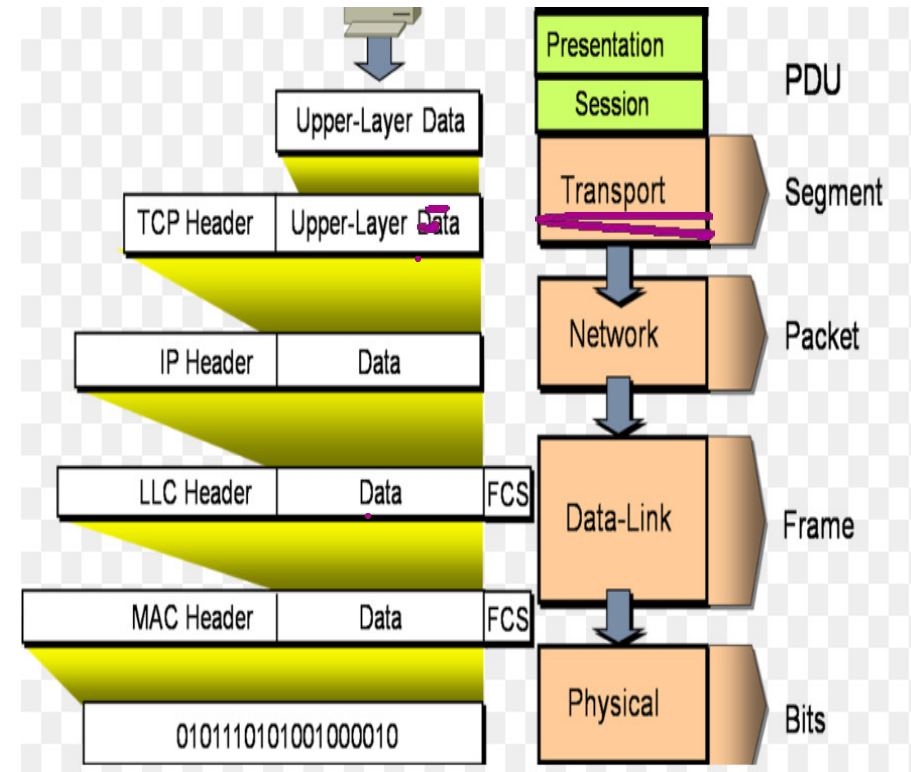
DIRECT VS INDIRECT COMMUNICATION



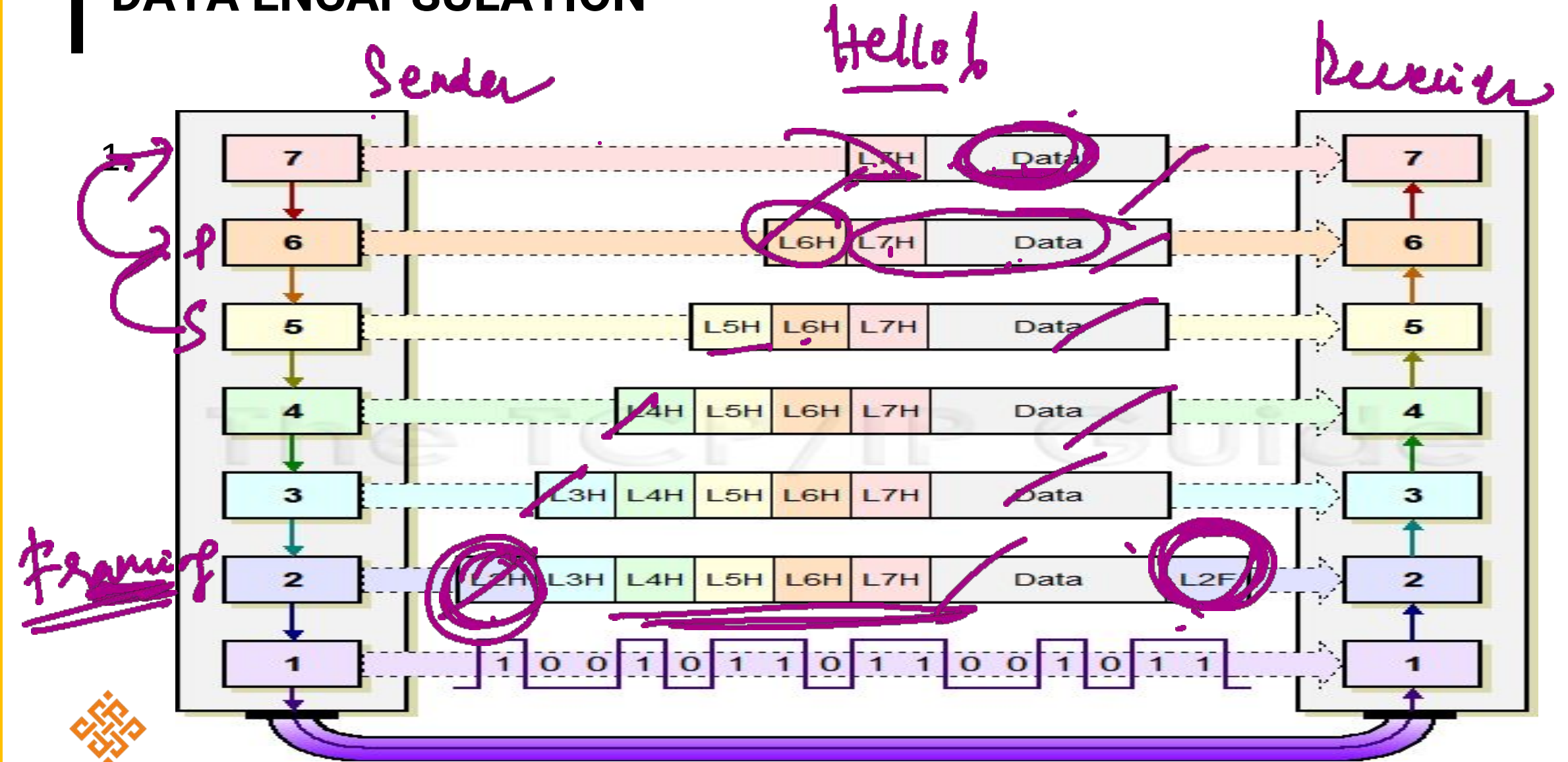
TYPES OF DATA UNITS

- Data Units:

- 1) Upper layer-> Data or message.
- 2) Transport layer-> Segments
- 3) Network Layer-> Packets
- 4) Data-link Layer-> Frames
- 5) Physical Layer-> Bits/signals

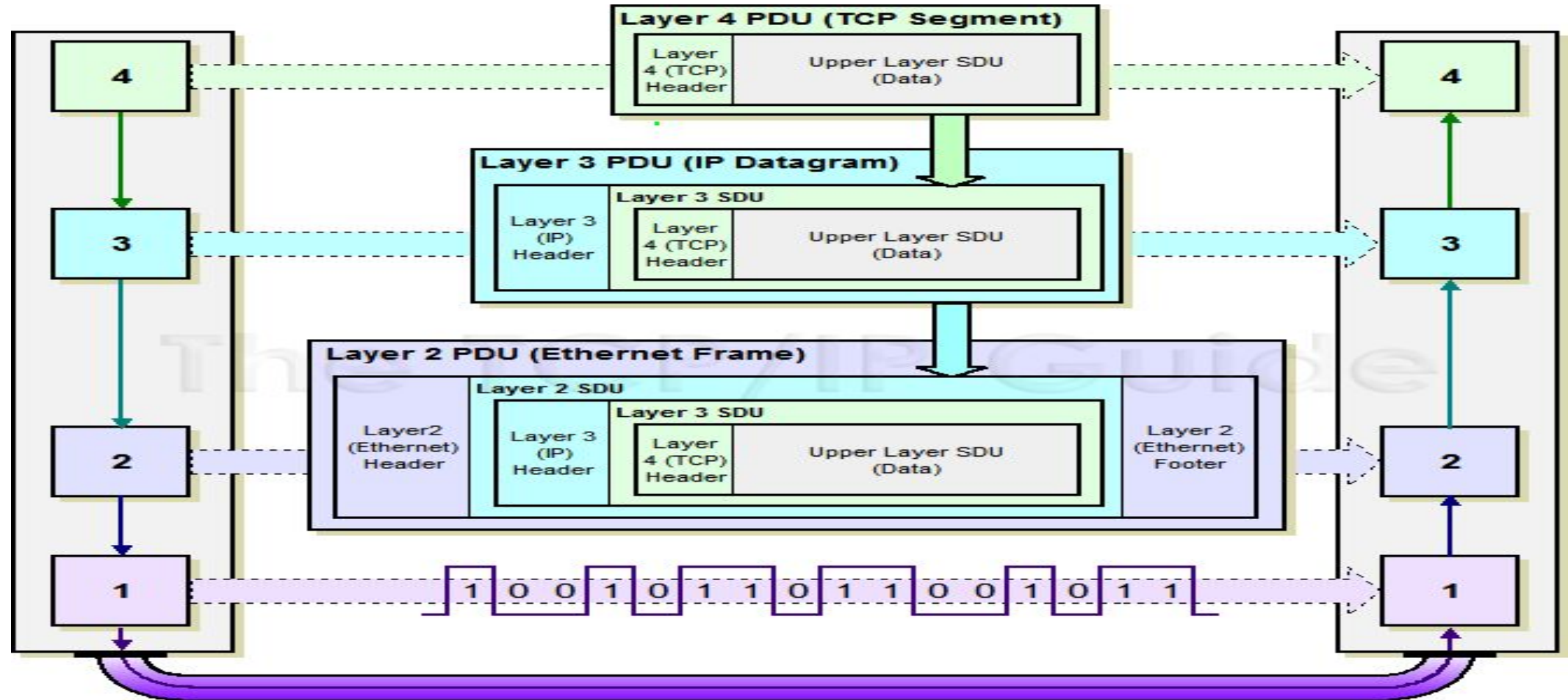


DATA ENCAPSULATION

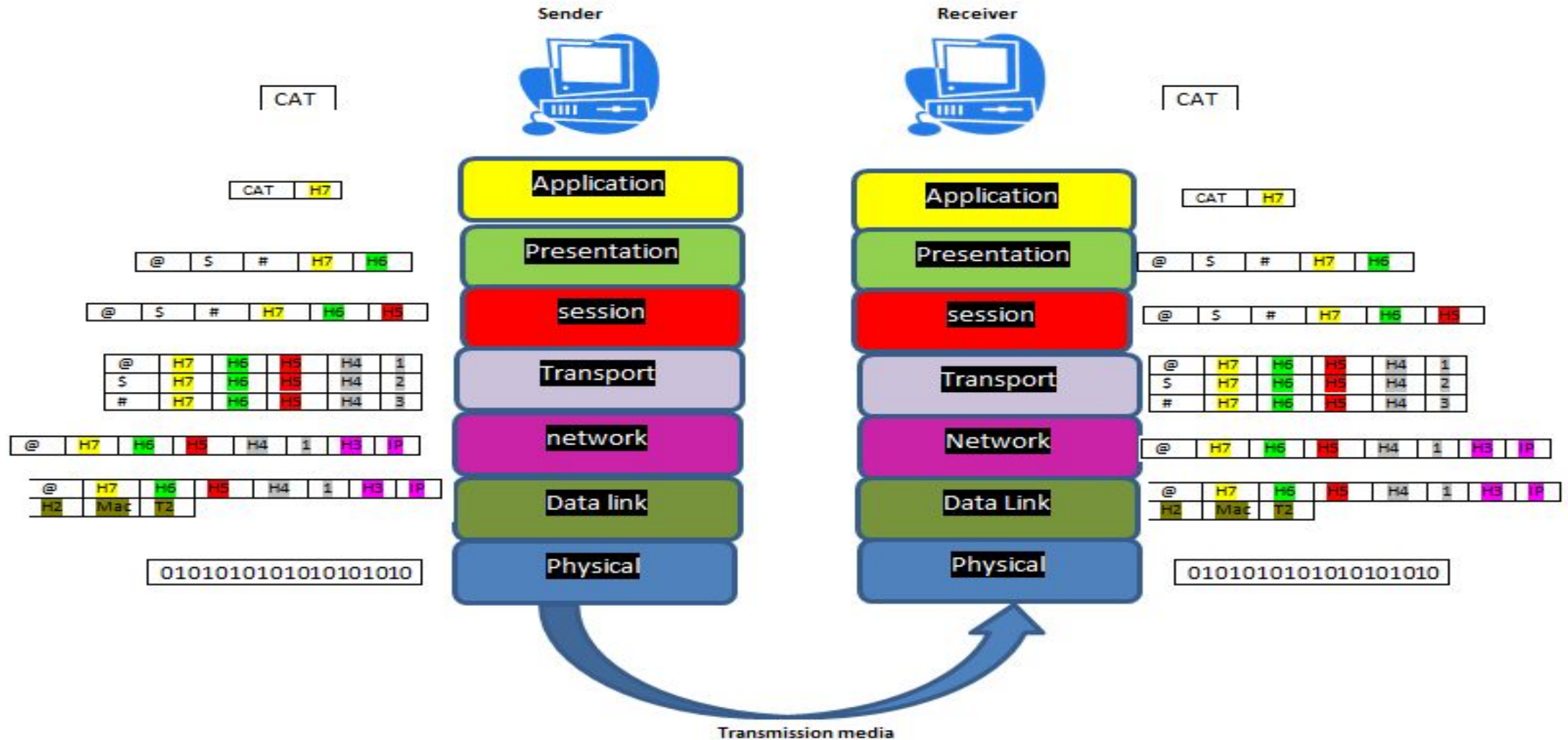


DATA ENCAPSULATION

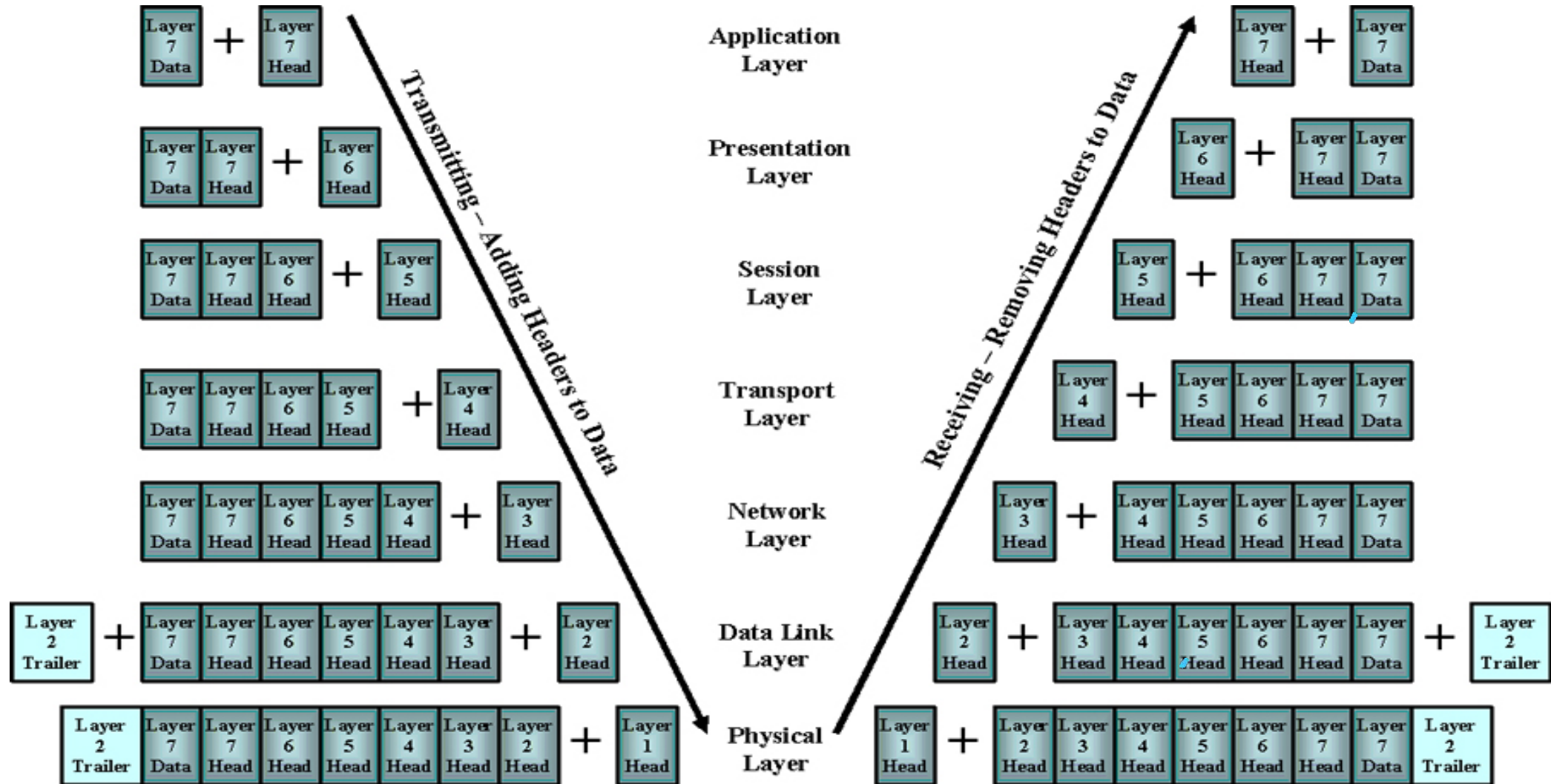
1.



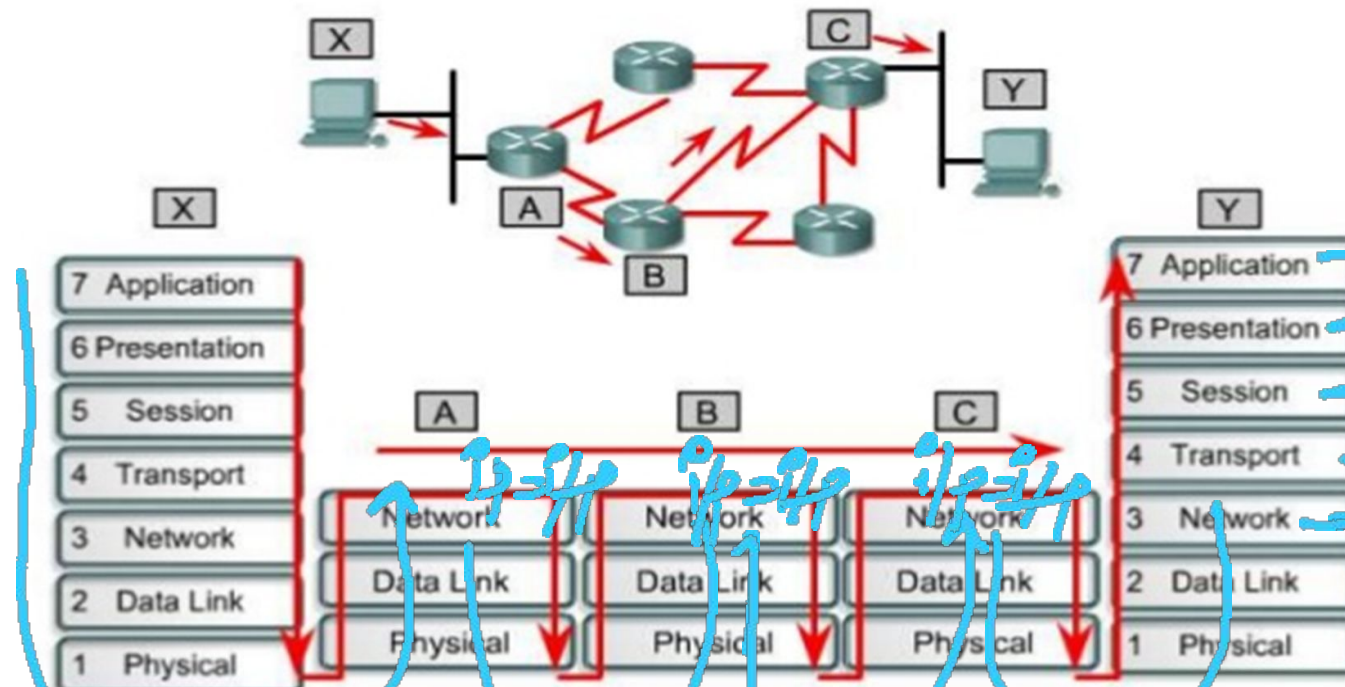
DATA ENCAPSULATION & STRIPPING



DATA ENCAPSULATION & STRIPPING -2 (SIZE)



DATA-FLOW IN NETWORK



Data flow in a network focuses on layers one, two and three of the OSI model. This is after being transmitted by the sending host and before arriving at the receiving host.



TYPES OF MESSAGE DELIVERY IN NA

1. Message delivery at peer-to-peer level

Sl.No	NA Layers	Message Delivery
1.	Data Link Layer	Node-to-Node Message delivery
2.	Internet/ Network Layer	End-to-End Message Delivery
3.	Transport Layer	Process-to-process Message Delivery.



TYPES OF ADDRESSES IN NA

Transport → port / process
Socket Address = port + IP

1. Addressing carried out in each layer

Sl.No	NA Layers	Message Delivery
1.	✓ Upper layer	✓ Domain Address
2.	Transport layer-	✓ Port Address (Socket address = port address + IP address)
3.	Network Layer	✓ IP address
4.	Data link layer-	✓ MAC address.



SIZE OF ADDRESSING SCHEME IN NA

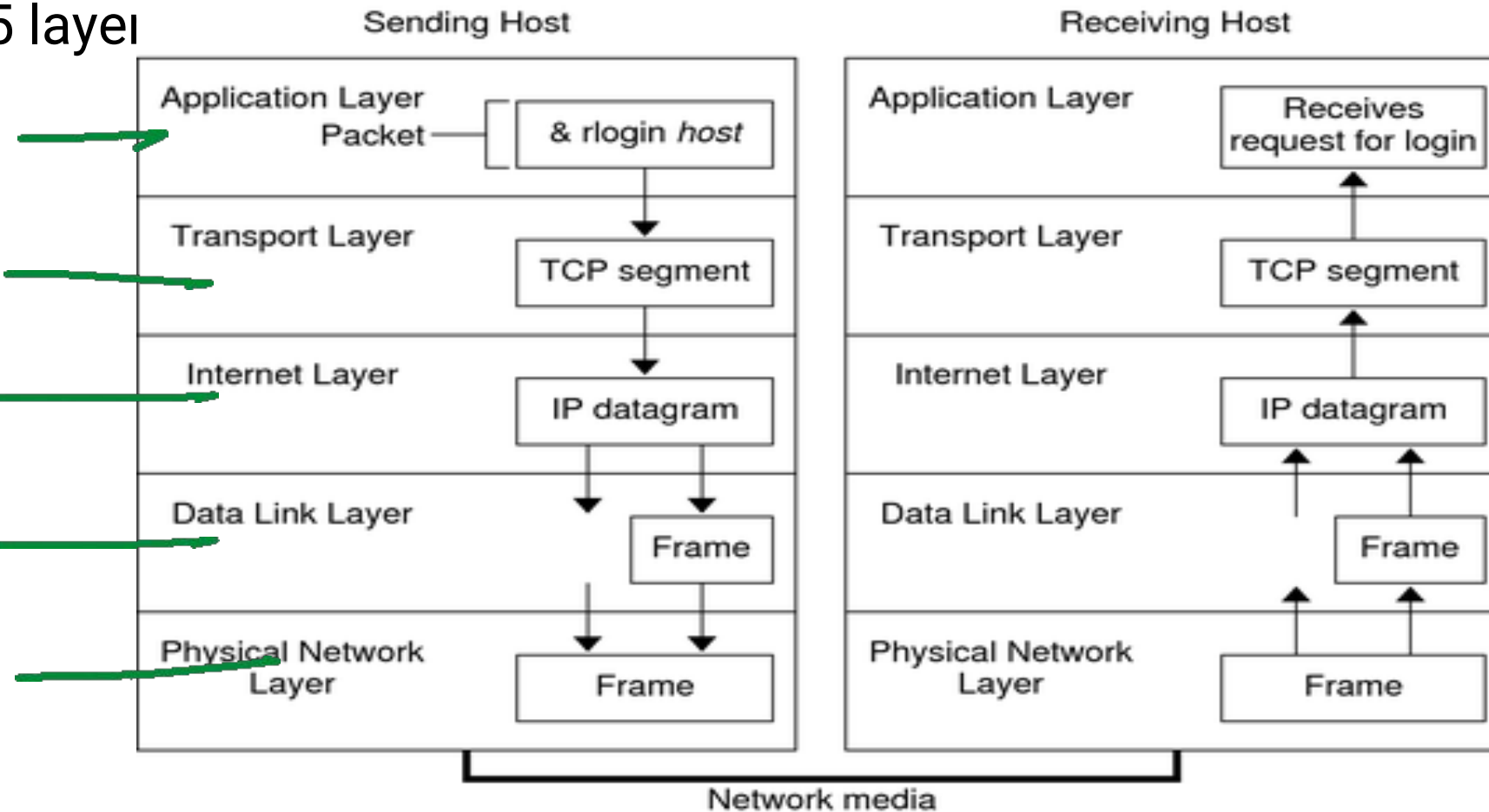
1. Bits level Address size

Sl.No	NA Layers	Message Delivery
1.	Domain address-	part of URL
2.	Port address-	16 bits
3.	IP address-	IPV4- 32 bits(4*8) , IPV6- 128 bits
4.	MAC address-	hardware address-> 48 bits.

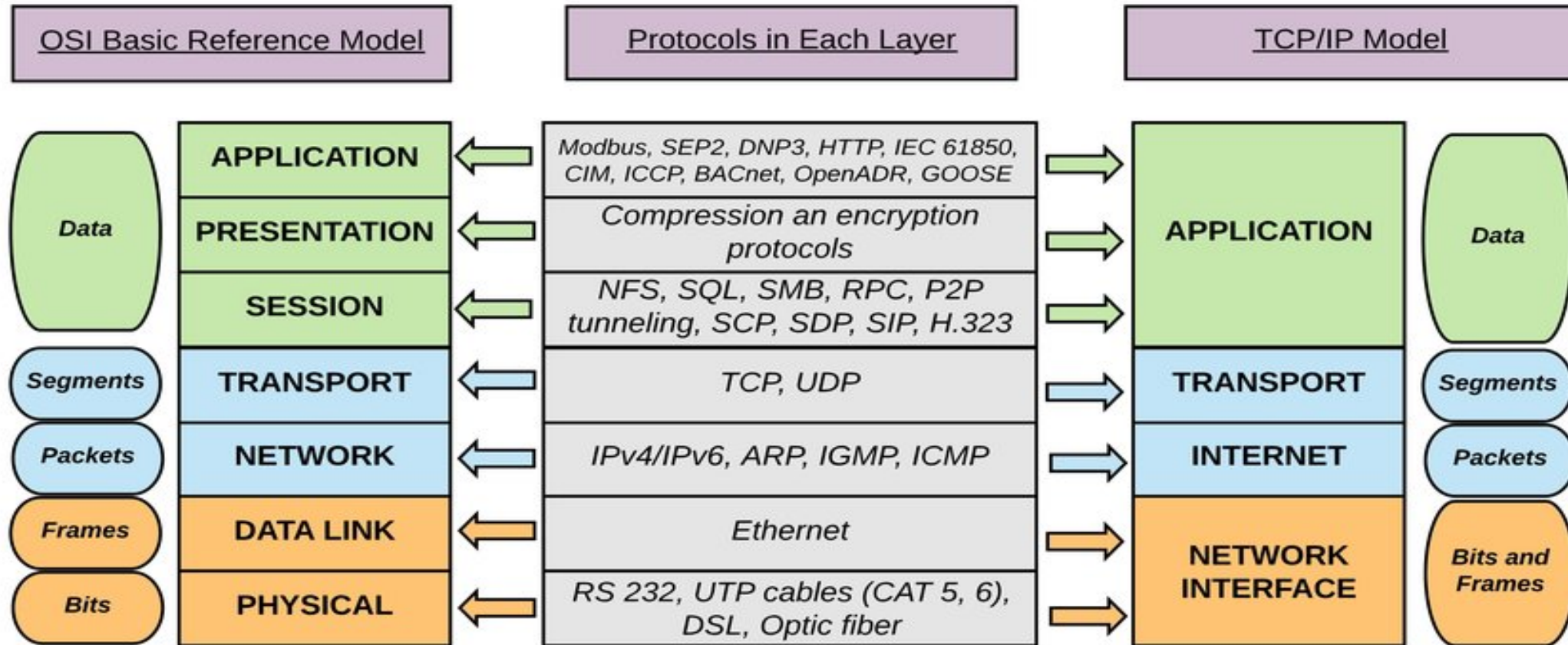


TCP/IP LAYER

1. 5 layer



COMPARISON BETWEEN TCP/IP & OSI



SUMMARY

- Lecture 8 : Network Reference Model

Q. 1 What is ISO OSI / 7 layered architecture?

Q. 2 What is TCP/IP or 5 layered architecture?

Q. 3 What are the differences between OSI and TCP/IP reference model?



ACTIVITY - 8

- Lecture 8 : Network Reference Model

Website:

<https://www.studytonight.com/computer-networks/uses-of-computer-networks>

Quiz:

http://highered.mheducation.com/sites/0072967722/student_view0/chapter_1_quiz.html



NETWORK | STANDARDS & ORGANIZATION

1. Apex Bodies



LECTURE –10

Network Standards & Organization

Recap



1. **OSI Reference model**
2. **TCP/IP Reference model**
3. **Direct vs Indirect communication**
4. **Physical/vertical vs Logical/horizontal communication**
5. **Data Encapsulation & Stripping**
6. **Data Units, Message delivery types, Address types**

Objectives

- **Network Standards**
- **Network Organizations**



NETWORK STANDARDS

Standards :

Define what is needed for interoperability: No more , No Less

Protocol Standards :

Defines the protocol over the wire but not the service interface inside the box

. (However, the real service interface are often proprietary).



NETWORK STANDARDS

Sl.No	Standards	Meaning
1.	De-Facto (Latin: “by fact” or “by convention”)	•Standards that have not been approved by an organized body but have been adopted as standards through widespread use . • De facto standards are often established originally by manufacturers who seek to define the functionality of a new product or technology. •These are also known as Market Standards.
2.	De-jure (Latin: “by Law” or “by organization”)	Those standards that have been legislated by an officially recognized body .



STANDARD ORGANIZATION

1. Standards developed through the cooperation of standards creation committees, forums, and government regulatory agencies.

Sl.No	Standards	Expansion	Area
1.	ISO	International Organisation for Standardisation	• Scientific , Technological, Economic actively.
2.	ITU-T	International Telecommunication Union-Telecommunication Standards	Telecommunication Ex: G.992,ADSL,H.264,MPEG4
3.	IETF	Internet Engineering Task Force	Internet Ex: 802.3,Ethernet,802.11,Wi-fi
4.	IEEE	Institute of Electrical & Electronics Engineers	Communication standards



STANDARD ORGANIZATION

Sl.No	Standards	Expansion	Area
5.	W3C	World Wide Web Consortium (Inventor: Tim Berners-Lee)	Web Standards Ex: HTML,CSS
6.	EIA	•Electronic Industries Association	
7.	ANSI	• American National Standards Institute	Development of voluntary consensus standards for products, services, processes, systems, and personnel in the United States.



SUMMARY

- Network Standard organization

Q. 1 What is De-facto?

Q. 2 What is De-jure?

Q. 3 How to convert de-facto to de-jure?

Q. 4 Name some of the international organisation in the world of communication and internet?



| DATA & SIGNALS

1. Base for communication



LECTURE - 11

Data & Signals

Recap

1. Network Standardisation
2. Standard Organisation



Objectives

- Data & Signals
- Characteristics of Signal
- Analogue & Digital Devices
- Analogue-Digital Conversion
- Digital-Analogue Conversion



DATA & SIGNALS

Data : These are characteristics collected through observation .

Data is a set of values of qualitative or quantitative variables about one or more persons or objects, while a datum (singular of data) is a single value of a single variable.

Signal: A signal is an electrical or electromagnetic current that is used for carrying data from one device or network to another.



Analog Signal



Digital Signal

Representation of Signals



CHARACTERISTICS OF SIGNAL

Analogue Signal

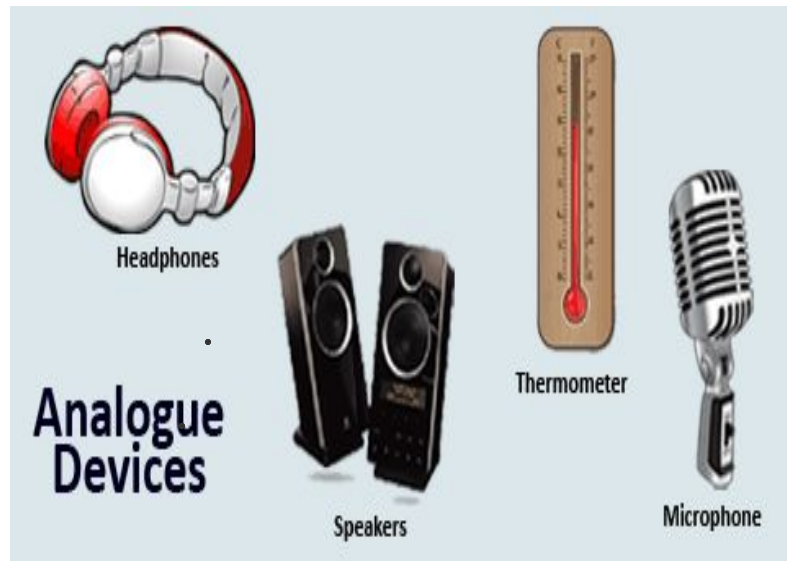
- Frequency
- Wavelength
- Amplitude
- Phase

Digital Signal

- Bit Rate
- Bit interval
- Bit length

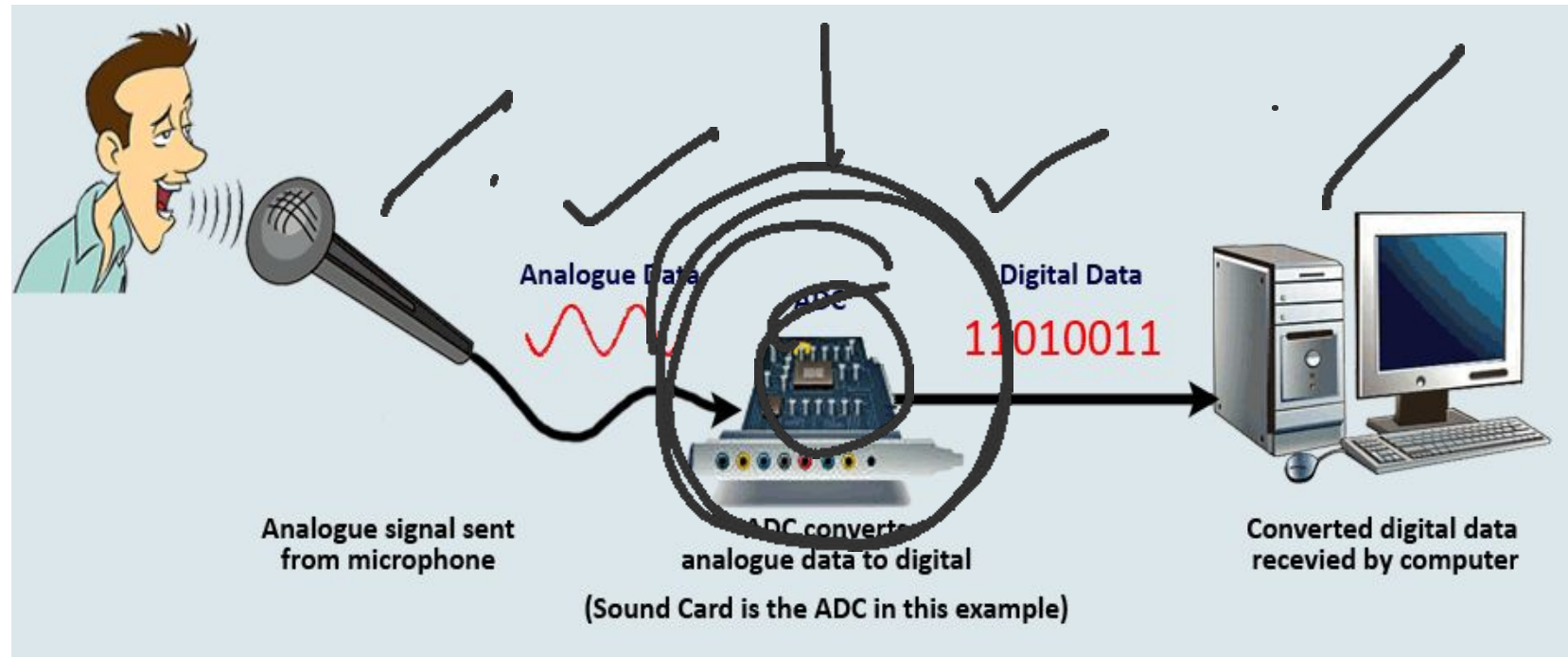


ANALOGUE & DIGITAL DEVICES



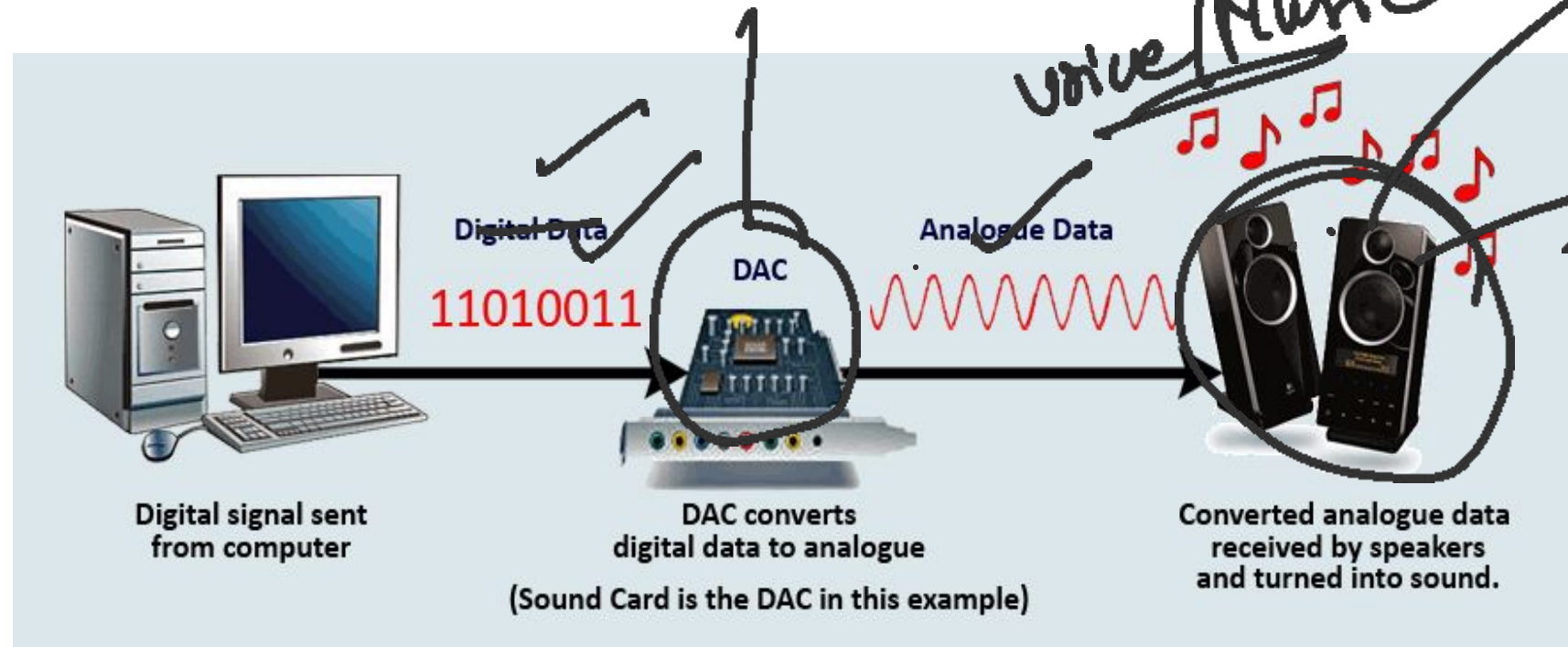
ANALOGUE-DIGITAL CONVERSION

1. Devices used to covert analogue signal to digital signal .



DIGITAL- ANALOGUE CONVERSION

1. Devices used to convert digital signal to analogue signal.



SUMMARY

- Data & Signals

Q.1 What is data?

Q.2 What is signal?

Q.3 What are the types of data & signal?

Q.4 How signals are represented?

Q.5 How data/signal conversion carried out?



ACTIVITY- 9

• Lecture 10 : Data , Signal & Basic of Waves

Reading:

Transmission impairment – attenuation, distortion, noise

Data Rate limits – bandwidth, signal level, quality of the channel

Performance – bandwidth, throughput, latency, bandwidth-delay product.

Waves:

1. <https://www.youtube.com/watch?v=R8kCskG7hKI> (Characteristics of waves).
2. <https://www.youtube.com/watch?v=UMC1EI-2sLo> (Characteristics of waves).
3. <https://www.youtube.com/watch?v=P798-zzEwT4> (Physics demonstration).



ELECTROMAGNETIC WAVE MODULATION

1 Guided and unguided



LECTURE - 12

Electromagnetic waves ,Bandwidth & Modulation

Recap

1. Data & Signal 
2. Characteristics of Signal
3. Analogue & Digital Devices
4. Analogue-Digital Conversion
5. Digital-Analogue Conversion

Objectives

- Electromagnetic waves
- Bandwidth
- Broadband & Baseband
- Modulation

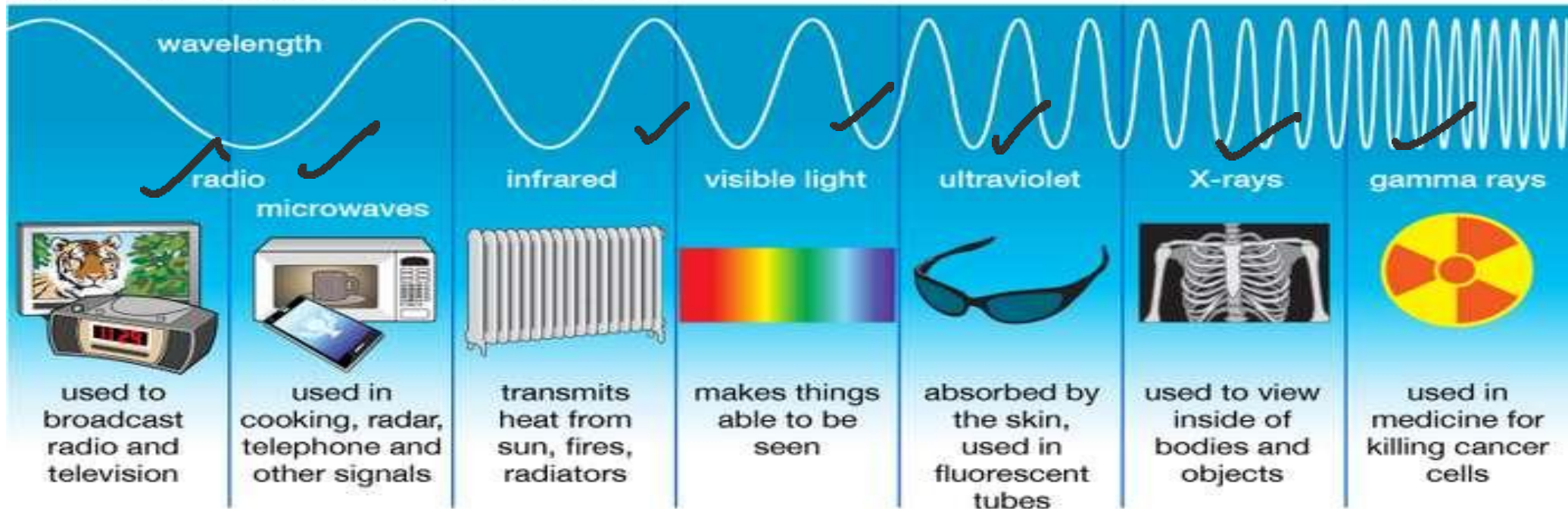


ELECTROMAGNETIC WAVES



1. - Longitudinal waves. compression rarefaction waves. vacuum waves

Types of Electromagnetic Radiation



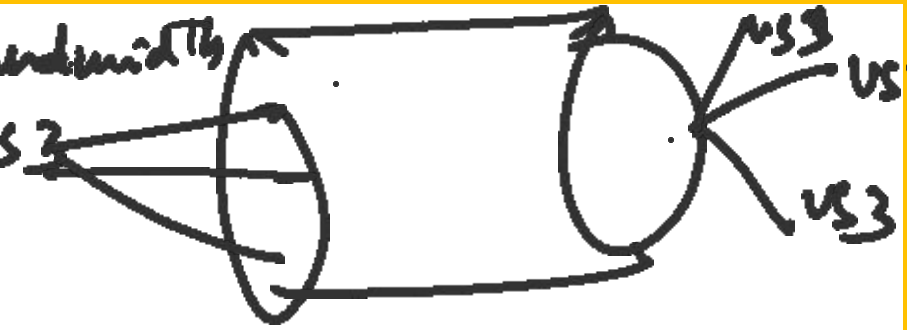
© 2013 Encyclopædia Britannica, Inc.



BANDWIDTH



Capacity < Bandwidth
 $\frac{1}{v_1 + v_2 + v_3}$



1. Network bandwidth is the amount of data from

- Baseband technology transmits a single data signal, stream, channel at a time
- **General Ex:** Railway track; **Technical ex:** Ethernet (TDM)
- Broadband technology transmits multiple data signals/streams/channels simultaneously at the same time.
- **General Ex:** Road ; **Technical Ex:** DSL, Cellular, Cable modern, Satellite (FDM, WDM)



Modulation $\Rightarrow$ ADS + CS $\Rightarrow$ Modulated



MODULATION

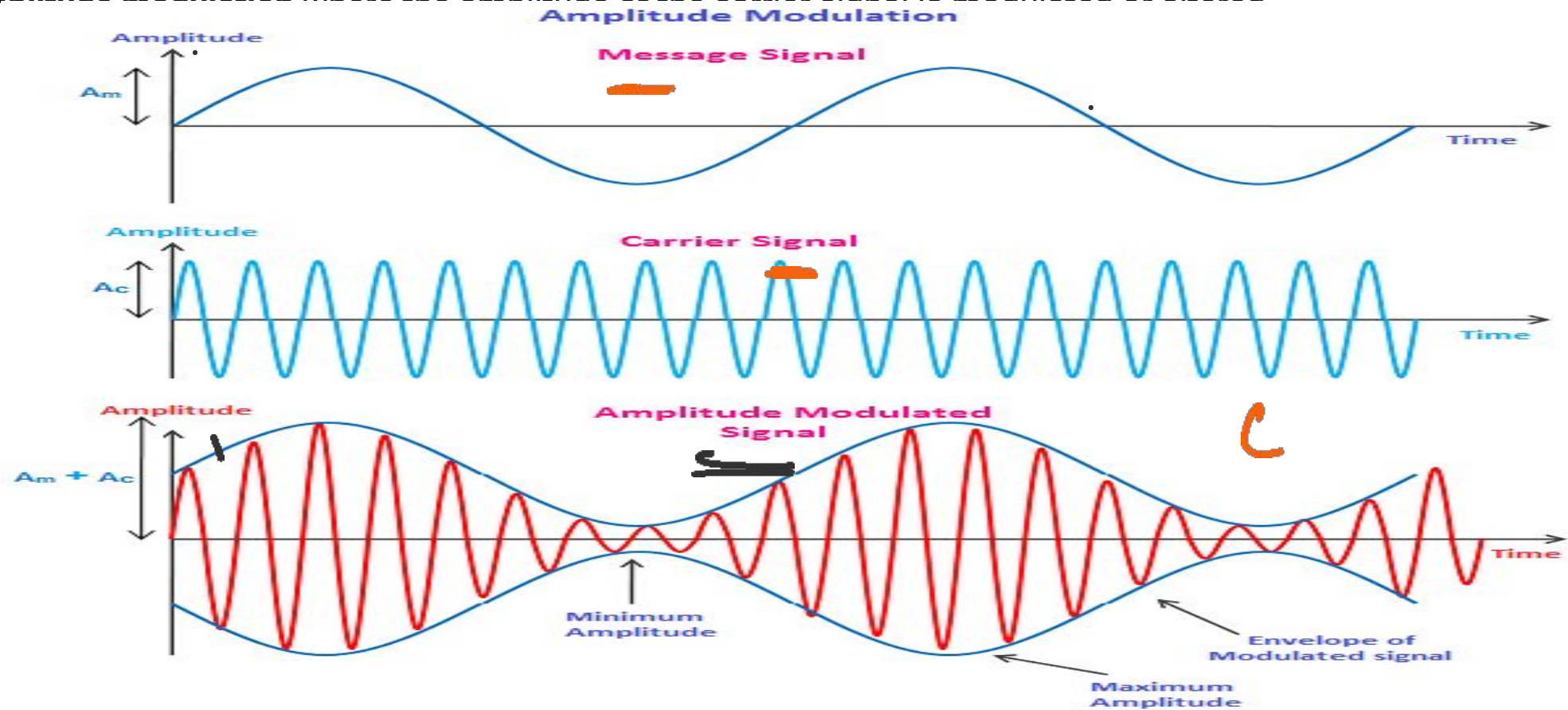
1. It is the process of mixing data signal to the carrier signal to form new signal. Mixing of low frequency signal with high frequency carrier signal is called **Modulation**.
2. **Modulation types :**
 1. Analog modulation
 2. Digital modulation

Message Signal + Carrier Signal = Modulation



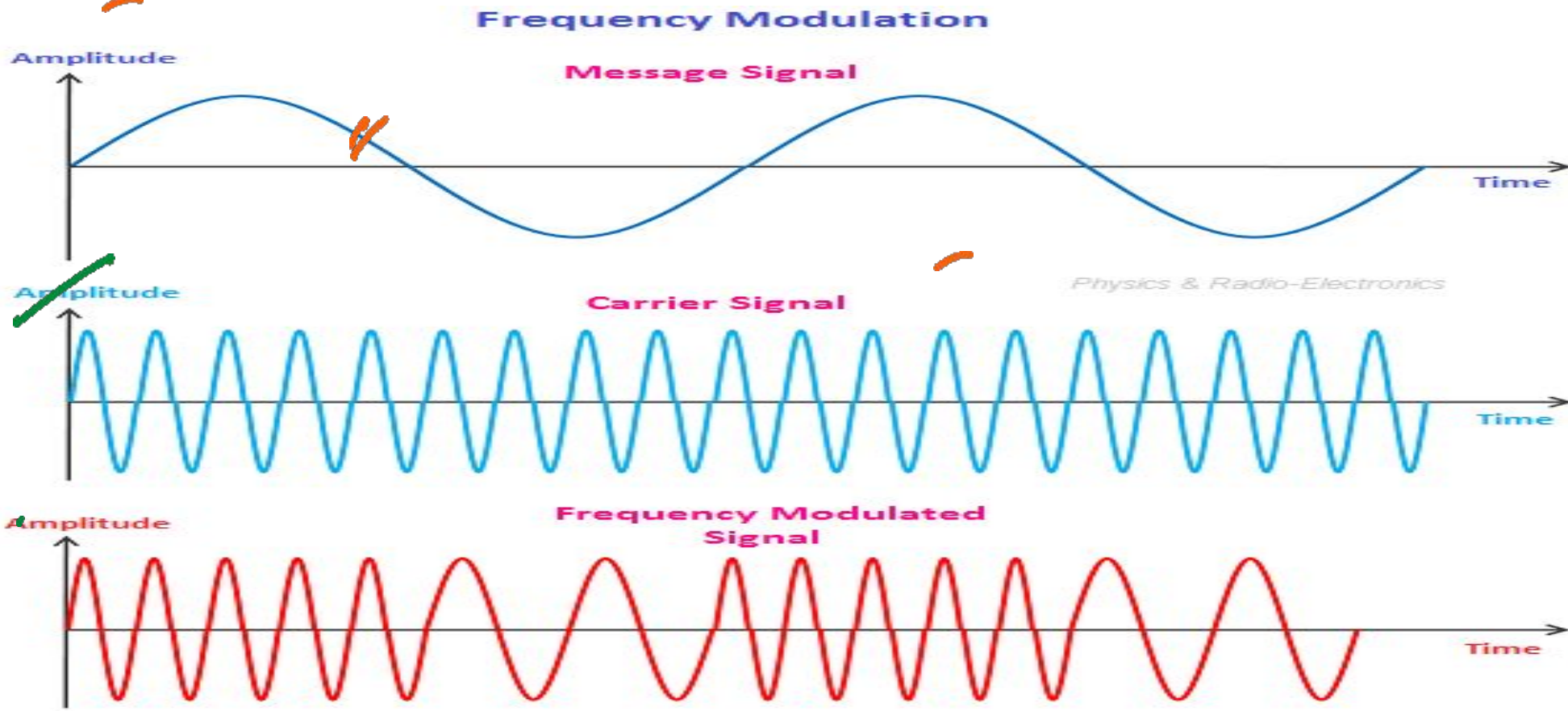
ANALOGUE MODULATION

1. Amplitude modulation where the amplitude of the carrier signal is modulated or altered

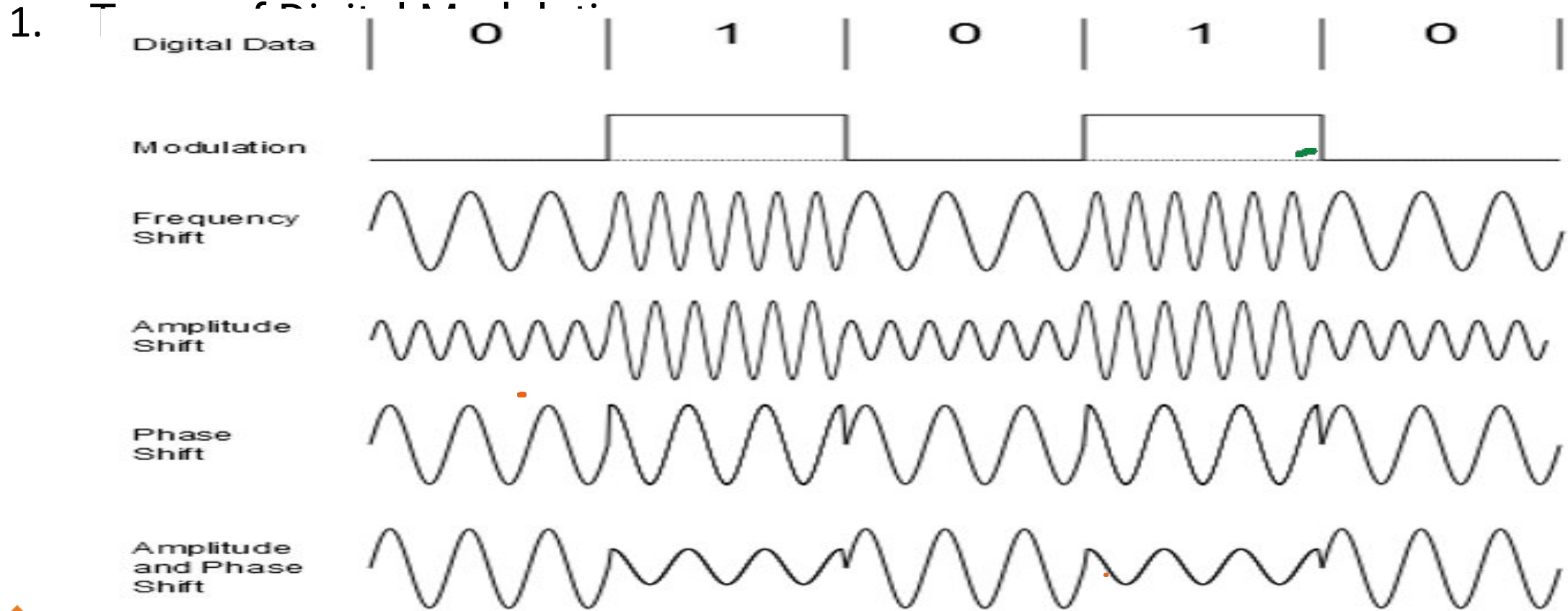


ANALOG MODULATION

1.



DIGITAL MODULATION



SUMMARY

- EMW, BW, Baseband & Broadband, Modulation

Q.1 What is Electromagnetic waves?

Q.2 What is Bandwidth?

Q.3 What is Baseband & Broadband?

Q.4 What is Modulation?



ACTIVITY -10

• Lecture 11: EMW, BW, Baseband & Broadband, Modulation

Electromagnetic Spectrum

https://science.nasa.gov/ems/01_intro

<https://www.youtube.com/watch?v=Elv3WpL32UE>

Bandwidth:

<https://www.youtube.com/watch?v=IS8pRayVIYc>

Speed vs Bandwidth

https://www.youtube.com/watch?v=A_-L-kn9biw

Bandwidth: Baseband vs Broadband

<https://www.youtube.com/watch?v=bNUlliQocTc>

Modulation:

<https://www.youtube.com/watch?v=dvGcCk1vbjk>



I TRANSMISSION MEDIA

1. Guided and unguided



LECTURE - 13

Transmission Media

Recap

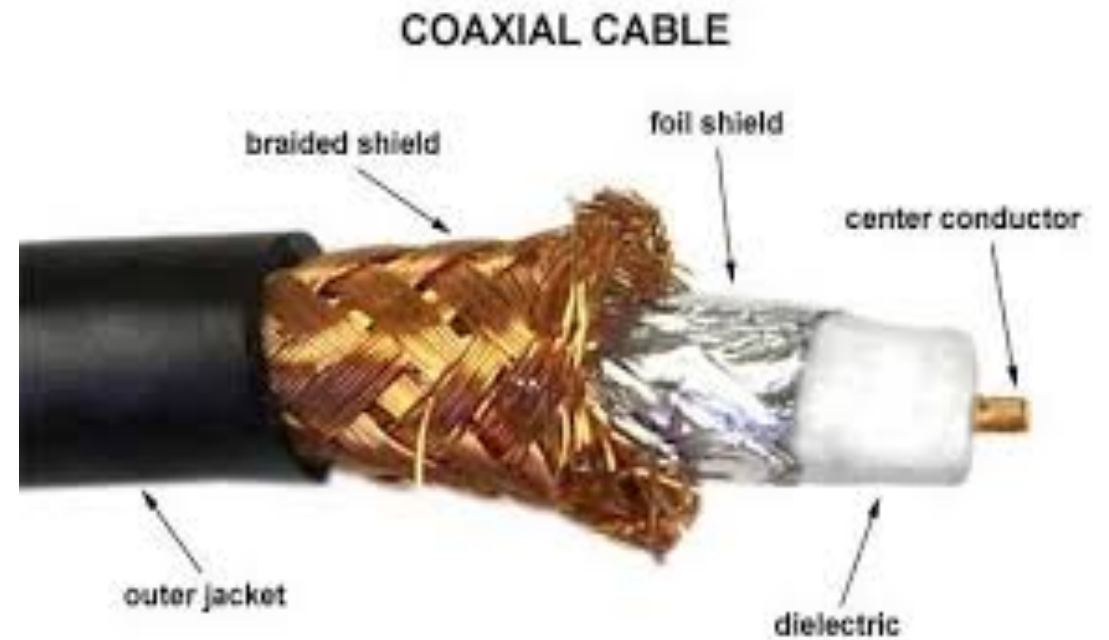
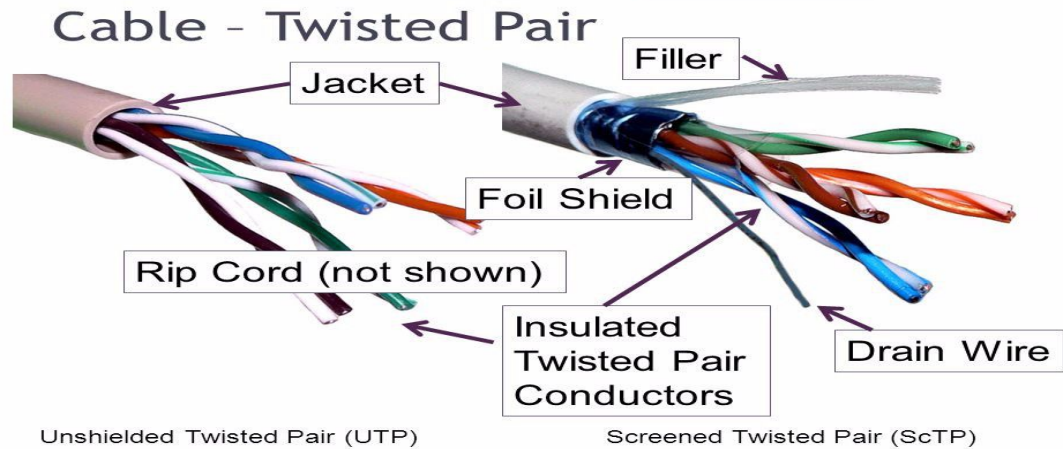
1. Electromagnetic waves
2. Bandwidth
3. Broadband & Baseband
4. Modulation

Objectives

- Transmission media or Guided media
- Non-Transmission media or Unguided media



TRANSMISSION MEDIA



COTIND.....

1. It is defined as the physical medium through which the signals are transmitted.
2. It is also known as **Bounded media**.
3. Types Of Guided media: Shielded Twisted Pair & Unshielded Twisted Pair
4. Twisted pair is a physical media made up of a pair of cables twisted with each other.
5. A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable.
6. The frequency range for twisted pair cable is from 0 to 3.5KHz.



COAXIAL CABLE IS OF TWO TYPES:

1. **Baseband transmission:** It is defined as the process of transmitting a single signal at high speed.
2. **Broadband transmission:** It is defined as the process of transmitting multiple signals simultaneously.

Advantages :

- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages :

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.



UNGUIDED TRANSMISSION

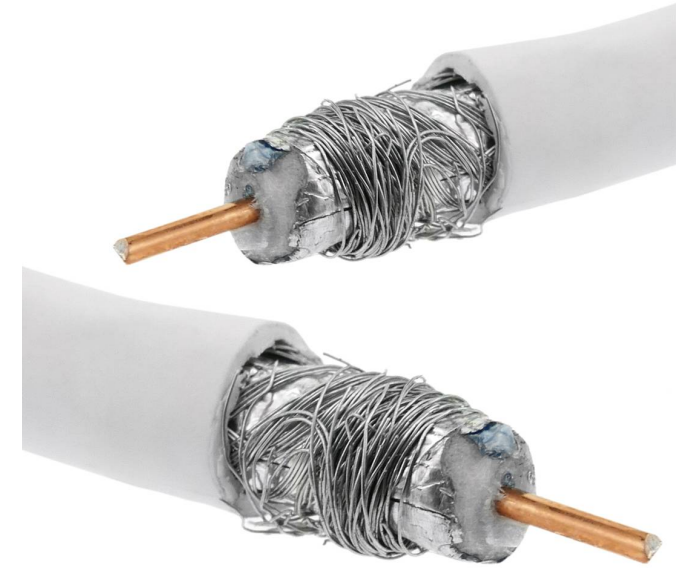
- An unguided transmission transmits the electromagnetic waves without using any physical medium. Therefore it is also known as **wireless transmission**.
- In unguided media, air is the media through which the electromagnetic energy can flow easily.

Radio waves

- Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.
- Radio waves are omnidirectional, the signals are propagated in all the directions.
- The range in frequencies of radio waves is from 3Khz to 1 khz.



DIFFERENT TYPES OF CABLES





RJ45 Plug



RJ45 Jack



GUIDED OR WIRED MEDIUM

Twisted
pair
cable

- Less data, Inference

Coaxial -
Cable

- Relatively more data, Inference

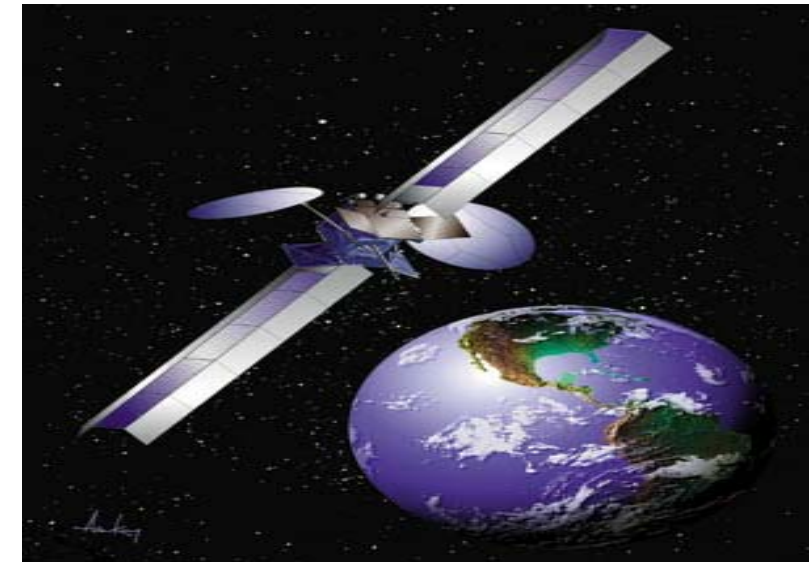
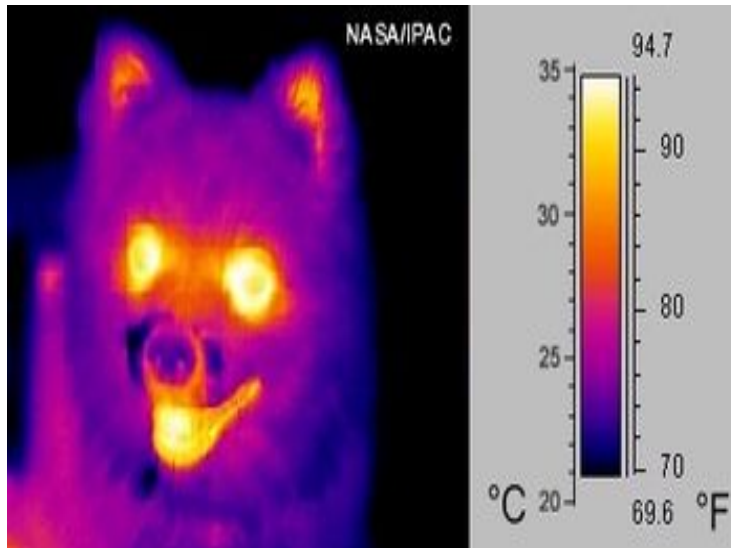
Fibre-
optics

- More data, Inference



TRANSMISSION MEDIA

1. - Unguided media



UNGUIDED OR WIRELESS MEDIA

Infrared Waves

- Light waves, Electromagnetic waves,
- Radiation waves, Average frequency carrier wave

Micro waves

- Low frequency carrier wave

Satellite & Radio waves

- Very frequency carrier wave



SUMMARY

- Lecture 12 : Transmission media

Q.1 What is guided media?

Q.2 What is unguided media?

Q.3 list the examples for wired medium and wireless medium.

Q. 4 What are the advantages of optical fibre over coaxial-cable?

Q.5 Explain the importance Radio and satellite waves.



ACTIVITY - 11

- Lecture 12 : Guided and Un-guided media

Video & Reading : Optical Fibre

1. <https://www.britannica.com/science/fiber-optics>
2. <https://www.youtube.com/watch?v=02wPSDOXMhc>



MULTIPLEXING

1. Bandwidth utilization technique



LECTURE - 14

Multiplexing

Recap

1. Transmission media or Guided media
2. Non-Transmission media or Unguided media

Objectives

- Multiplexing
- Analogue Vs Digital multiplexing
- Frequency division multiplexing
- Wavelength division multiplexing
- Time division multiplexing



MULTIPLEXING

1. Bandwidth utilization technique for Efficiency

Analog Signal

- Frequency division multiplexing (FDM)
- Wavelength division multiplexing (WDM)

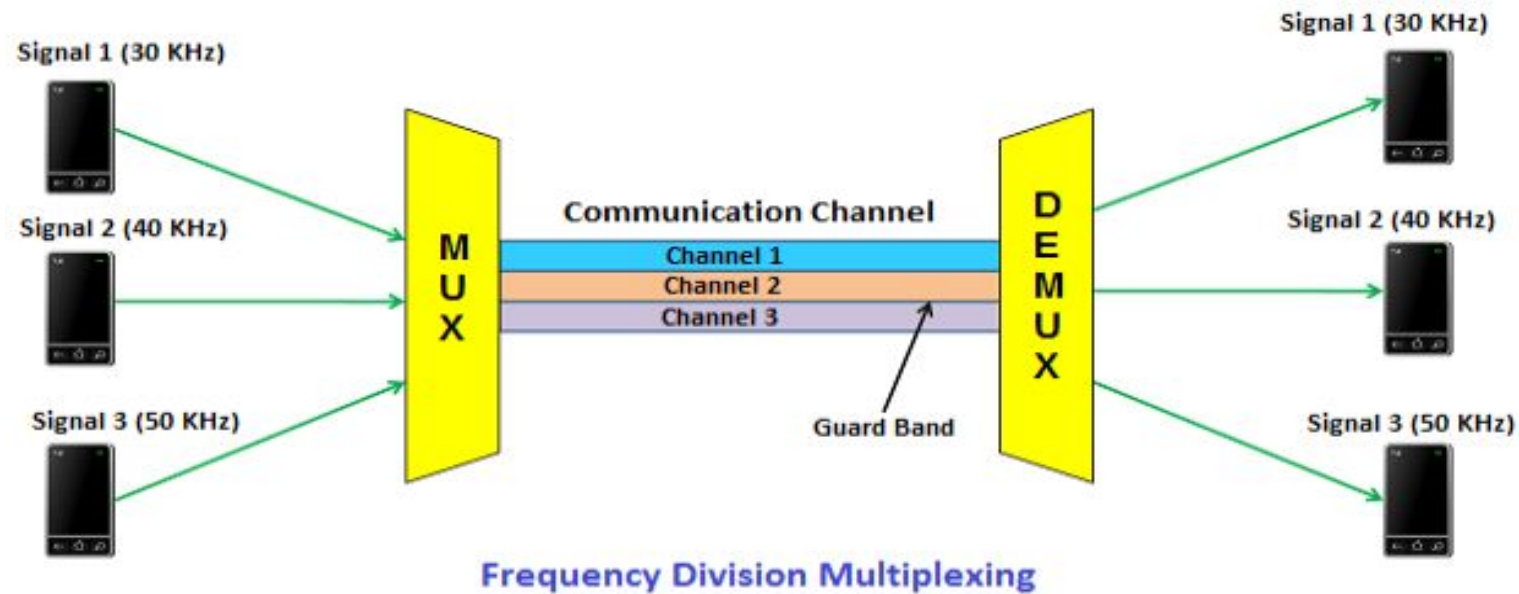
Digital Signal

- Synchronous Time division multiplexing
- Statistical Time division multiplexing



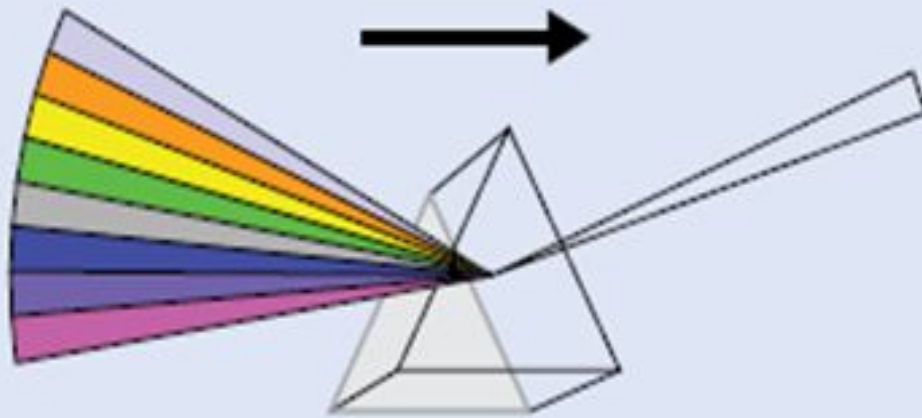
FDM : FREQUENCY DIVISION MULTIPLEXING

1. Analogue Multiplexing

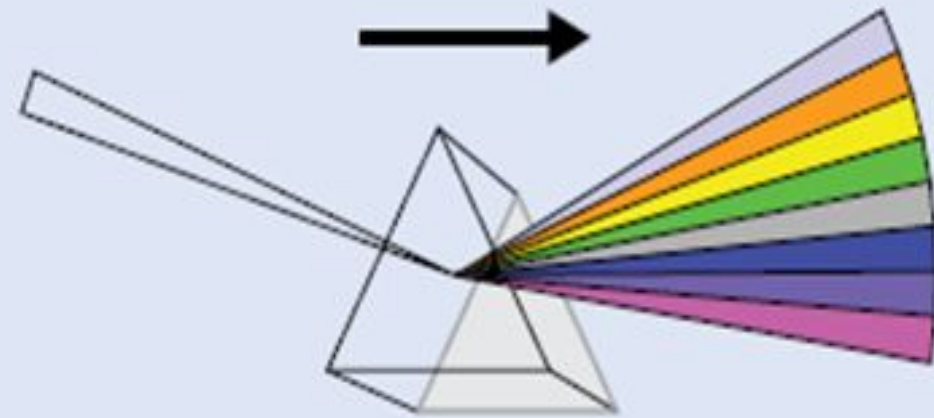


WDM : WAVELENGTH DIVISION MULTIPLEXING

1.



At Ingress: Multiple optical signals of differing wavelengths are combined to form a single optical signal.



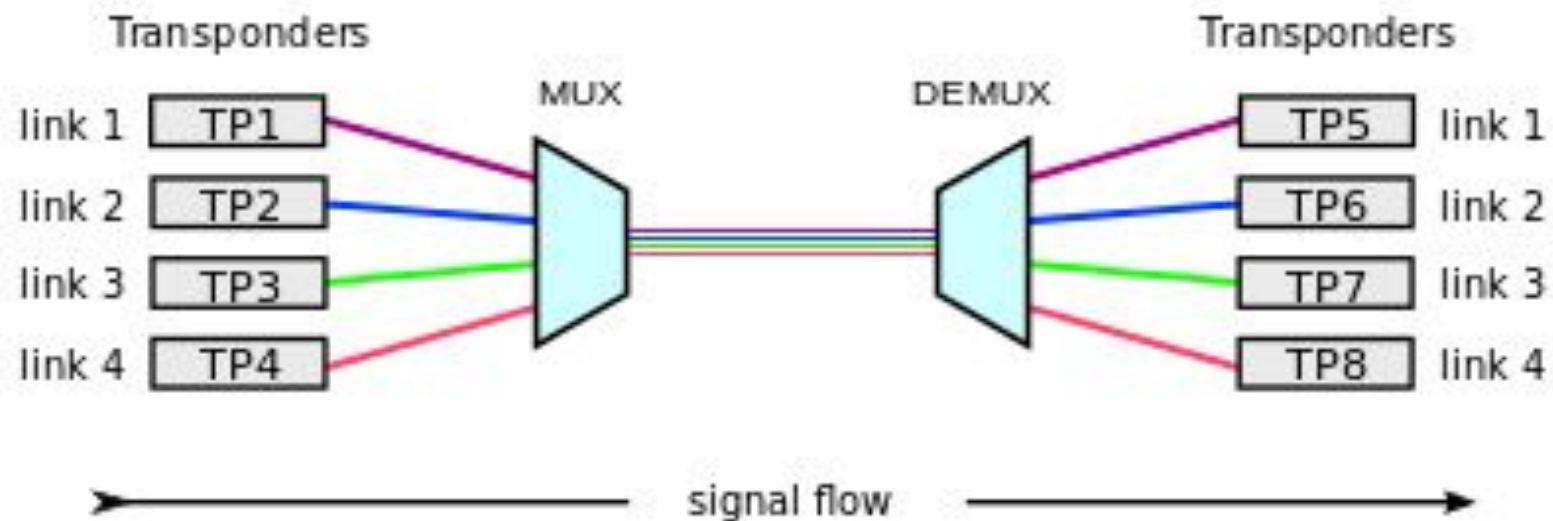
At Egress: A single optical signal is refracted to separate multiple optical signals of differing wavelengths.



WDM : WAVELENGTH DIVISION MULTIPLEXING

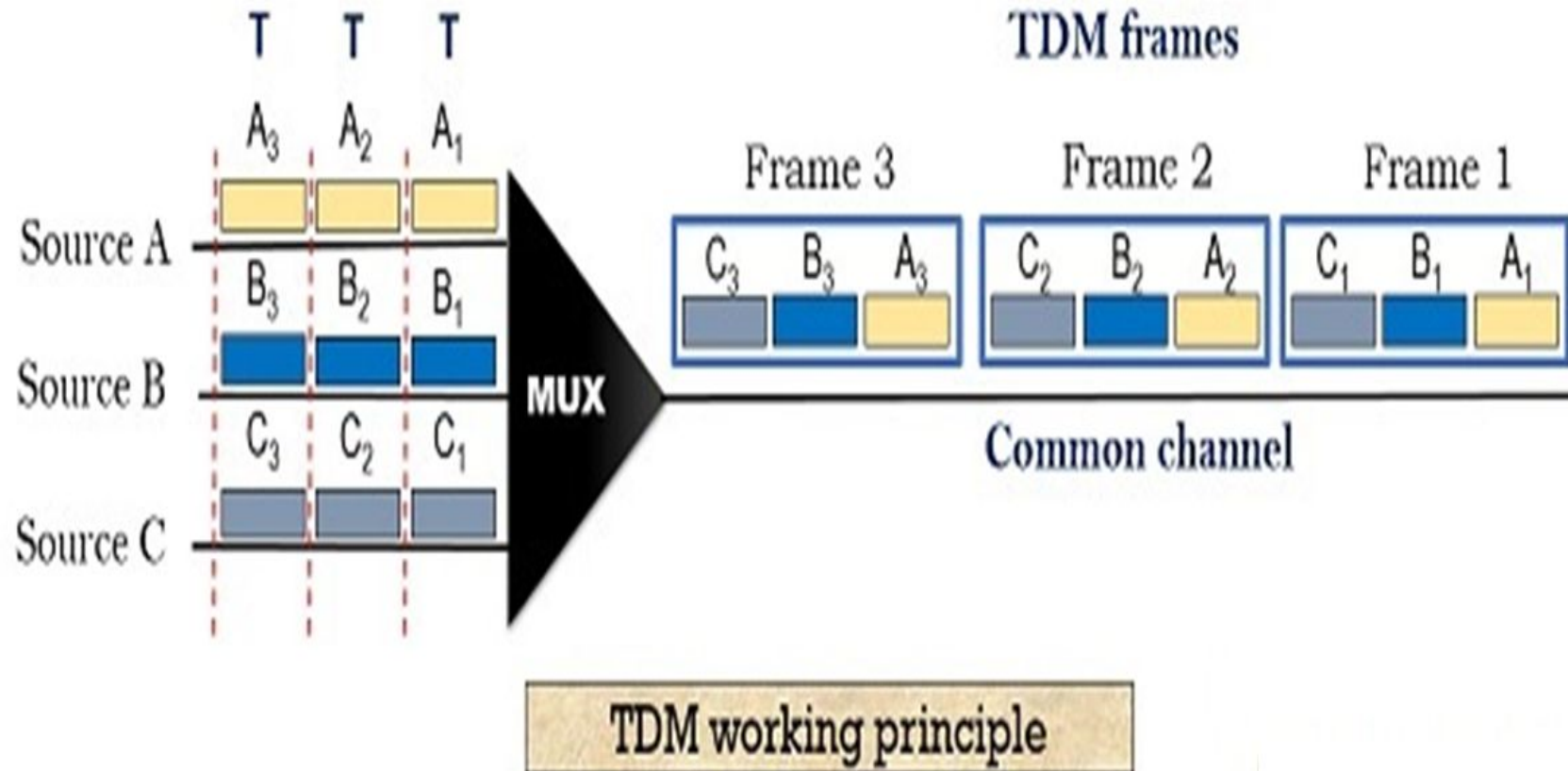
1. Mechanism

wavelength-division multiplexing (WDM)



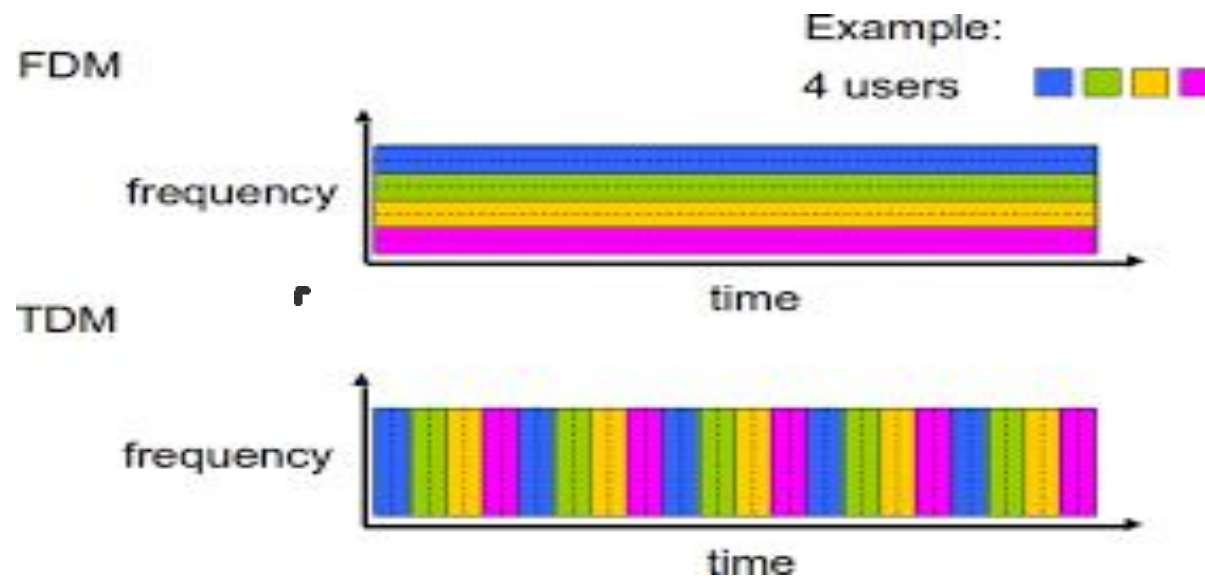
TDM : TIME DIVISION MULTIPLEXING

1. Wo

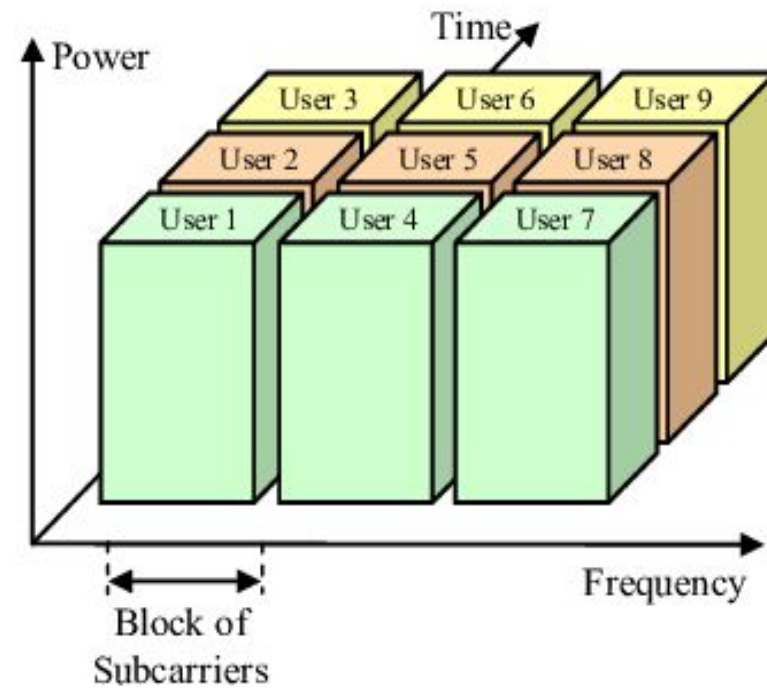
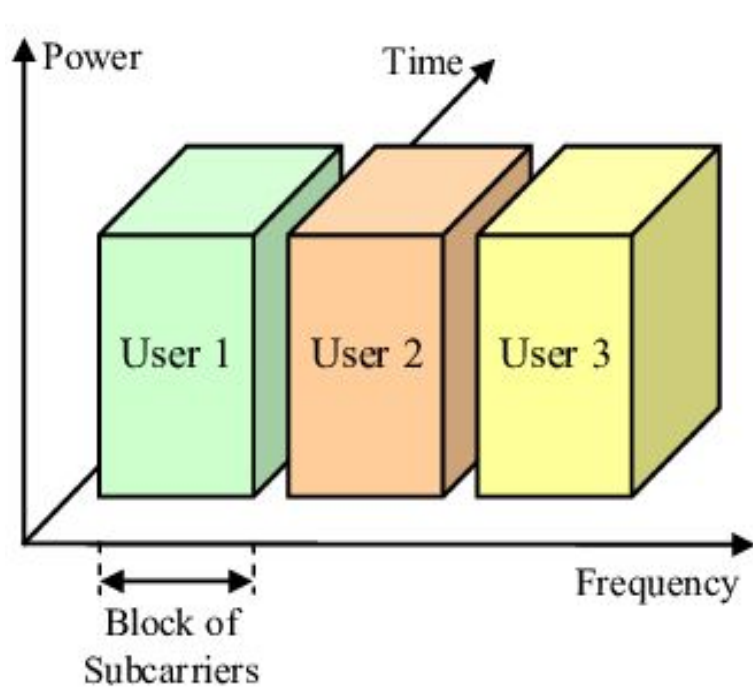


FDM & TDM

1. Comparison



FDM & TDM



SUMMARY

- Multiplexing

Q.1 What is multiplexing and why?

Q.2 How channels in link are created?

Q.3 How multiplexing is carried down?

Q. 4 Explain the types of multiplexing, its features and limitation?

Q.5 What are the differences between FDM & TDM?



ACTIVITY - 12

- Lecture 13 : Multiplexing

Reading : WDM

1. <http://www.fiber-optic-transceiver-module.com/wavelength-division-multiplexing-wdm-technology.html>



| SWITCHING

1. Technique



LECTURE - 15

Switching

Recap

1. **Multiplex** 
2. **Analogue Vs Digital multiplexing**
3. **Frequency division multiplexing**
4. **Wavelength division multiplexing**
5. **Time division multiplexing**

Objectives

- Switching
- Message switching
- Circuit switching
- Virtual circuit packet switching
- Datagram packet switching



SWITCHING

1. Technique to reduce # of devices and links in network

Traditional Switching

- Message switching
- Circuit switching

Packet Switching

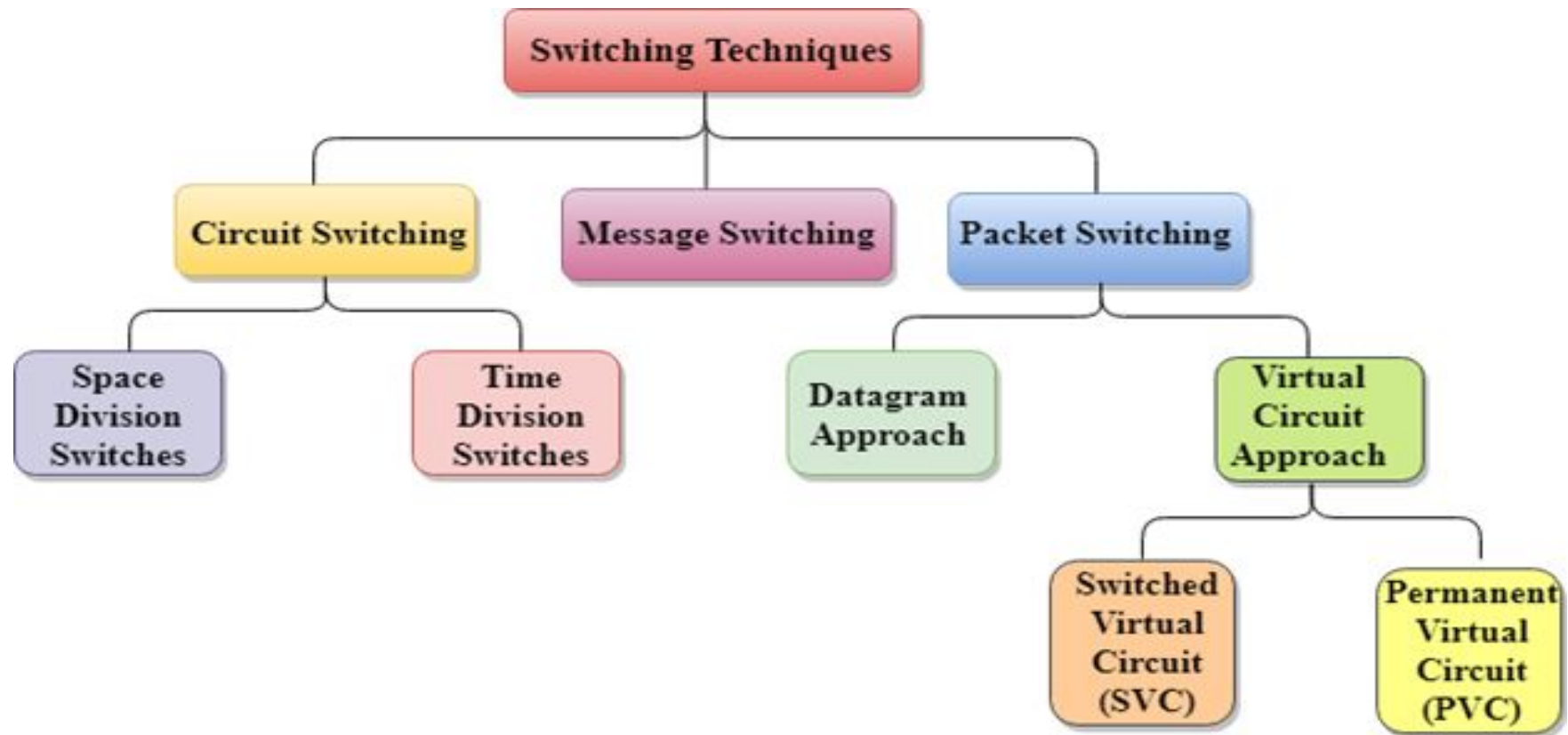
- Circuit packet switching
- Datagram packet switching

Mobile Switching

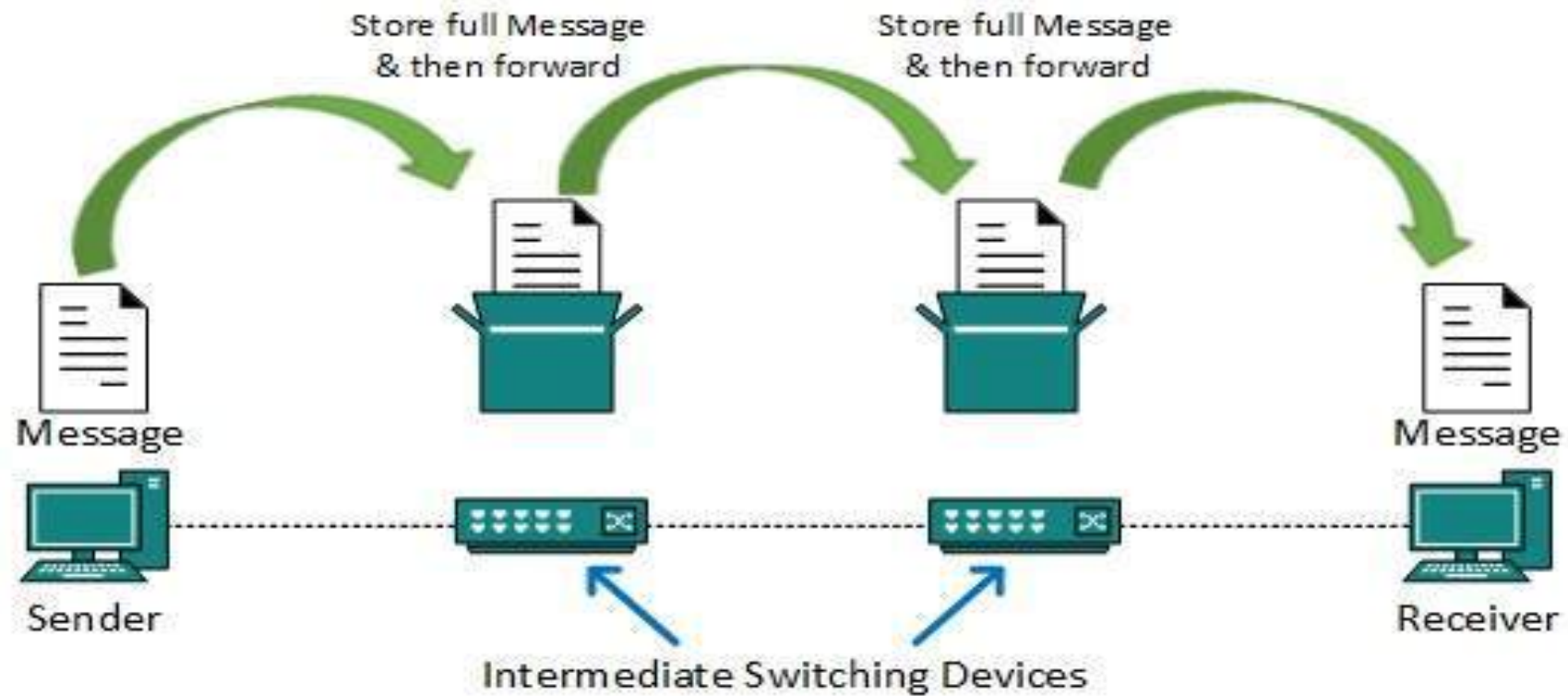
- Cellular switching



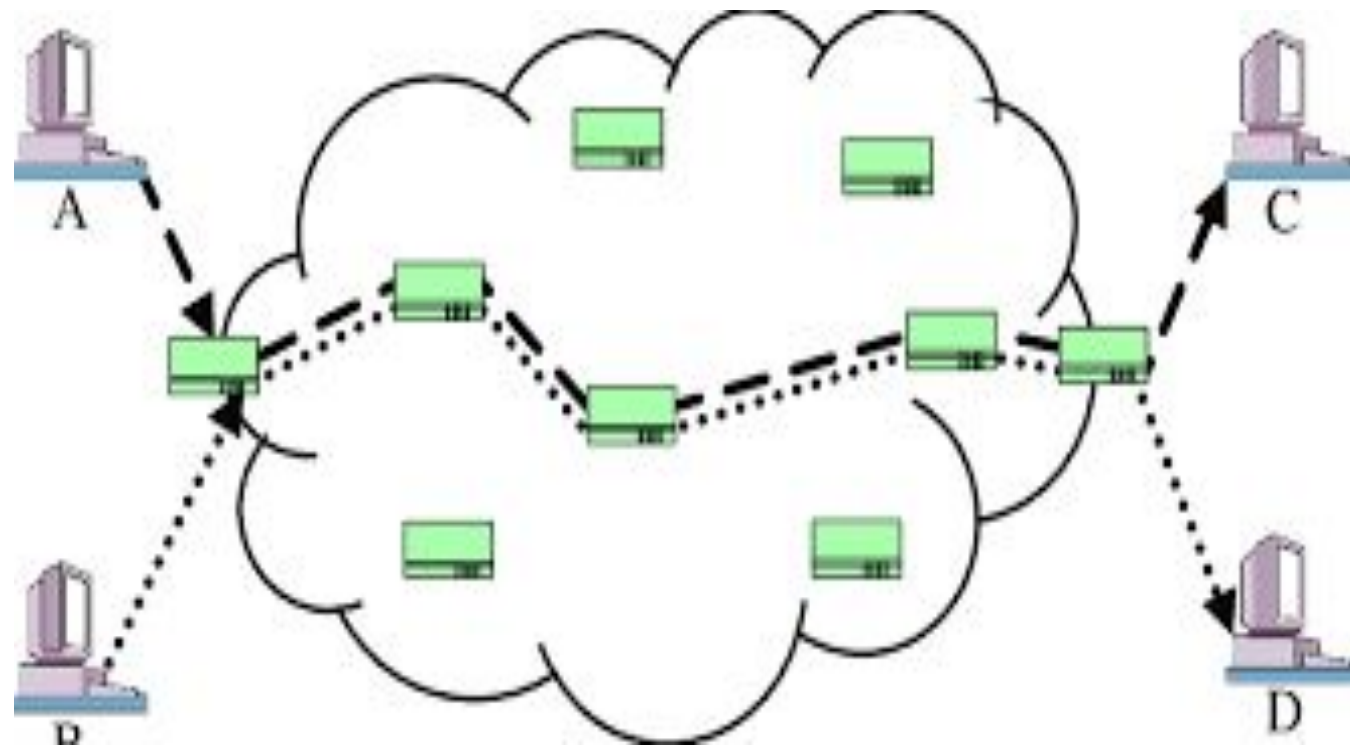
SWITCHING TECHNIQUE CLASSIFICATION



MESSAGE SWITCHING

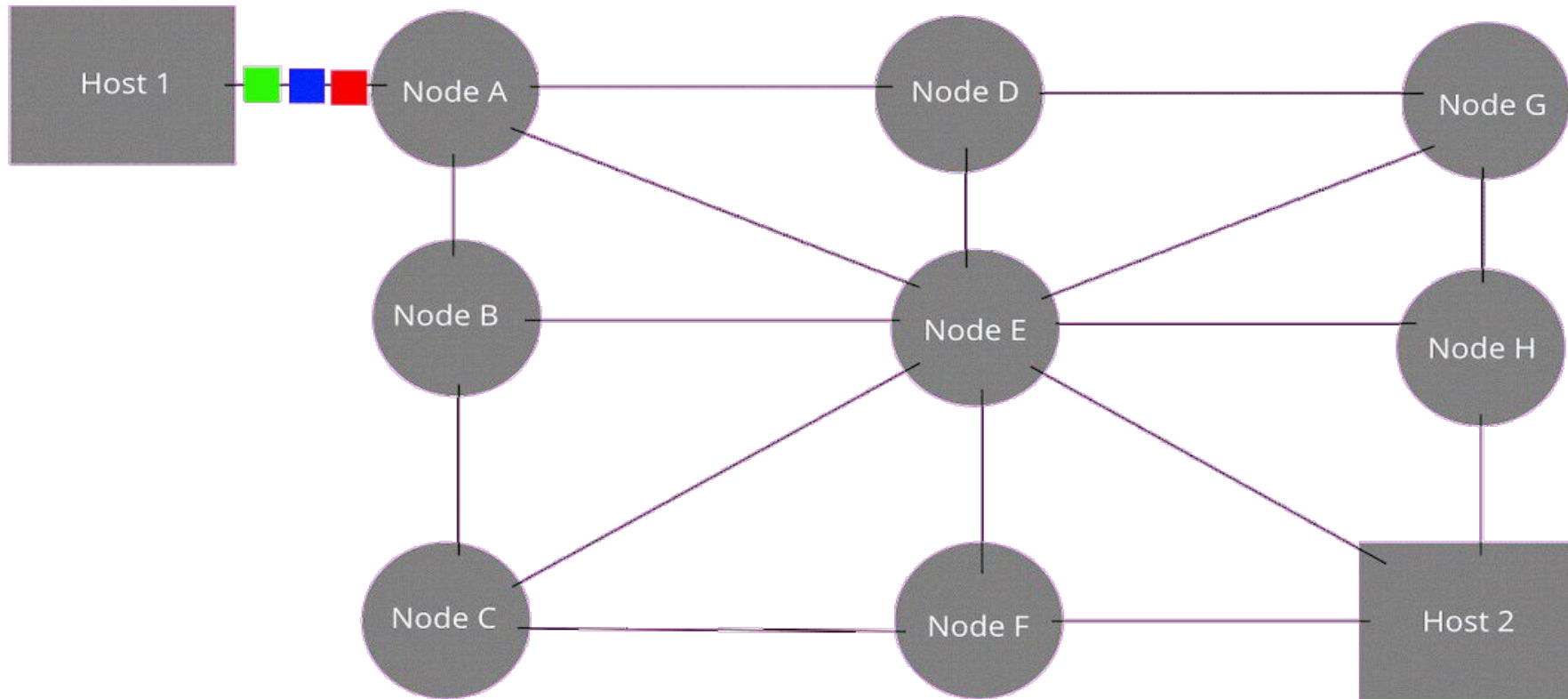


CIRCUIT SWITCHING → Telephone



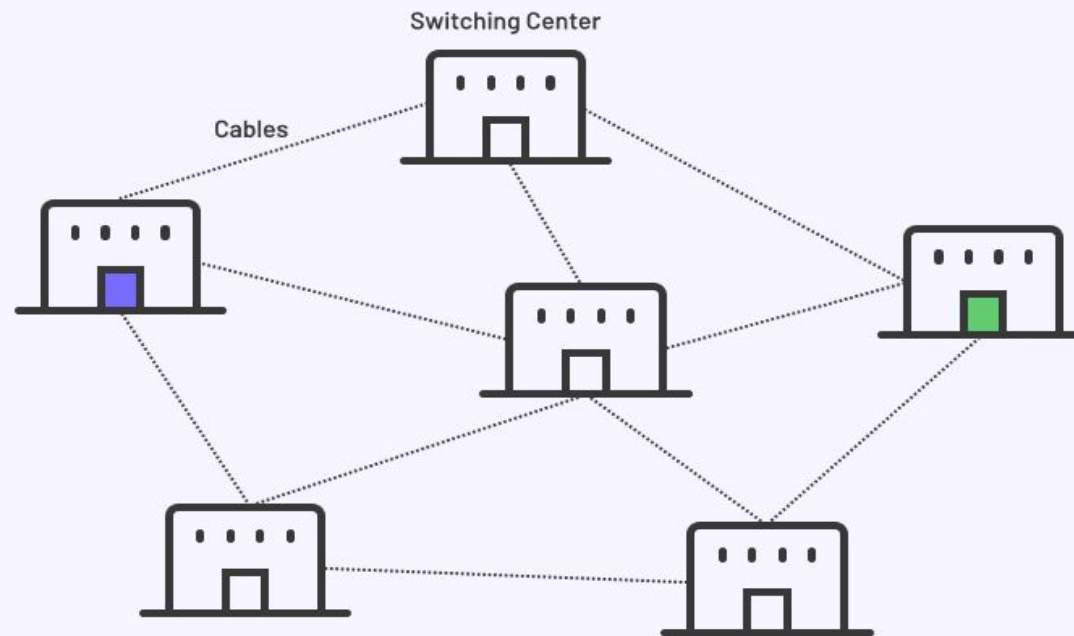
REPRESENTATION

The original message is **Green**, **Blue**, **Red**.



Circuit Switching

BALA'S
PHONE



LATA'S
PHONE



PACKET SWITCHING

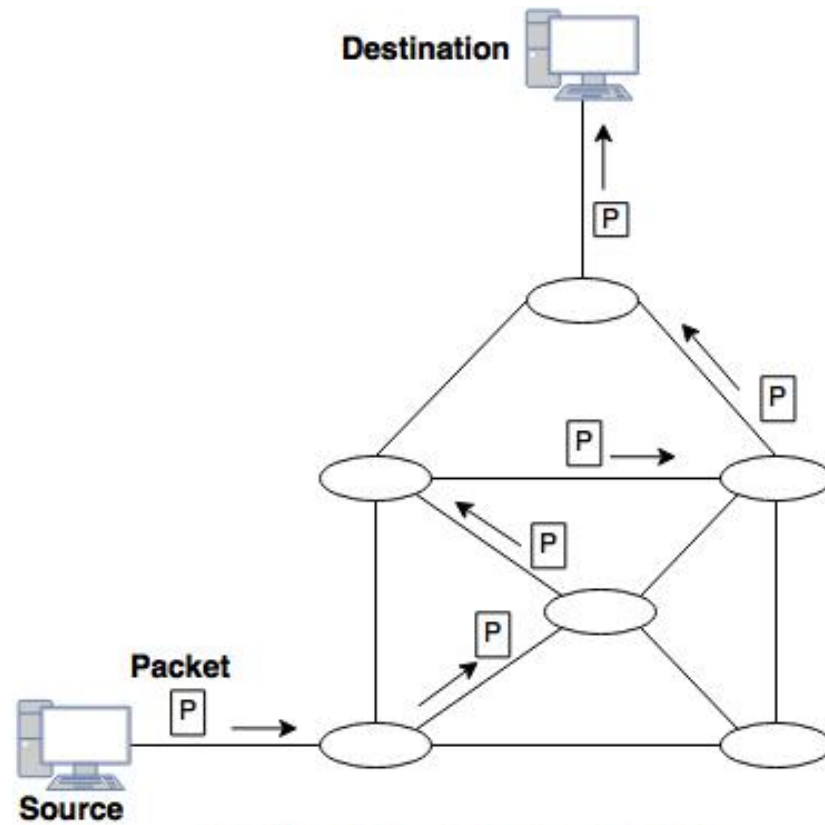


Fig: Virtual Circuit Packet Switching

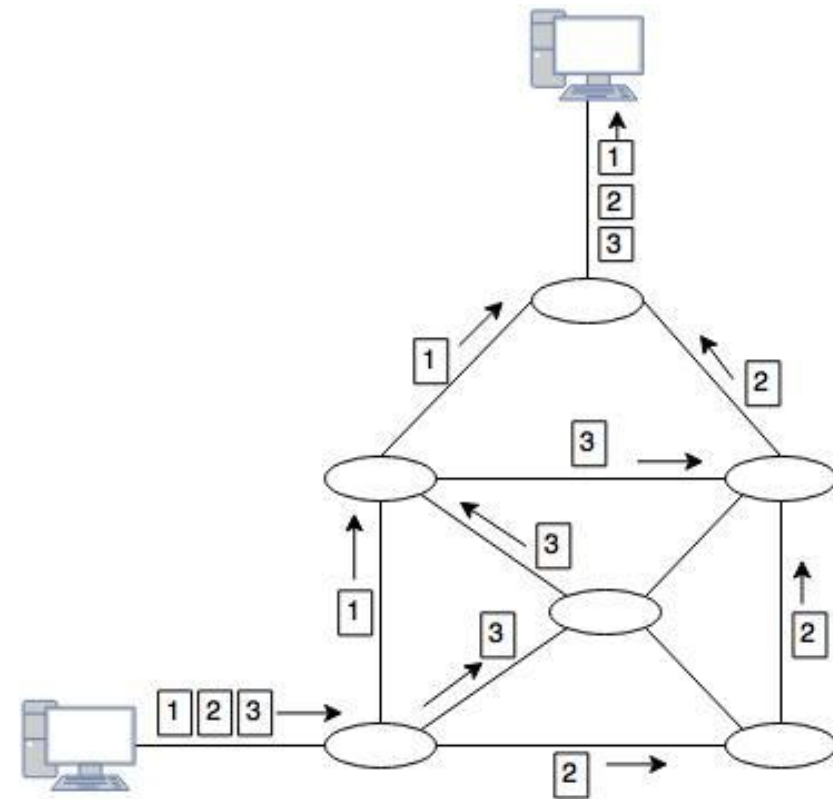


Fig: Datagram Packet Switching



SUMMARY

- Switching

Q. 1 What is switching and why?

Q. 2 What device used for switching?

Q. 3 What are the types of switching?

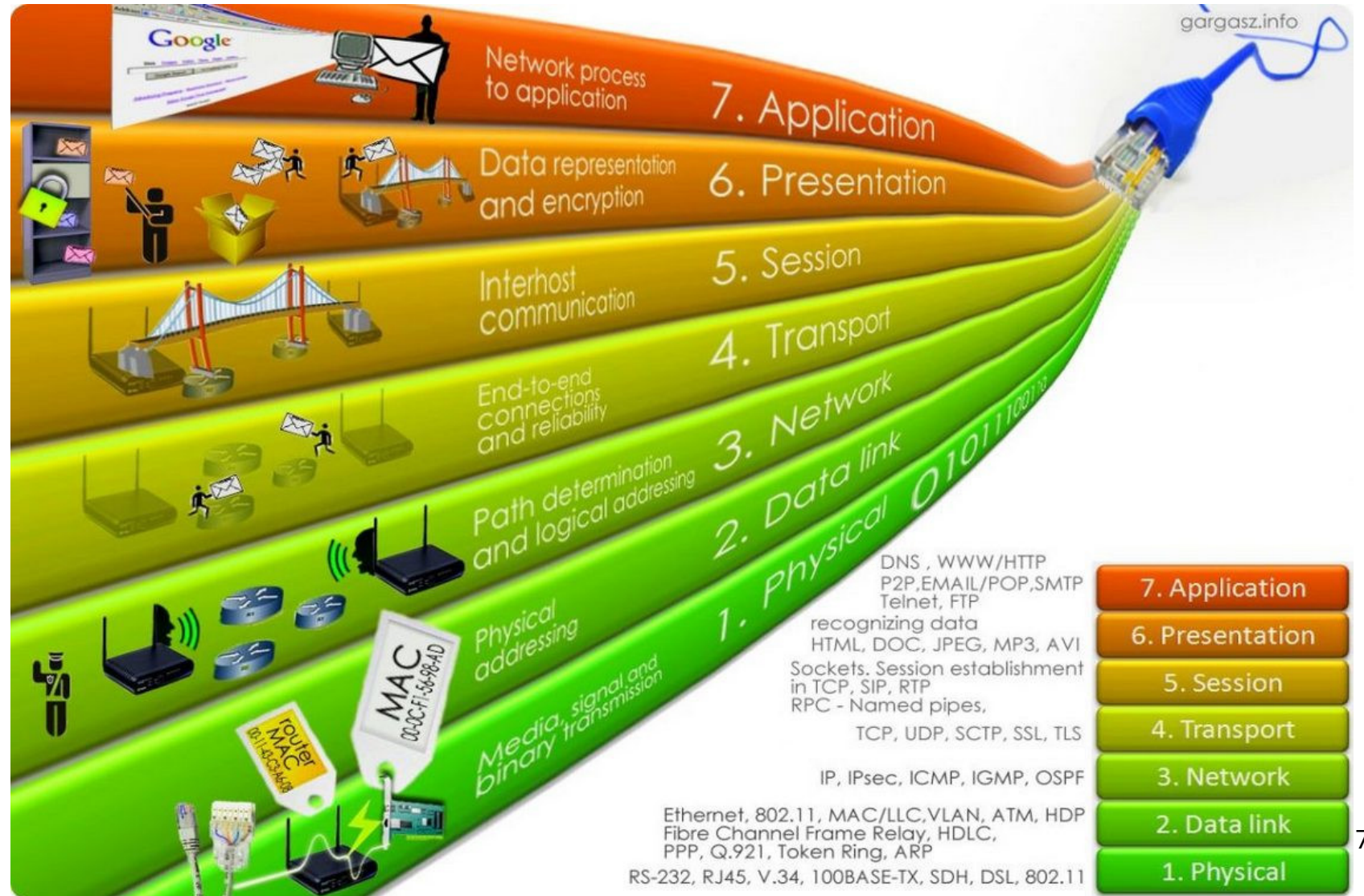
Q. 4 What are the differences between MS,CS, PS?

Q.5 What are the differences between VCPS and DPS?



REFER THE BELOW LINK

1. [All about MAC](#)



THANK YOU

